Lecture 1 (revised): Course Outline. Free Particle. Motion?

Quantum Mechanics is cruel. We cannot look directly at either intramoleclar structure or dynamics. But, as chemists or physicists, we seek and build beautiful detailed pictures of structure, dynamics, and mechanism.

How do we do this? How do we get what we want (more than mere description) from experiment or calculation? Reductionist.

5.73: Quantum Mechanics for **use**, not admiration, history, or philosophical challenges.

Insight and Intuition

Confidence to make free-hand drawn pictures (cartoons) of ψ and level structure (matrix structure) and wavepacket motion.

What are we allowed to know and how are we allowed to measure (or calculate) it?

Tricks for evaluating integrals
Tricks for reducing Quantum Mechanics to Classical Mechanics
The Periodic Table re-emerges.

5.73 Lecture #1

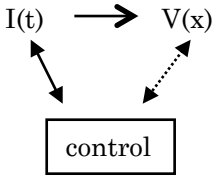
Course Outline

increasingly complex, mostly time-independent problems

* 1-Dimensional in $\Psi(x)$ picture (Schrödinger)

- spectrum $\{E_n\} \leftrightarrow$ potential $V(x)$ | central problem in Physical Chemistry, until recently

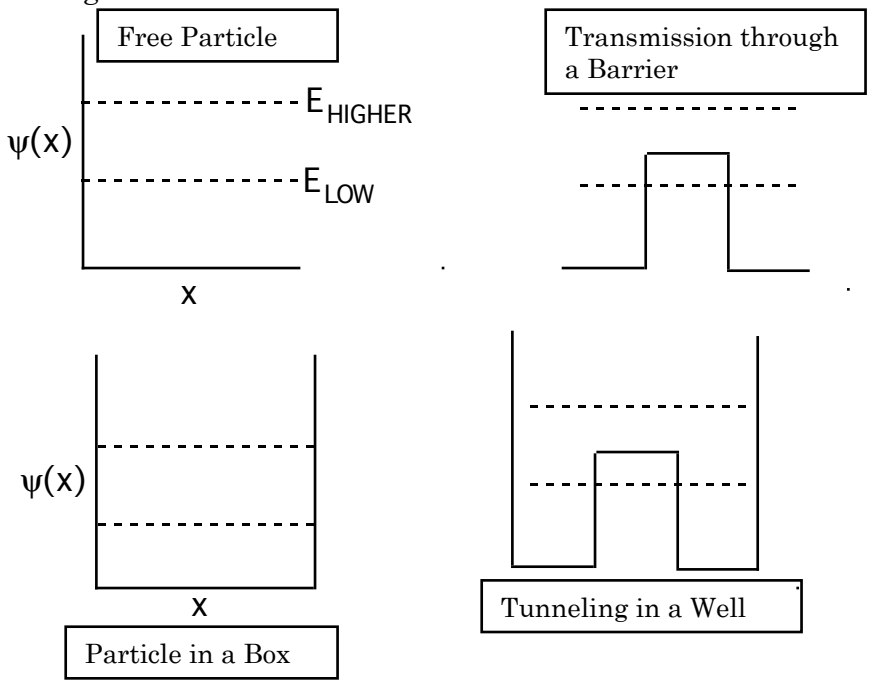
- femtochemistry: wavepackets exploring V(x), information about V(x) from timing experiments.



How is a wavepacket encoded for x_c, Δx, p_c, Δp? (c = center)
What to look for?

- stationary phase → trick to evaluate integrals
  → where does stuff "happen"

- Confidence to draw cartoons of $\Psi(x)$, even for problems that you have solved once but no longer remember the details.



Semi-classical: WKB. Connection of ψ across E = V(x) turning point. Quantization condition, RKR [Wentzel, Kramers, Brillouin, and Rydberg, Klein, Rees]

* Matrix Picture (Linear algebra)

- $\Psi(x)$ replaced by a large collection of numbers called "matrix elements"

- Main tool: **perturbation theory**
  - * small distortions from exactly solved problems
  - * f(quantum numbers) ↔F(potential parameters)

representation of spectrum — representation of potential

Vibration-Rotation Energy Levels:

e.g. Dunham Expansion

$$E_{vJ} = \sum_{\ell,m} Y_{\ell m}\left(v+\frac{1}{2}\right)^{\ell}\left[J\left(J+1\right)\right]^{m}$$

$$V(\xi) = \sum_{n=0} a_n \xi^n \qquad \xi \equiv \frac{R-R_e}{R_e}$$

The $\{a_n\}$ determine the $\{Y_{\ell m}\}$. Energy levels determine the potential energy curve.

- Linear Algebra: "Diagonalization"↔Eigenvalues and Eigenvectors

- How to set up and "read" a matrix.
- **Density Matrices**: specify general state of system ($\boldsymbol{\rho}$) and operators (**Op**) that correspond to a specific type of measurement, "populations" and "coherences".

***Problems in Three Dimensions**

- Central Forces: Radial x Angular factorization

Specific Problem — Universal Symmetry, exactly soluble

- The Hydrogen Atom: eigenstates and perturbations
- Rigid Rotor, $\mathbf{J}^2$, $\mathbf{J}_z$, $\mathbf{J}_\pm$ matrix elements and selection rules
- Vector Model
- Transformation between "coupled" and "uncoupled" basis sets
- Scattering : quantum mechanics for processes that involve unbound states.

***Many Particle Systems**

- many-electron atoms
- Slater determinants satisfy anti-symmetrization requirement for Fermions
- matrix elements of Slater determinantal wavefunctions
- orbitals→configurations→states ("terms"): L–S–J states
- molecular constants for many-electron systems↔orbital integrals

[Periodicity "recovered"]

* Many-Boson systems: anharmonically coupled vibrations
Intramolecular Vibrational Redistribution (IVR)
* Periodic Lattices: band structure of metals

Some warm-up exercises

Particle in a constant V(x)

* standard Quantum Mechanics problem ~ warm up

* practice with complex numbers

* building of intuition – "encoding" of motion

Hamiltonian $$H = T + V = \frac{p^2}{2m} + V(x)$$ Classical Mechanics

special QM prescription $$\hat{p}_x = \frac{\hbar}{i}\frac{\partial}{\partial x}$$ Why? Could have started with $\left[\hat{x}, \hat{p}_x\right] = i\hbar$

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)$$

Schrödinger Equation $$\left(\hat{H} - E\right)\psi = 0$$

Free particle in constant V(x)

$$\left(-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V_0 - E\right)\psi = 0$$ Schrödinger Equation

What kind of Ψ satisfies this differential equation?

$$\frac{d^2\psi}{dx^2} = -\left[\left(E - V_0\right)\frac{2m}{\hbar^2}\right]\psi$$

call this k^2

k real if $E \geq V_0$

k imaginary if $E < V_0$ (classically forbidden region)

Thus $k = \left[\left(E - V_0\right)\frac{2m}{\hbar^2}\right]^{1/2}$

general solution (more general than sin kx and cos kx)

$$\psi(x) = Ae^{ikx} + Be^{-ikx}$$

Two cases:

$E > V_0$ Classicallly allowed

$E < V_0$ Forbidden. $k \equiv i\kappa$ $\quad \kappa = \left[\left(V_0 - E\right)\frac{2m}{\hbar^2}\right]$ real

General solutions:

$$E > V_0 \quad \psi(x) = \underbrace{Ae^{ikx} + Be^{-ikx}}_{\text{oscillatory}} \quad k \text{ real}$$

$$E < V_0 \quad \psi(x) = \underbrace{ce^{\kappa x} + De^{-\kappa x}}_{\text{exponential}} \quad \kappa \text{ real}$$

Neat! form of $\psi(x)$ depends on sign of $E - V_0$

Some refresher for complex numbers:

$$z = x + iy \qquad i^2 = -1$$

(Re(z) = x, Im(z) = y)

$z^* = x - iy$
$2\mathrm{Re}(z) = (z + z^*)$
$2i\mathrm{Im}(z) = (z - z^*)$

$|z|^2 = zz^* = x^2 + y^2$ real and positive

$e^{\pm ikx} = \cos kx \pm i \sin kx$

$$\cos kx = \frac{1}{2}\left[e^{ikx} + e^{-ikx}\right]$$

$$\sin kx = \frac{1}{2i}\left[e^{ikx} - e^{-ikx}\right]$$

What happens when we apply $\hat{p}$ to e^{ikx}?

$$\hat{p}e^{ikx} = \frac{\hbar}{i}\frac{d}{dx}e^{ikx} = \hbar k e^{ikx}$$

eigenfunction of $\hat{p}$

eigenvalue of $\hat{p}$

$\hbar k = p$ — a number, not an operator

$p = +\hbar k$ motion in $+x$ direction
$p = -\hbar k$ motion in $-x$ direction

OK, we have some connection to Classical Mechanical motion. We want to ask: how is motion encoded in $\Psi(x)$?

* Intuition
* Before we get to the TDSE $\hat{H}\Psi(x,t) = i\hbar\frac{\partial\Psi}{\partial t}$ (mostly in 5.74)

How is $\psi(x)$ **encoded** for motion?

My favorite word
in Quantum Mechanics

e^{ikx} is an eigenfunction of $\hat{p}_x$. It is also a complex function of a real variable.

k is called the "wave vector" (or wave number). Why is it called wave vector?

- in 3-D where $e^{i\vec{k}\cdot\vec{r}}$ $\vec{k}$ points in direction of motion
- $e^{i(kx+2\pi)} = e^{ikx}$ e^{ikx} is periodic $x \rightarrow x+\lambda$
- $e^{ik(x+\lambda)} \equiv e^{ikx}$ $\therefore$ $k\lambda = 2\pi$ $k = \frac{2\pi}{\lambda}$

advance x by one full oscillation cycle $\equiv \lambda$ "wavelength"

k is # of waves per 2π unit length

Now go back to $\Psi(x)$.

ψ is probability amplitude density $\quad \psi = Ae^{ikx} + Be^{-ikx}$ (travels to left?)

probability distribution $\quad \psi^*\psi = |A|^2 + |B|^2 + A^*Be^{-2ikx} + AB^*e^{2ikx}$

this sum had better be real!

Why? It is $\mathrm{Re}(A^*B\,e^{-2ikx})$!

(AB^*e^{2ikx}: travels to right?)

simplify: $\quad z = \mathrm{Re}(z) + i\,\mathrm{Im}(z)$

$$2\,\mathrm{Re}\left(A^*B\right) = A^*B + AB^*$$

$$2i\,\mathrm{Im}\left(A^*B\right) = A^*B - AB^*$$

$$e^{\pm i\alpha} = \cos\alpha \pm i\sin\alpha$$

We know $|\Psi|^2$ is real because it has the form $z + z^*$.

Look at it in more detail. The cross-terms:

$$A^*B(\cos 2kx + i\sin 2kx) + AB^*(\cos 2kx - i\sin 2kx) =$$
$$(A^*B + AB^*)\cos 2kx + (A^*B - AB^*)i\sin 2kx =$$
$$2\mathrm{Re}(A^*B)\cos 2kx + 2i\,\mathrm{Im}(A^*B)i\sin 2kx =$$
$$2\mathrm{Re}(A^*B)\cos 2kx - 2\mathrm{Im}(A^*B)\sin 2kx.$$

(note $i \times i = -1$)

This is real for all x. It has x-regions of *positive* and *negative* value.

$$|\psi|^2 = \psi^*\psi = \underbrace{|A|^2 + |B|^2}_{\text{constant}} + \underbrace{2\mathrm{Re}(A^*B)\cos 2kx + 2\mathrm{Im}(A^*B)\sin 2kx}_{\text{wiggly – only if both A and B are nonzero}}$$

(delocalized particle)

standing wave, real, neither complex nor imaginary

If $|A| > |B|$ wiggles encode net $+x$ motion. What is fraction $\frac{\text{to right?}}{\text{to left?}}$

A,B are determined by problem-specific boundary conditions.

• Can't directly see any motion in x unless we go to time-dependent Schrödinger Equation

• Need superposition of +k and –k parts in $\Psi(x)$ to get wiggles.



= superpositon of waves with different values of k

= "localization" requires another kind of superposition (wave-packet)

• Motion becomes really clear when we do two things:

* time-dependent $\Psi(x,t)$

* create localized states called wavepackets by superimposing several e^{ikx} with *different* |k|'s.

Now for the $E < V_0$ case.

$e^{\pm \kappa x}$ (with $E < V$ below) does not have wiggles, no motion in classically forbidden region.

To convince yourself, go back to

$$\psi(x) = Ce^{\kappa x} + De^{-\kappa x}$$

$$|\psi|^2 = |C|^2 e^{2\kappa x} + |D|^2 e^{-2\kappa x} + \underbrace{CD^* + C^* D}_{\text{real and independent of } x}$$

To *really* see **motion**, need to use time-dependent Schrödinger Equation.

To see **localization**, need superposition of several e^{ikx} with different values of $|k|$.

NEXT LECTURE: CTDL, pages 21-24, 28-31 (motion, infinite box, δ-function potential, scattering off δ-function barrier).

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5.73 Quantum Mechanics I
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Infinite Box, $\delta(x)$ Well, $\delta(x)$ Barrier.

Last Time: free particle $V(x)=V_0$

$\psi = Ae^{ikx} + Be^{-ikx}$ general solution

A,B are complex constants, determined by "boundary conditions"

$k = \frac{p}{\hbar}$ (from e^{ikx}, an eigenfunction of $\hat{p}$ for a free particle, and the real number, $\hbar k = p$, is the eigenvalue of $\hat{p}$)

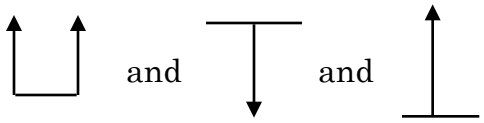
$$k = \left[(E - V_0) \frac{2m}{\hbar^2} \right]^{1/2} \quad \text{for } E \geq V_0$$

probability distribution

$$P(x) = \psi^* \psi = \underbrace{|A|^2 + |B|^2}_{\text{const.}} + \underbrace{2\text{Re}(A^*B)\cos 2kx + 2\text{Im}(A^*B)\sin 2kx}_{\text{wiggly, real at all } x}$$

only get wiggly stuff when ψ contains a superposition of 2 or more different values of k are superimposed. In this special case we had the two values of k: $+k$ and $-k$.

TODAY



1. infinite box
2. δ(x) well
3. δ(x) barrier (non-lecture)

What do we know about a $\psi(x)$ for a physically realistic V(x)?

$\psi(\pm\infty) = ?$

$\psi^*(x)\psi(x)$ for all x?

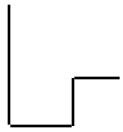
$\int_{-\infty}^{\infty} \psi^*(x)\psi(x)dx$?

Continuity of ψ and $d\psi/dx$?

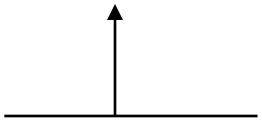
Computationally convenient potentials have steps and flat regions.



infinite step



finite step



infinitely high but infinitely thin step,"δ-function"

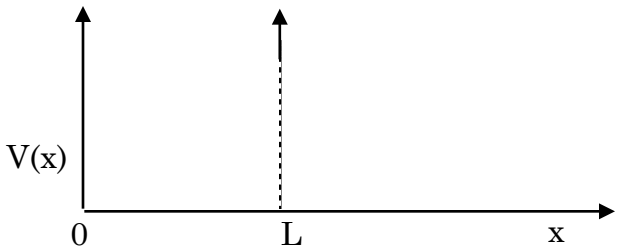
ψ continuous

$\frac{d\psi}{dx}, \frac{d^2\psi}{dx^2}$ not continuous for *infinite* step, and not for δ-function

$\frac{d\psi}{dx}$ is continuous for *finite* step

<u>More warm up exercises</u>

1. Infinite box



$\psi(x) = Ae^{ikx} + Be^{-ikx} = C\cos kx + D\sin kx \qquad [C = A + B, D = iA - iB]$

Where do these 2 equations come from? Be sure you can derive (and never forget) these 2 equations for C and D.

Boundary conditions:

$\psi(0) = 0 \Rightarrow C = 0$

$\psi(L) = 0 \Rightarrow kL = n\pi \qquad n = 1,2,\ldots \qquad (\text{why not n} = 0?)$

recall $k^2 = \left(E - V_0\right)\frac{2m}{\hbar^2} = \boxed{\frac{n^2\pi^2}{L^2}}$ $V_0 = 0$ here.

Insert $kL = n\pi$ boundary condition

$$E_n = n^2 \frac{\hbar^2\pi^2}{2mL^2} = n^2\left[\frac{h^2}{8mL^2}\right]$$

∞ # of bound levels

E_1

$n = 0$ would be empty box!

E_n is integer multiple of common factor, E_1. Important for many wavepacket problems!

normalization (P=1 for 1 particle in well)

$$1 = |D|^2 \int_0^L dx \sin^2(n\pi x / L)$$

$$\psi_n(x) = (2/L)^{1/2}\sin(n\pi x / L)$$

$\Longrightarrow$ $|D| = (2/L)^{1/2}$ because $\int_0^L \sin^2(n\pi x/L)dx = L/2$

$$D = (2/L)^{1/2}\underbrace{e^{ia}}_{\substack{\text{arbitrary}\\ \text{phase}\\ \text{factor}}}$$

cartoons of $\psi_n(x)$: what happens to $\{\psi_n\}$ and $\{E_n\}$ if we move the well:

left or right in x?

up or down in E?

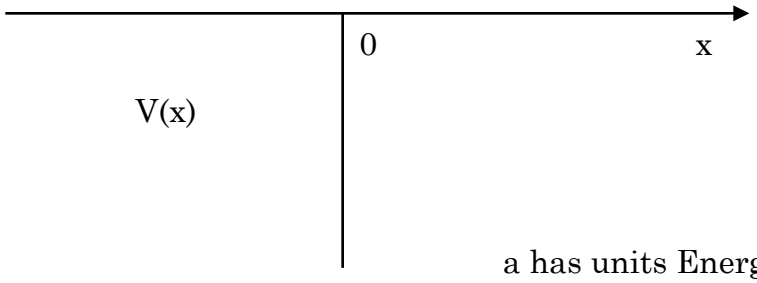
There is always a short-cut. A picture is often more informative than equations.

Infinite well was easy: 2 boundary conditions plus a normalization requirement.

Generalize to stepwise constant potentials: in each V(x)=constant region, need to know 2 complex coefficients and, if the particle is confined within a finite range of x, there is quantization of energy.

* boundary and joining conditions
* normalization
* overall phase arbitrariness

So next step is to deal with case where boundary conditions are not so obvious. $\delta(x)$ well and barrier.



$V(x) = -a\,\delta(x)$ $\quad a > 0$

a has units Energy x Length (because, as we will see, δ(x) has units of reciprocal length)

= 0 everywhere except V(0) = –a "∞", a is the "strength" of the δ-function well

Schrödinger Equation $\quad \dfrac{d^2\psi}{dx^2} = -\left(\underbrace{(E + a\delta(x)}_{E-V(x)}\right)\dfrac{2m}{\hbar^2}\psi$

Integrate: $\quad \lim_{\varepsilon\to 0}\int_{-\varepsilon}^{+\varepsilon}\dfrac{d^2\psi}{dx^2}dx = -\lim_{\varepsilon\to 0}\left[\int_{-\varepsilon}^{+\varepsilon}dx\left(\dfrac{2mE}{\hbar^2}\psi(x) + \dfrac{2ma}{\hbar^2}\delta(x)\psi(x)\right)\right]$

$LHS = \left.\dfrac{d\psi}{dx}\right|_{x=+\varepsilon} - \left.\dfrac{d\psi}{dx}\right|_{x=-\varepsilon}$ = size of discontinuity in $\dfrac{d\psi}{dx}$ at x = 0

$$RHS = \left[0 \quad - \quad \frac{2ma}{\hbar^2}\psi(0)\right]$$

because $\dfrac{2mE}{\hbar^2}\psi(0)$ is finite and the integral is over region of length $2\varepsilon \approx 0$.

because, by the definition of a δ –fn,

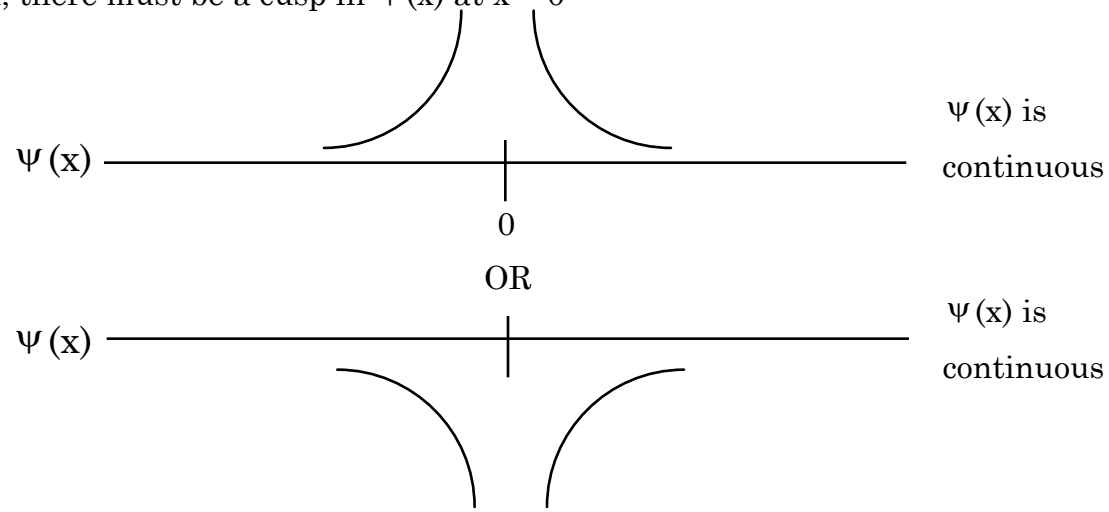
$\int \delta(x)\psi(x)dx = \psi(0)$

or, more generally

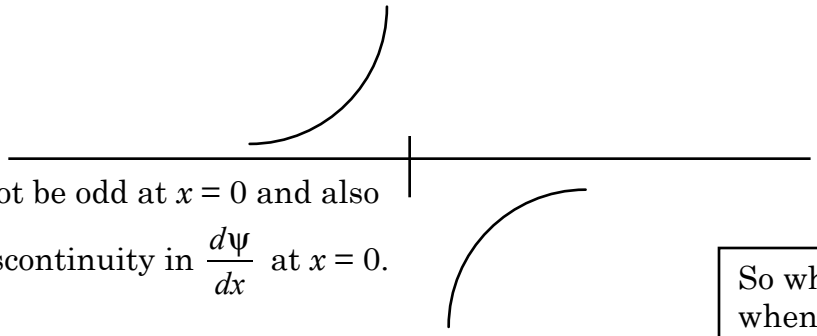
$\int_{-\infty}^{\infty} \delta(x - a)\psi(x)dx = \psi(a)$

This is a really important derivation. You will want to remember it!

Since the potential has even symmetry with respect to $x \to -x$, $\psi(x)$ must be even or odd (not a mixture) with respect to $x \to -x$, thus $\psi(x) = \pm\psi(-x)$. If $\psi(x)$ is an even function, there must be a cusp in $\psi(x)$ at $x = 0$



BUT NOT



ψ(x) cannot be odd at $x = 0$ and also have a discontinuity in $\frac{d\psi}{dx}$ at $x = 0$.

So what happens when Ψ(x) is an odd function?

$$\frac{d\psi(+)}{dx} - \frac{d\psi(-)}{dx} = -\frac{2ma}{\hbar^2}\psi(0)$$

Thus we have a new connection condition on $\frac{d\psi}{dx}$

since there must be + reflection symmetry for an even $\psi(x)$

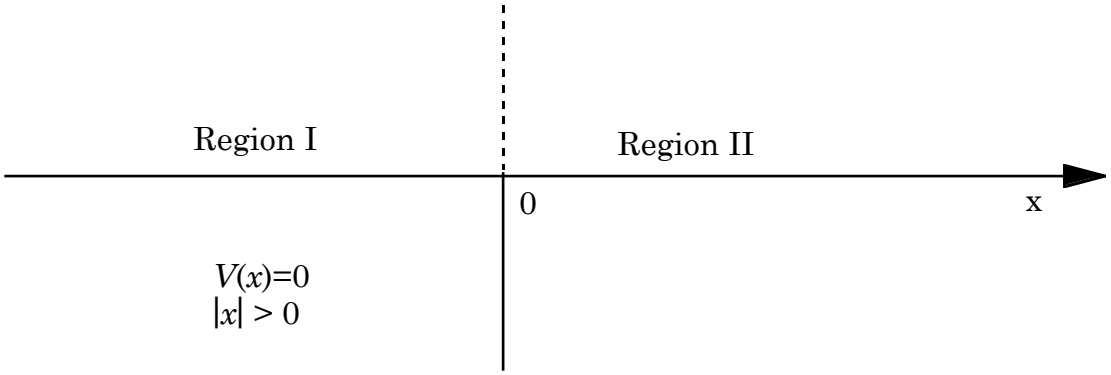
$$\frac{d\psi(+)}{dx} = -\frac{d\psi(-)}{dx}$$

$$\frac{d\psi(\pm)}{dx} = \mp\frac{ma}{\hbar^2}\psi(0)$$

$$\frac{d\psi(+)}{dx} - \frac{d\psi(-)}{dx} = -\frac{2ma}{\hbar^2}\psi(0)$$

This is the δ-function joining condition for a symmetric potential.

Now find the eigenfunctions and eigenvalues. Standard procedure: divide space into regions and match ψ and dψ/dx across boundaries.



Let E < 0 $E = -|E|$

$\psi_L \equiv \psi_I = A_L e^{+\kappa x} + B_L e^{-\kappa x}$

$\psi_R \equiv \psi_{II} = A_R e^{+\kappa x} + B_R e^{-\kappa x}$

$\kappa = \left[\frac{|E|2m}{\hbar^2}\right]^{1/2}$

$\kappa = ik$

(8 unknowns, because A and B can be complex numbers)

(THIS IS WHAT WE ALWAYS DO WHEN k IS IMAGINARY)

		unknowns determined	
$\psi(+\infty) = 0$	$A_R = 0$	(2)	$\psi_L = Ae^{\rho x}$
$\psi(-\infty) = 0$	$B_L = 0$	(2)	$\psi_R = Ae^{-\rho x}$
$\psi_L(-\varepsilon) = \psi_R(+\varepsilon)$	$A_L = B_R \equiv A$	(2)	
	arbitrary phase	(1)	
	normalization	(1)	**Done!**
		(8)	**TOTAL**

$$\frac{d\psi_R(+)}{dx} = -\kappa A e^{-0} = \frac{-ma}{\hbar^2}\underbrace{|\psi(0)|}_{A}$$

required discontinuity in $d\psi/dx$ at $x = 0$.

$$\therefore \kappa = \frac{ma}{\hbar^2}$$

$$\frac{d\psi_L(-)}{dx} = +\kappa A e^{+0} = \frac{+ma}{\hbar^2}\underbrace{|\psi(0)|}_{A}$$

again $\kappa = \frac{ma}{\hbar^2}$

Only one acceptable value of $\kappa \rightarrow$ one value of E < 0

$$\kappa = \frac{ma}{\hbar^2}$$

$$|E| = \frac{\kappa^2\hbar^2}{2m} = \frac{ma^2}{2\hbar^2} = -E$$

$$\mathrm{E} = -\frac{\mathrm{ma}}{2\hbar^2}$$

Actually, the above solution was specifically for an even ψ(x). What about an odd ψ(x) for a V(*x*) with a $\delta(x)$ at x = 0? *No calculation is needed.* Why?

Normalization of ψ

$$1 = \int_{-\infty}^{\infty} |\psi|^2 \, dx$$

$$\psi_R = Ae^{-\mathrm{max}/\hbar^2}$$

$$1 = 2\int_0^{\infty} |A|^2 \, e^{-\left(2ma/\hbar^2\right)x} dx = 2|A|^2 \left(\frac{\hbar^2}{2ma}\right)$$

see Gaussian Handout

$$A = \pm\left(\frac{ma}{\hbar^2}\right)^{1/2}$$

$$\psi_\delta = \pm\left(\frac{\mathrm{ma}}{\hbar^2}\right)^{1/2} \mathrm{e}^{-\mathrm{ma|x|}/\hbar^2}$$

only one bound level, regardless of magnitude of a

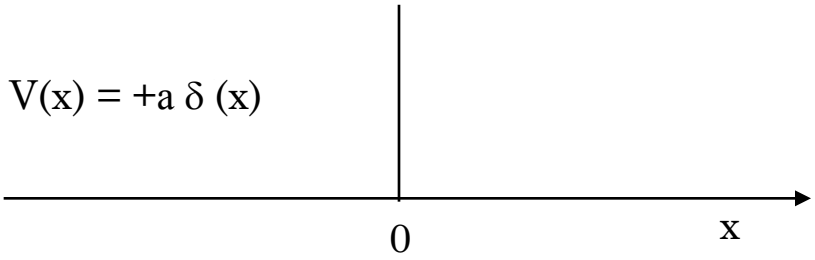
large a, narrower and taller ψ

There is a continuum of ψ's possible for E > 0. Since the particle is free for E > 0, specific form of ψ must reflect specific problem:

e.g., particle probability incident from x < 0 region. It is even more interesting to turn this into the simplest of all barrier scattering problems. See Non-Lecture pp. 2-8, 9, 10.

Nonlecture

Consider instead scattering off of V(x) = + a δ(x) a > 0



$\psi_L = A_L e^{ikx} + B_L e^{-ikx}$

$\psi_R = A_R e^{ikx} + B_R e^{-ikx}$

$$k = \left(\frac{2mE}{\hbar^2} \right)^{1/2}$$

In this problem let's assume that we have flux entering exclusively from the left. The entering probability flux is $|A_L|^2$.

Two things can happen:

1. transmit through barrier $\propto |A_R|^2$
2. reflect at barrier $\propto |B_L|^2$

There is no way that $|B_R|^2$ can become different from 0. Why? (Hint: where does the flux enter the system and in what direction is it flowing?)

Our goal is to determine $|A_R|^2$ and $|B_L|^2$ vs. E.

$\psi_L(0) = \psi_R(0)$ ⇩ continuity of ψ

$A_L + B_L = A_R + B_R$ but $B_R = 0$ $A_L + B_L = A_R$

$$\left[\frac{d\psi_R(+0)}{dx} - \frac{d\psi_L(-0)}{dx} - \right] = + \frac{2ma}{\hbar^2} \psi(0)$$

discontinuity of $\frac{d\psi}{dx}$ at δ-function

$$ikA_R - (ikA_L - ikB_L) = \frac{2ma}{\hbar^2} A_R \quad \leftarrow \psi_R(0)$$

$A_R = A_L + B_L$

$$ik(A_L + B_L) - ik(A_L - B_L) = \frac{2ma}{\hbar^2}(A_L + B_L)$$

$\psi_L(0)$

$$2ikB_L = \frac{2ma}{\hbar^2}\left(A_L + B_L\right)$$

$$B_L\left(2ik - \frac{2ma}{\hbar^2}\right) = \frac{2ma}{\hbar^2}A_L$$

$$\frac{A_L}{B_L} = \frac{\hbar^2}{2ma}\left(2ik - \frac{2ma}{\hbar^2}\right) = \frac{ik\hbar^2}{ma} - 1 \equiv \alpha$$

$$\alpha + 1 = \frac{ik\hbar^2}{ma}$$

$$A_R = A_L + B_L = A_L\frac{B_L}{B_L} + B_L = \alpha B_L + B_L = B_L(\alpha + 1)$$

$\alpha = A_L/B_L$

$$\boxed{A_R = B_L\left(\frac{ik\hbar^2}{ma}\right)}$$

Transmission is $T = \dfrac{\left|A_R\right|^2}{\left|A_L\right|^2}$: $\dfrac{(\text{moving to right in R region})}{(\text{incident from left in L region})}$

Reflection is $R = \dfrac{\left|B_L\right|^2}{\left|A_L\right|^2}$: $\dfrac{(\text{moving to left in L region})}{(\text{incident from left in L region})}$

What is T(E), R(E)?

$$\boxed{\left|A_R\right|^2 = \left|B_L\right|^2\frac{k^2\hbar^4}{m^2a^2} = \left|B_L\right|^2\frac{2mE}{\hbar^2}\frac{\hbar^4}{m^2a^2} = \left|B_L\right|^2\frac{2\hbar^2E}{ma^2}}$$

$$\left(\frac{A_L}{B_L}\right)\left(\frac{A_L}{B_L}\right)^* = \left(\frac{ik\hbar^2}{ma} - 1\right)\left(-\frac{ik\hbar^2}{ma} - 1\right)$$

$$\frac{\left|A_L\right|^2}{\left|B_L\right|^2} = \frac{k^2\hbar^4}{m^2a^2} + 1 = \frac{2\hbar^2E + ma^2}{ma^2} \qquad \left[k = \left(\frac{2mE}{\hbar^2}\right)^{1/2}\right]$$

$$R(E) = \frac{ma^2}{2\hbar^2E + ma^2} = \left[\frac{2\hbar^2E}{ma^2} + 1\right]^{-1}$$ decreasing to zero as E increases

$$T(E) = \frac{2\hbar^2E}{2\hbar^2E + ma^2} = \left[\frac{ma^2}{2\hbar^2E} + 1\right]^{-1}.$$ increasing to one as E increases

$$R(E) + T(E) = 1$$

Note that: $R(E)$ starts at 1 at $E = 0$ and goes to 0 at $E \to \infty$

$T(E)$ starts at 0 and increases monotonically to 1 as E increases.

Note also that extending the equations for $R(E)$ and $T(E)$ to $E < 0$, we see at

$\boxed{E = -\frac{ma^2}{2\hbar^2}}$ $R \to \infty$ as E approaches $-ma^2/2\hbar^2$ from above and then changes sign as E passes through $-ma^2/2\hbar^2$!

This is the energy of the bound state in the δ (x)-function well problem.

This tells you that something special happens when you "extend" the scattering calculation to scattering off a V(x) at the energy of a bound state. This is strange because it is difficult to imagine scattering at E < 0.

See CTDL Chapter 1 Problem #3b (page 87) for a related problem

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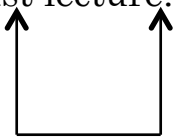
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Lecture #3: $|\psi(x,t)|^2$: Motion, Position, Spreading, Gaussian Wavepacket

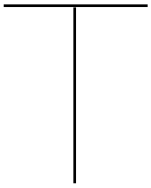
Reading Chapter 1, CTDL, pages 9-39, 50-56, 60-85

Last lecture:



Infinite well

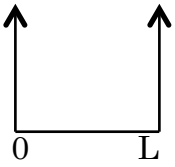
and



Delta-function well

What are the key points?

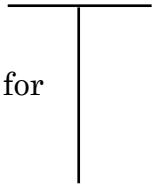
E_n, ψ_n for



$$E_n = n^2\left[\frac{h^2}{8mL^2}\right]$$

$$\psi_n = (2/L)^{1/2}\sin(n\pi x)$$

E, ψ for



$$E = -\frac{ma^2}{2\hbar^2}$$

$$\psi = \left(\frac{ma}{\hbar^2}\right)^{1/2} e^{-ma|x|/\hbar}$$

Do E and ψ for delta function well behave as you expect?

TODAY: Can we construct a Ψ(x,t) for which $|\Psi|^2$ acts like a CM particle, but with correct QM characteristics?

* stationary phase point and its motion
* stationary phase approximation for evaluating an integral with wiggly integrand

Motion *requires* $\Psi(x,t)$ from TDSE! Motion is *encoded* in $\psi(x)$, but we will need to actually observe motion (pages 3-4 thru 3-12).

Our goal is to use a well understood function that appears frequently in quantum mechanics, a normalized Gaussian, as a particle-like quantum mechanical state function, a "Gaussian Wavepacket."

What we want is to know how the time-evolving center position, center amplitude, center velocity, and the width of this wavepacket are encoded in the mathematical expression. This will guide us in constructing particle-like states with chosen properties and in knowing how to recognize these properties in an unfamiliar state function.

From a stationary Gaussian G(x; x_0, Δx) to a moving Gaussian wavepacket $|\Psi(x,t)|^2$

1. $G(x;x_0,\Delta x) = (2\pi)^{-1/2}(\Delta x)^{-1} e^{-(x-x_0)^2/2(\Delta x)^2}$

Normalized: $\int_{-\infty}^{\infty} G(x;x_0,\Delta x)\,dx = 1$

You can show by evaluating the integral that $G(x; x_0, \Delta x)$ is normalized to 1.

$\langle x^2 \rangle \equiv \int_{-\infty}^{\infty} G(x;x_0,\Delta x) x^2 dx$ and a similar equation for $\langle x \rangle$.

The width, Δx, the standard deviation of $G(x)$, is the square root of the variance

$$\Delta x = \left[\langle x^2 \rangle - \langle x \rangle^2 \right]^{1/2}$$

Finally, we want a function that is normalized to 1 at $t = 0$

$$1 = \int_{-\infty}^{\infty} |\Psi(x,0)|^2 dx$$

$G(x;x_0,\Delta x) = \Psi(x,0)^* \Psi(x,0) = |\Psi(x,0)|^2$

normalized as $\int_{-\infty}^{\infty} |\Psi(x,0)|^2 dx = 1$

$\Psi(x,0) = (2\pi)^{-1/4}(\Delta x)^{-1/2} e^{-(x-x_0)^2/4(\Delta x)^2}$

This is also a Gaussian. Ψ(x,0) is broader and not as tall as $G(x;x_0,\Delta x)$ at $x = x_0$.

2. How do we get to the following complicated-looking textbook function?

$$|\Psi(x,t)|^2 = \underbrace{\left(\frac{2}{\pi a^2}\right)^{1/2}\left(1+\frac{4\hbar^2 t^2}{m^2 a^4}\right)}_{\text{time-dependent normalization and magnitude. Since probability is conserved, the normalization factor must be t-dependent because the denominator of the exponential factor is t-dependent.}} e^{\underbrace{-\left[\frac{2a^2\left(x-\frac{\hbar k_0}{m}t\right)^2}{a^4+\frac{4\hbar^2 t^2}{m^2}}\right]}_{\text{spreading and moving with minimum width at } t=0}}$$

$$|\Psi(x,0)|^2 = \left(\frac{2}{\pi a^2}\right)^{1/2} e^{-2x^2/a^2}$$

at $t = 0$, by comparison to the normalized Gaussian

$$\left(\frac{2}{\pi a^2}\right)^{1/2} = \left(\frac{1}{2\pi(\Delta x)^2}\right)^{1/2}$$

we have $\Delta x(t=0) = a/2$.

Now, for motion of the center of the wavepacket, $x_0(t)$, we expect that

$$x_0(t) = x_0(0) + \frac{p_0}{m}t \qquad p_0 \text{ is momentum at } t=0$$

$$v_0(0) = \frac{p_0(0)}{m} = \frac{\hbar k_0(t)}{m} \qquad v_0 \text{ is velocity at } t=0,\ k_0 \text{ is the wavenumber, } k = p/\hbar \text{ at } t=0$$

$$x_0(t) = x_0 + \frac{\hbar k_0(t)}{m}t$$

$\Delta x(0) = a/2.$ Width increases as $|t|$ increases from $t = 0$.

Wavepacket is moving and changing its width. Minimum width is at $t = 0$.

Could shift the t at which minimum width occurs by replacing t by $t' = t + \delta$ in the formula for $\Psi(x,t)$.

How do we know that the width of the wavepacket is t-dependent? If the value of Ψ at the t-dependent center is changing and $\Psi(x,t)$ is normalized, then the wavepacket must be spreading or contracting. We will have to look at the t-dependent Schrödinger equation to see how the momentum depends on x.

NON-LECTURE

The Fourier transform of a Gaussian is another Gaussian. This means that if you have a wavepacket, $|\Psi(x,0)|^2$, with a Gaussian shape, the momentum distribution of this wavepacket, $|\Phi(p,0)|^2$, will also be a Gaussian. This Gaussian distribution of the momentum will cause the time-dependent spatial shape of the wavepacket to be either stretching or compressing. If the wavepacket shape, $|\Psi(x,t)|^2$, expands as t advances, it compresses as t decreases until it reaches the minimum possible width and then re-expands. The widths, Δx and Δp, are reciprocally related and the minimum uncertainty wavepacket, at the t when $\Delta x \Delta p$ reaches its minimum value, is of particular interest. It is at this instant that the quantum mechanical wavepacket maximally resembles a classical particle.

How do we get from $G(x; x_0, \Delta x)$ to $|\Psi(x,t)|^2$?

Time Dependent Schrödinger Equation (TDSE)

$$\mathbf{H}\Psi = i\hbar\frac{\partial\Psi}{\partial t}$$

Time Independent Schrödinger Equation (TISE)

$$\mathbf{H}\Psi = E\psi \qquad \text{or} \qquad \mathbf{H}\psi_n = E_n\psi_n$$

Special very useful case: if **H** is independent of time and if we know the solutions to the TISE, hen it is trivial to go from $\{E_n, \psi_n\}$ to $\Psi(x,t)$.

Suppose we create an arbitrary state at $t = 0$. It is always possible to express this arbitrary state as a linear combination of eigenstates of **H**,

$$\psi(x) = \sum_n a_n \psi_n$$

because the set of $\{\psi_n\}$ is "complete". We can convert this $\psi(x)$ to $\Psi(x,t)$ very simply:

$$\Psi(x,t) = \sum_n c_n \psi_n e^{-iE_n t/\hbar}$$

Show that this satisfies the TDSE:

$$i\hbar\frac{\partial\Psi}{\partial t} = i\hbar\sum_n (-iE/\hbar)c_n\psi_n e^{-iE_nt/\hbar}$$

$$= \sum E_n c_n \psi_n e^{-iE_nt/\hbar}$$

$$\mathbf{H}\Psi = \mathbf{H}\sum_n c_n\psi_n e^{-iE_nt/\hbar}$$

$$= \sum_n E_n c_n \psi_n e^{-iE_nt/\hbar}$$

same, so the TDSE is satisfied

It is clear that $\Psi(x,t)$ "moves", but we still need help in understanding that motion.

(i) $\int_{-\infty}^{\infty}\Psi^*(x,t)\Psi(x,t)dx = 1$

no motion because all $\Delta n \neq 0$ integrals involving $\int_{-\infty}^{\infty}\psi_n^*\psi_{n'}dx = 0$ by orthogonality.

(ii) $\Psi^*(x,t)\Psi(x,t)$ evolves in time if eigenstates that belong to *at least two* different E_n are included.

For example,

$$\Psi_{1,2} = c_1\psi_1 e^{-iE_1t/\hbar} + c_2\psi_2 e^{-iE_2t/\hbar}$$

$$\left|\Psi_{1,2}\right|^2 = \left|c_1\right|^2\left|\psi_1\right|^2 + \left|c_2\right|^2\left|\psi_2\right|^2 + c_1^*\psi_1^*c_2\psi_2 e^{iE_1t/\hbar}e^{-iE_2t/\hbar}$$
$$+c_1\psi_1c_2^*\psi_2^*e^{-iE_1t/\hbar}e^{iE_2t/\hbar}$$

$$\omega_{12} \equiv \left(E_1 - E_2\right)/\hbar$$

$$\left|\Psi_{1,2}\right|^2 = \left|c_1\right|^2\left|\psi_1\right|^2 + \left|c_2\right|^2\left|\psi_2\right|^2 + c_1^*c_2\psi_1^*\psi_2 e^{i\omega_{12}t}$$
$$+c_1c_2^*\psi_1\psi_2^*e^{-i\omega_{12}t}$$

The first two terms are *t*-independent and the second two terms are *t*-dependent and their sum is definitely a real number:

$$2\,\mathrm{Re}\left(c_1^* c_2 \psi_1^* \psi_2 e^{i\omega_2 t}\right).$$

Now let us consider the particle in a constant potential.

eigenfunctions $\left\{\psi_k = e^{ikx}, \psi_{-k} = e^{-ikx}\right\}$

$$E_{|k|} = \frac{p^2}{2m} + \underbrace{E_0}_{\substack{\text{arbitrary} \\ \text{zero of} \\ \text{energy}}} = \frac{\hbar^2 k^2}{2m} + E_0$$

$$\frac{E_{|k|} - E_0}{\hbar} \equiv \omega_k$$

$$\Psi_{|k|}(x,t) = e^{-i\omega_k t}\left[Ae^{ikx} + Be^{-ikx}\right]$$
$$= Ae^{i(kx-\omega t)} + Be^{-i(kx+\omega t)}$$

Stationary phase

$$kx_\phi - \omega t = 0$$

(could choose any constant instead of 0)
x_ϕ is the constant phase point.

$$x_\phi = \frac{\omega t}{k} \qquad \text{A-term}$$

$$\mathrm{x}_\phi = -\frac{\omega t}{k} \qquad \text{B-term}$$

phase velocity

$$\frac{dx_\phi}{dt} = v_\phi = \pm\omega / k$$

$$x_\phi(t) = x_\phi(0) \pm \frac{\omega t}{k}$$

Some arbitrarily chosen constant phase point on $\Psi(x,t)$ moves at a velocity ω/k.

What about $\int_{-\infty}^{\infty}|\Psi(x,t)|^2 dx$?

The t-dependent term integrates to zero due to $\int_{-\infty}^{\infty} e^{\pm 2ikx} dx = 0$.

So there is no motion in $|\Psi(x,t)|^2$, only a constant term and standing waves.

But $\Psi(x,t)$ encodes motion through $\langle \hat{p} \rangle$ and $\langle \hat{x} \rangle$. For example:

$$\langle \hat{p}_x \rangle = \int_{-\infty}^{\infty} \Psi^* \frac{\hbar}{i} \frac{\partial}{\partial x} \Psi dx$$

$$\frac{\hbar}{i}\frac{\partial}{\partial x}\Psi(x,t) = \frac{\hbar}{i} e^{-i\omega_k t}\left[Aike^{ikx} - Bike^{-ikx} \right] = \frac{\hbar}{i} ike^{-i\omega_k t}\left[Ae^{ikx} - Be^{-ikx} \right]$$

Now the whole thing:

$$\frac{\hbar}{i}\Psi^*(x,t)\frac{\partial}{\partial x}\Psi = \hbar k\left[A^* e^{-ikx} + B^* e^{ikx} \right]\left[Ae^{ikx} - Be^{-ikx} \right]$$

Now integrate $\int_{-\infty}^{\infty} dx$

$$\int e^{\pm 2ikx} dx = 0$$

$$\langle p \rangle = \hbar k\left[|A|^2 - |B|^2 \right]$$

as expected! Motion, just like Classical Mechanics!

To get motion, it is necessary that $|A| \neq |B|$

Now for the payoff.

Consider a superposition of e^{ikx} for many values of k:

$$\Psi(x,0) = \int g(k)e^{ikx}\,dk$$

We can experimentally produce any $g(k)$ we want.

Let $g(k)$ be a Gaussian in k

$$g(k) = e^{-(a^2/4)(k-k_0)^2}$$

But $\int_{-\infty}^{\infty} g(k)e^{ikx}dk$ is the Fourier Transform of a Gaussian in k.

Fourier Transform and Inverse Fourier Transform

$$f(x) = (2\pi)^{-1/2}\int_{-\infty}^{\infty} g(k)e^{ikx}dk \qquad \text{get rid of } k$$

$$g(k) = (2\pi)^{-1/2}\int_{-\infty}^{\infty} f(x)e^{ikx}dx \qquad \text{get rid of } x$$

So let us build $\Psi(x,0)$ as a superposition of e^{ikx}. We can write $g(k)$ in amplitude, argument form:

$$g(k) = |g(k)|e^{i\alpha(k)}$$

complex function of real variable

We want $|g(k)|$ to be sharply peaked near $k = k_0$, so use a Gaussian

$$|g(k)| = e^{-(a^2/4)(k-k_0)^2}$$

center $k = k_0$
width $\Delta k = 2^{1/2}a$

$$\alpha(k) = \underbrace{\alpha(k_0)}_{\alpha_0} + (k-k_0)\left.\frac{d\alpha}{dk}\right|_{k=k_0}$$ power series expansion

$$g(k) = e^{-(a^2/4)(k-k_0)^2} e^{i\alpha_0} e^{i(k-k_0)\frac{d\alpha}{dk}}$$

$$g(k)e^{ikx} = \underbrace{e^{-(a^2/4)(k-k_0)^2} e^{i\alpha_0}}_{\text{independent of } x} e^{i\left[(k-k_0)\frac{d\alpha}{dk}+kx\right]}$$ rapidly oscillating in x except at a special region of x

To find the value of x at which the *phase is stationary*, we want

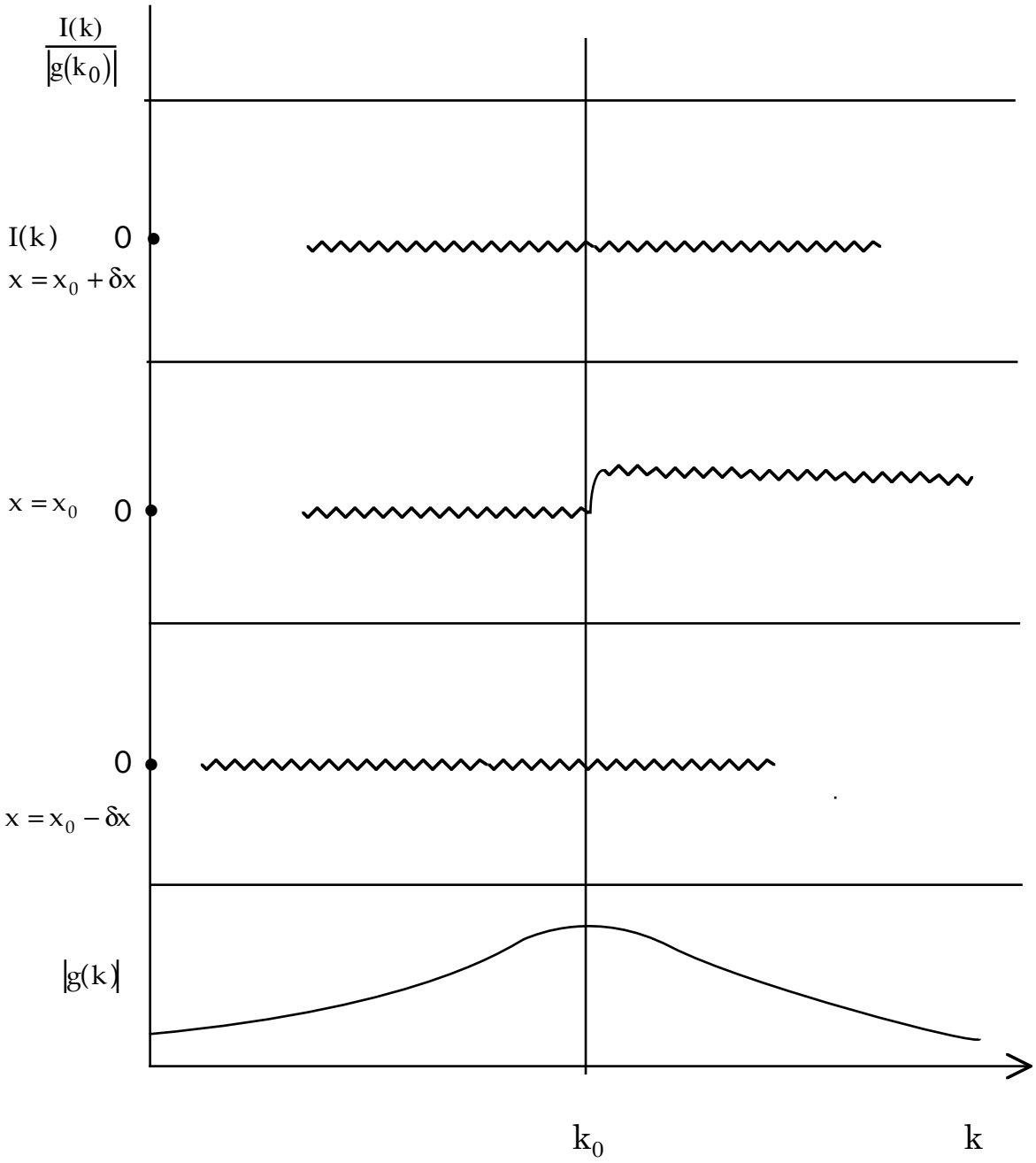
at which the phase is stationary, we want

$$\frac{d}{dk}\left[(k-k_0)\frac{d\alpha}{dk}+kx\right] = 0$$

$$\frac{d\alpha}{dk} + x = 0$$

so if we choose $\left.\frac{d\alpha}{dk}\right|_{k=k_0} = -x_0$ we have stationary phase in k near k_0 and near $x = x_0$. This means that the $\int_{-\infty}^{\infty} g(k)e^{idx}dx$ integral accumulates to its exact value near $x = x_0$.

How does an integral over a rapidly oscillating integrand accumulate? It accumulates near the stationary phase point, x_0.



Integral accumulates near $k = k_0$ but only when $x \approx x_0$.

$$I(k) = \int_{-\infty}^{k} f(x,k)\,dk.$$

If you examine the integrand and can identify the stationary phase region, you can determine the value of the integral without actually evaluating the integral.
Amaze your friends!

NON-LECTURE

Joel Tellinghuisen, "Reflection and Interference Structure in Diatomic Franck-Condon Factors," J. Mol. Spectrosc. **103**, 455-465 (1984). The figures in this paper show how an integral accumulates at a stationary phase point of the integrand. The stationary phase point, x_{sp}, is the coordinate at which the vibrational wavefunctions for states 1 and 2 have the same classical momentum, $p_{classical}=\left[2m(E-V(x_{sp})\right]^{1/2}$. The stationary phase point is located at the crossing of the V_1 and V_2 potential curves, $V_1(x_{sp})=V_2(x_{sp})$. The semiclassical approximation for calculating vibrational overlap integrals is discussed on pages 278-285 of H. Lefebvre-Brion and R. W. Field, The Spectra and Dynamics of Diatomic Molecules.

$$\Psi(x,0)=\frac{a^{1/2}}{(2\pi)^{3/4}}\int_{-\infty}^{\infty}e^{-(a^2/4)(k-k_0)^2}\underbrace{e^{-i(k-k_0)x_0}}_{e^{-k(x-x_0)}e^{ik_0x_0}}e^{ikx}\,dk$$

This is a δ-function. It causes $\Psi(x,0)$ to be localized near x_0.

So we get $|\Psi|^2$ localized at $x_0(t)$, k_0, $\Delta x(t)$, Δk if $g(k)$ is Gaussian.

$\Delta x = 2^{-1/2}a$
$\Delta k = 2^{1/2}/a$
$\Delta x \Delta k = 1$ at $t = 0$

We have constructed a Gaussian wavepacket, $\Psi(x,t)$, from $\Psi(x,0)$ with localization of $x_0(t)$, $\Delta x(t)$ minimum at $t = 0$, Gaussian in x, Gaussian in k.

We can now ask how this $\Psi(x,t)$ can be modified by features of any time-independent $V(x)$.

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Lecture #4: Stationary Phase and Gaussian Wavepackets

Last time:

tdSE → motion, motion requires non-sharp E
phase velocity
began Gaussian Wavepacket

goal: $\langle x\rangle$, Δx, $\langle p\rangle = \hbar\langle k\rangle$, $\Delta p = \hbar\Delta k$ by construction or inspection

$\Psi(x,t)$ is a complex function of real variables. Difficult to visualize.

What are we trying to do here?

Techniques for solving series of increasingly complex problems. Illustrate philosophical points along the way toward solutions of problems.

So far: free particle, infinite well, δ-function } very artificial

* nothing particle-like
* nothing molecule-like
* no spectra

Minimum Uncertainty (Gaussian) Wavepacket -- QM version of particle. We are going to construct a $\Psi(x,t)$ for which $|\Psi(x,t)|^2$ is a Gaussian in x and the FT of $\Psi(x,t)$, gives $\Phi(k,t)$, for which $|\Phi(k,t)|^2$ is a Gaussian in k.

center of wavepacket follows Newton's Laws

extra stuff: spreading
interference
tunneling

Today: (improved and embellished repeat of material in pages 3–4 through 3–11

1. Infer Δk by comparing g(k) to $G(x; x_0, \Delta x)$
2. $g(k) = |g(k)|e^{i\alpha(k)}$ for k near k_0
3. $\left.\frac{d\alpha}{dk}\right|_{k=k_0} \equiv -x_0$ STATIONARY PHASE
4. $|\Psi(x,t)|^2$ *moving*, spreading wavepacket
5. $v_G \neq v_\phi$ { how is it possible that the center of the wavepacket moves at a different velocity than its center k-component

revised 9/25/20 9:33 AM

1. Here is a normalized Gaussian (see Gaussian Handout)

$$G(x;x_0,\Delta x)=(2\pi)^{-1/2}\frac{1}{\Delta x}e^{-(x-x_0)^2}\Big/\left[2(\Delta x)^2\right]$$

$$\left.\begin{array}{l}\left[\text{normalized } \int_{-\infty}^{\infty} G(x;x_0,\Delta x)\,dx=1\right]\\ \text{center } \langle x\rangle = x_0\\ \text{width = standard deviation} \quad \Delta x\equiv\left[\langle x^2\rangle-\langle x\rangle^2\right]^{1/2}\end{array}\right\}\text{by construction}$$

Now compare this special form to

$$\Psi(x,0)=\frac{a^{1/2}}{(2\pi)^{3/4}}\int_{-\infty}^{\infty}\underbrace{e^{-(a^2/4)(k-k_0)^2}}_{g(k)}\underbrace{e^{ikx}}_{\text{free particle}}\,dk$$

Fourier Transform of a Gaussian in k

a Gaussian in k, but what are the width and $\langle k\rangle$?

by analogy, a Gaussian in k

$$G(k;k_0,\Delta k)=(2\pi)^{-1/2}\underbrace{\left(\frac{a}{2^{1/2}}\right)}_{1/\Delta k}g(k)$$

$$\frac{a^2}{4}=\frac{1}{2(\Delta k)^2}$$
$$\therefore\quad \Delta k=\frac{2^{1/2}}{a}$$

by analogy with $G(x;x_0,\Delta x)$

So casual inspection of this form of $\Psi(x,0)$ gives us $\langle k\rangle$ and Δk. Not quite so easy to get $\langle x\rangle$ and Δx.

If we actually carry out the F.T. specified in the definition of $\Psi(x,0)$ above, we get

$$\Psi(x,0)=\left(\frac{2}{\pi a^2}\right)^{1/4}e^{ik_0x}e^{-x^2/a^2}$$

$$\langle x\rangle = x_0 = 0$$

$$\frac{1}{2(\Delta x)^2}=\frac{1}{a^2}$$
$$\Delta x = a/2^{1/2}$$

$$\Delta x = 2^{-1/2}a, \quad \text{previously } \langle k\rangle = k_0,\ \Delta k=\frac{2^{1/2}}{a};$$

But the square of a Gaussian is a Gaussian and the Δx or Δk of the squared Gaussian is a factor of $2^{-1/2}$ smaller than the original Gaussian.

$$\Delta x \text{ for } \Psi(x,0) \text{ is } 2^{-1/2}a, \ \Delta\text{x for } \left|\Psi(x,0)\right|^2 \text{ is } \frac{a}{2}.$$

$$\Delta k \text{ for } \Phi(k,0) \text{ is } \frac{2^{1/2}}{a}, \ \Delta\text{k for } \left|\Phi(k,0)\right|^2 \text{ is } \frac{1}{a}.$$

For the squared Gaussian:

$$\Delta x \Delta k = \frac{a}{2}\frac{1}{a} = \frac{1}{2}$$

This is a very special Gaussian wavepacket

* minimum uncertainty
* $x_0 = 0$

3. What about more general Gaussian wavepackets?

g(k) is a <u>complex function</u> of k <u>sharply peaked</u> near $k = k_0$

$g(k)=|g(k)|e^{i\alpha(k)}$ amplitude, argument form

If $|g(k)|$ is sharply peaked near $k = k_0$, then the *only relevant part* of $\alpha(k)$ is the part for k near k_0

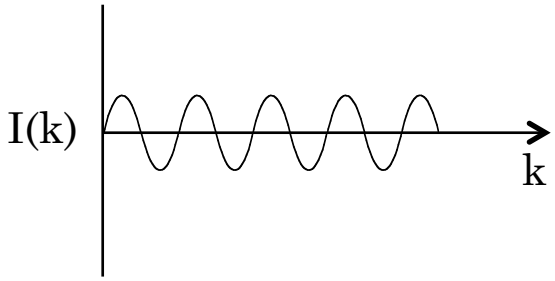
$$\text{Expand } \alpha(k)=\underbrace{\alpha(k_0)}_{\alpha_0}+(k-k_0)\left.\frac{d\alpha}{dk}\right|_{k=k_0}+ \quad \text{higher terms neglected}$$

$$\Psi(x,0)=\frac{a^{1/2}}{(2\pi)^{3/4}}\int_{-\infty}^{\infty} \underbrace{|g(k)|e^{i\alpha(k)}e^{ikx}}_{|g(k)|e^{i\alpha_0}e^{i\left[(k-k_0)\left.\frac{d\alpha}{dk}\right|_{k=k_0}+kx\right]}} dk$$

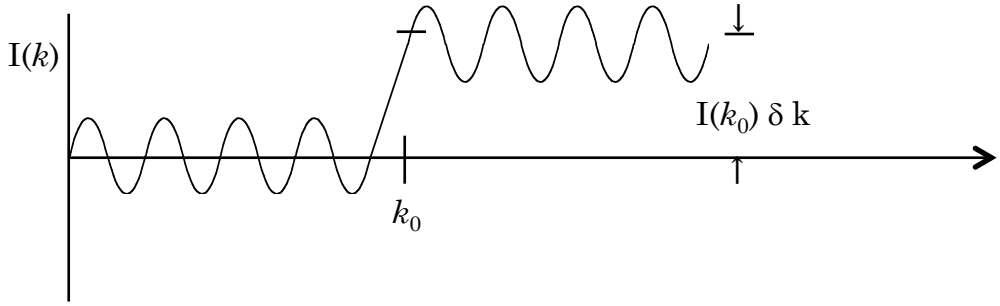
We want to "cook" $\Psi(x,0)$ so that it is localized near $x = x_0$. In order for this to happen, the factor $\left[(k-k_0)\left.\frac{d\alpha}{dk}\right|_{k=k_0}+kx\right]$, must be indpendent of k near $k = k_0$. <u>Stationary Phase</u>!

How does the integral of a wiggly function accumulate? All of the "accumulation" to the $\int_{-\infty}^{\infty}$ value occurs near the "stationary phase points".

e.g., $I(k) = \int_{-\infty}^{k} e^{ik'x} dk'$



remains near zero, but if the phase factor stops wiggling near $k = k_0$



where δk is the range of k over which the phase factor changes by π.

So, arrange for the phase factor to become stationary near $k = k_0$. Expand $\alpha(k)$ in a power series and keep only the first term in the expansion:

$$0 = \frac{d}{dk}\left[(k - k_0)\frac{d\alpha}{dk} + kx\right]$$

$$0 = \frac{d\alpha}{dk} + x$$

satisfied if $\boxed{\left.\frac{d\alpha}{dk}\right|_{k=k_0} \equiv -x_0}$! The phase is stationary near $x = x_0$

Thus

$$\Psi(x,0)=\frac{a^{1/2}}{(2\pi)^{3/4}}e^{i\alpha_0}\int_{-\infty}^{\infty}\underbrace{e^{-\left(a^2/4\right)\left(k-k_0\right)^2}}_{|g(k)|}\underbrace{e^{-i\left(k-k_0\right)x_0}e^{ikx}}_{e^{ik(x-x_0)}e^{ik_0x_0}}dk$$

$-\left.\frac{d\alpha}{dk}\right|_{k=k_0}$ ↓ (stops wiggling only when $x \approx x_0$)

↓ $\delta(x-x_0)$ shifts the center of Ψ to any desired value of x_0

(insertion of $e^{-i(k-k_0)x_0}$ phase factor to cause the wavepacket to be centered at x_0.)

---

4. Now put in time-dependence by adding

$e^{-i\omega_k t}$ factor

$$\omega_k=\frac{E_k}{\hbar}=\left(\frac{\hbar^2k^2}{2m}\right)\frac{1}{\hbar}$$

$$\omega_k=\frac{\hbar k^2}{2m}$$

$$\Psi(x,t)=\frac{a^{1/2}}{(2\pi)^{3/4}}\int_{-\infty}^{\infty}\underbrace{|g(k)|e^{-i\left(k-k_0\right)x_0}}_{g(k)}\underbrace{e^{ikx}}_{\text{eigenstate of free particle } \mathbf{H}}e^{-i\omega_k t}dk$$

This FT is evaluated and simplified in CTDL, page 64

$$|\Psi(x,t)|^2 = \left(\frac{2}{\pi a^2}\right)^{1/2} \underbrace{\left(1+\frac{4\hbar^2 t^2}{m^2 a^4}\right)}_{\text{time dependent normalization}} \exp - \underbrace{\left[\frac{2a^2\left(x-\frac{\hbar k_0}{m}t\right)^2}{a^4+\frac{4\hbar^2 t^2}{m^2}}\right]}_{\text{Gaussian with time dependent width and center position}}$$

Maximum of Gaussian occurs when numerator of exp –[ ] is 0.

MOTION: $0 = x - \frac{\hbar k_0}{m}t$ $\qquad x_0(t) = \frac{\hbar k_0}{m}t$ (center of moving wavepacket is at $x_0(t)$)

$$v_G = \frac{d}{dt}x_0(t) = \frac{\hbar k_0}{m} = \frac{p_0}{m} = v_{classical}$$

↑ velocity of center of Gaussian

This is 2× larger than v_ϕ (the phase velocity).
Classically we expect a free particle to move at constant $v = \frac{p}{m}$

WIDTH: compare coefficient of $(x\text{-}x_0(t))^2$ in exp–[ ] to standard $G(x;x_0,\Delta x)$ in handout

$$\Delta x = \left[\frac{a^4+4\hbar^2t^2/m^2}{4a^2}\right]^{1/2} \approx \underbrace{\frac{a}{2}}_{\text{minimum width at t = 0}} + \underbrace{\left|\frac{\hbar t}{ma}\right|}_{\text{width increases linearly in t at long time (quadratically at early time).}}$$

long time limit

$$\frac{1}{2(\Delta x)^2} = \frac{2a^2}{a^4+\frac{4\hbar^2t^2}{m^2}}$$

$\langle x\rangle$ and Δx are time-dependent, but what about $\langle k\rangle$ and Δk?

Recall original definition of $\Psi(x,0)$ (page 4-2), where $\Psi(x,0)$ is written as the FT of a Gaussian in k

$$g(k,t) = e^{-i\omega_k t} g(k,0)$$

$$\therefore |\Phi(k,t)|^2 \text{ has } \left.\begin{array}{l}\langle k\rangle = k_0 \\ \Delta k = \dfrac{1}{a}\end{array}\right\} \textit{time} \text{ independent}$$

We know free particle must have time independent k_0 and Δk
(because there are no forces acting on the w.p. — divide w.p. into Δk wide slices)

$\Delta x \Delta k = \dfrac{1}{2}\left[1+\dfrac{4\hbar^2 t^2}{m^2 a^4}\right]^{1/2}$ minimum uncertainty at t = 0 (and linearly increasing at long t).

For free particle, build w.p. with any desired x_0, k_0, Δk starting from

1. $\Psi(x,t) = \int_{-\infty}^{\infty} g(k) e^{ikx} e^{-i\omega_k t} dk$ and $\omega_k = \dfrac{\hbar k^2}{2m}$

2. find x_0 from $-\left.\dfrac{d\alpha}{dk}\right|_{k=k_0}$

3. $x_0(t) = x_0 + v_G t$ $\quad v_G = \dfrac{\hbar k_0}{m}$

4. $\Delta x = \dfrac{a}{2}\left[1+\dfrac{4\hbar^2 t^2}{m^2 a^4}\right]^{1/2}$

5. If we want a value of Δx other than a/2 at t = 0, replace x by $x' = x + \delta$ such that when the center of w.p. reaches x_0 at t = 0 it has the desired width.

Could have started with $\bar{\Psi}(k,0) = \int_{-\infty}^{\infty} \underbrace{\bar{g}(x)}_{\substack{\text{Gaussian} \\ \text{in x}}} \underbrace{e^{-ikx}}_{\substack{\text{inverse} \\ \text{F.T.}}} dx$

and then encoded k_0 in $\bar{g}(x)$ thru $\left.\dfrac{d\alpha}{dx}\right|_{x=x_0} = +k_0$

where $\alpha(x)$ is the argument of $\bar{g}(x) = \left|\bar{g}(x)\right| e^{i\alpha(x)}$

For next class read C-TDL pages 103-107, 1468-1476.

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Fall 2018

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Lecture #5: Continuum Normalization

Last time: Free Wavepacket
encoding of x_0, Δx, p_0, Δp

* use of the Gaussian functional form, $G(x; x_0, \Delta x)$, to avoid calculating integrals

* use of stationary phase to encode x_0 in $|g(k)|e^{i\alpha(k)}$

* use g(k) because it is automatic to put in $e^{-iE_k t/\hbar}$

For moving and spreading free wavepacket:
Δx is time dependent
Δp is not (because free wavepacket is not subject to any force)

Today: Normalization of eigenfunctions which belong to continuously (as opposed to discretely) variable eigenvalues.

- convenience of ortho-normal basis sets: generalization for continua
- we often talk about "density of states", but in order to do that we need to define what we mean by "state"
- computation of absolute probabilities — cannot depend on how we choose to define "state".

1. Identities for δ -functions.
2. $\Psi_{\delta k}$, $\Psi_{\delta p}$, $\Psi_{\delta E}$ for eigenfunctions that correspond to continuously variable eigenvalues.
3. finite box with countable number of discrete states taken to the limit $L \to \infty$. Normalization independent quantity:

$$P(x,\theta) = \left(\frac{\#\text{ states}}{\delta\theta}\right)\left(\frac{\#\text{ particles}}{\delta x}\right)$$

θ is the argument of the delta-function. So if we integrate over a region of θ and x, we have the absolute probability, $\iint d\theta dx\, P(x,\theta)$.

4. two examples — "predissociation" rate and smoothly varying spectral density.

In Quantum Mechanics, there are two very different classes of systems.

* SPATIALLY CONFINED:
- E is quantized
- can count states, easy to compute density of states $\frac{dn}{dE} = \rho_E$
- can normalize to $1 = \int_{-\infty}^{\infty} \psi_E^* \psi_E dx$

what is ρ_E good for?

T: classical period of oscillation

* # of encounters/sec: $\frac{1}{T}$

* fraction of time in region of length L: $\frac{L/|v|}{T}$ (v, classical velocity, is dependent on x)

* SPATIALLY UNCONFINED:
- E continuously variable
- can't count states, so how to compute $\frac{dn}{dE}$?

**
- can ask what is the absolute probability of finding the system between E, E + dE *and* x, x + dx

For confined systems, we can express ortho-normalization in terms of Kronecker- δ

$$\delta_{ij} = \int_{-\infty}^{\infty} \psi_i^* \psi_j \, dx$$

$\delta_{ij} = 0$ $\quad i \neq j$ $\quad$ orthogonal

$\delta_{ij} = 1$ $\quad i = j$ $\quad$ normalized

ψ has dimension of $L^{-1/2}$

δ_{ij} has dimension of pure number. (Kronecker-δ)

For unconfined systems, we are going to ortho-normalize states to Dirac δ - functions

In order to do this we need to know better what a δ - function is and what some of its mathematical properties are.

One of several equivalent definitions of a δ-function:

$$\delta(x - x') = \delta(x, x') = \frac{1}{2\pi} \int e^{-iu(x-x')} \, du.$$

What is it good for?

$\int \delta(x,x')\psi(x)dx = \psi(x')$. ← shifts a function evaluated at x to the same function evaluated at x′.

$\delta(x, x')$ has dimension of 1/x. (Dirac-δ function)

Some useful δ–function identities:

We do this so that we will be able to transform between δk, δp, and δE (where E = f(k)) delta-function normalization schemes.

1. $\delta(ax,ax') = \frac{1}{|a|}\delta(x,x')$ e.g., $\delta(p-p') = \delta\left(\hbar(k-k')\right) = \frac{1}{\hbar}\delta(k-k')$

dimension of p^{-1} — dimension of 1/k

nonlecture proof of #1 above

$$\delta(ax,ax') = \frac{1}{2\pi}\int e^{-iu(ax-ax')}du$$ change variables

$v = au$

$dv = a\,du$

$$\delta(ax,ax') = \frac{1}{2\pi}\frac{1}{a}\int e^{-iv(x-x')}dv = \frac{1}{a}\delta(x,x')$$

but, since $\delta(ax,ax') \equiv \delta(ax-ax') = \delta(ax'-ax) = \delta\left([-a](x-x')\right)$

(δ is an even function), $\delta(ax,ax') = \frac{1}{|a|}\delta(x,x')$

2. $$\delta\left(g(x)\right) = \sum_{\underbrace{i}_{\substack{\text{zeros}\\ \text{of g(x)}}}} \left|\frac{dg(x_i)}{dx}\right|^{-1}\delta(x,x_i)$$ provided that $\frac{dg(x_i)}{dx} \neq 0$

expand g(x) in the region near each 0 of g(x),

i.e., x near x_i $\qquad g(x) \cong \left.\frac{dg}{dx}\right|_{x=x_i}(x-x_i).$

If there is only 1 zero, then identity #1 above gives the required result. It is clear that δ(g(x)) will only be nonzero when g(x) = 0. Otherwise we need to carry out the sum in identity #2.

g(x) = (x – a)(x – b) has zeroes at x = a and x = b.

$$\frac{dg}{dx} = \frac{d}{dx}\left[x^2 - x(a+b) + ab\right] = 2x - (a+b)$$

$$\left.\frac{dg}{dx}\right|_{x=a} = a - b \qquad \left.\frac{dg}{dx}\right|_{x=b} = b - a$$

$$\delta(g(x)) = \sum_i \left|\frac{dg(x_1)}{dx}\right|^{-1} \delta\left(x, x_i\right) \qquad \text{(zeroes of } g(x)\text{)}$$

$$= \left|\frac{1}{a-b}\right|\left[\delta(x,a) + \delta(x,b)\right]$$

Other examples:

$$\delta\left(x^2 - a^2\right) = \frac{1}{2|a|}\left[\delta(x,a) + \delta(x-a)\right]$$

$$\delta\left(x^{1/2} - a^{1./2}\right) = 2a^{1/2}\delta(x-a) \quad (a > 0)$$

See Merzbacher, Quantum Mechanics, 3rd Edition, pages 630-632.

EXAMPLES

A. g(x) = (x–a)(x–b) This has zeroes at x = a, and x = b.

You should show that $\delta(g(x)) = \frac{1}{|a-b|}[\delta(x,a) + \delta(x, b)]$.

B. $\delta\left(E^{1/2}, E'^{1/2}\right)$

$g(E) = E^{1/2} - E'^{1/2}$ has one zero at E = E′, expand g(E) about E = E′, thus for E near E′

$g(E) \pm \frac{1}{2} E'^{-1/2}(E - E')$.

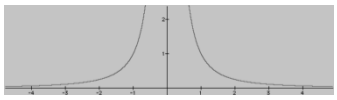
you should show that $\delta\left(E^{1/2}, E'^{1/2}\right) = 2\left|E'^{1/2}\right|\delta(E, E')$

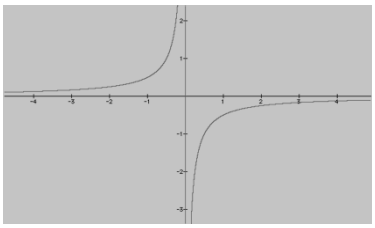
This is useful because k $\propto E^{1/2}$ $\delta(E - E') = \left(\frac{m}{2h^2(E' - V_0)}\right)^{1/2}$ [$\delta(k - k') + \delta(k + k')$] for a free particle

or $\delta\left(k_E(x) - k_{E'}(x)\right) = \left(\frac{2h^2}{m}\right)^{1/2} (E' - V(x))^{1/2}\delta(E - E')$

Another property of δ-functions: $\frac{d}{dx}\delta(x, x')$

$\delta(x, x')$ is an even function:





$\therefore$ expect $\frac{d}{dx}\delta(x, x') \equiv \delta'(x, x')$ to be an odd function:

This is useful because application of $\frac{d}{dx}\delta(x, x')$ to $f(x)$ is capable of picking out $\frac{df}{dx}$ evaluated at x′.

Non-lecture:

Use definition of derivative to prove that

$$\int_{-\infty}^{\infty} \delta'(x, x') f(x) dx = -f'(x')$$

$$\frac{d}{dx}\delta(x, x') = \lim_{\varepsilon \to 0} \frac{\left[\delta(x + \varepsilon, x') - \delta(x, x')\right]}{\varepsilon}$$

$$\int \delta(x + \varepsilon, x') f(x) dx = f(x' - \varepsilon)$$

$$\int \delta(x, x') f(x) dx = \quad f(x')$$

$$\therefore \int \lim_{\varepsilon \to 0} \frac{\left[\delta(x + \varepsilon, x') - \delta(x, x')\right]}{\varepsilon} f(x) dx = \lim_{\varepsilon \to 0} \frac{f(x' - \varepsilon) - f(x')}{\varepsilon} = -f'(x')$$

There are several useful ways to normalize wavefunctions.

Bound states. Particle is confined in space (with tunneling tails outside the box). Space normalized. 1 particle (mostly) in box.

Box normalized $\psi_{L,E_i}(x)$ (box of length L)

$$\int_{-\infty}^{\infty} dx\, \psi^*_{L,E_i} \psi_{L,E_j} = \delta_{ij} \qquad \text{Kronecker - delta}$$

OK for bound states, but not continua.

dimension of $\psi_{L,E}$ is $L^{-1/2}$
dimension of δ_{ij} or $\delta_{E_iE_j}$ is 1.

Continua. We need some other form of normalization.

e.g. $\int_{-L/2}^{L/2} dx \left(e^{ikx}\right)^* e^{ik'x} = \int_{-L/2}^{L/2} dx e^{-i(k-k')x}$

if $k = k'$ we get L
if $k \neq k'$ we expect to get 0 (in limit $L \to \infty$)

So we can normalize to a delta function in E, p, or k.

δ E : $\int_{-\infty}^{\infty} dx \psi^{*}_{\delta E, E_i} \psi_{\delta E, E_j} \equiv \delta\left(E_i - E_j\right)$

$\delta\,(E_i - E_j)$ has the useful **δ**-function property:

$$\int dE\, \delta\left(E - E_j\right) \psi_{\delta E, E} = \psi_{\delta E, E_j}$$

This implies that $\delta(E - E_j)$ has the dimension of 1/E and that $\psi_{\delta E,E}$ has dimension of $L^{-1/2}E^{-1/2}$

δ p : $p_E(x) = \left[2m\left(E - V(x)\right)\right]^{1/2}$ $\qquad P_E^2/2m = E - V(x)$

$$\int_{-\infty}^{\infty} dx \psi^{*}_{\delta p, p_{E_i}} \psi_{\delta p, p_{E_j}} = \delta\left(p_{E_i}(x) - p_{E_j}(x)\right)$$

$\delta\left(p - p'\right)$ has dimension of 1/p

$\psi_{\delta p, p_E}$ has dimension of $L^{-1/2} p^{-1/2}$

δ k : $k_E(x) = \left[\frac{2m}{\hbar^2}\left(E - V(x)\right)\right]^{1/2}$ $\qquad \frac{\hbar^2 k_E^2}{2m} = E - V(x)$

$$\int_{-\infty}^{\infty} dx \psi^{*}_{\delta k, k_{E_i}} \psi_{\delta k, k_{E_j}} = \delta\left(k_{E_i}(x) - k_{E_j}(x)\right)$$

$\delta\left(k - k'\right)$ has units of 1/k

$\psi_{\delta k, k}$ has units of $L^{-1/2} k^{-1/2}$

What are all of these normalization schemes good for?

When you make a measurement on a continuum (unbound) system, you ask

What is the probability of finding a particle between

x, x + dx
and θ, $\theta + d\theta$? θ **can be E, $p_E(x)$, or $k_E(x)$**

The probability is $P(x,\theta)dxd\theta$

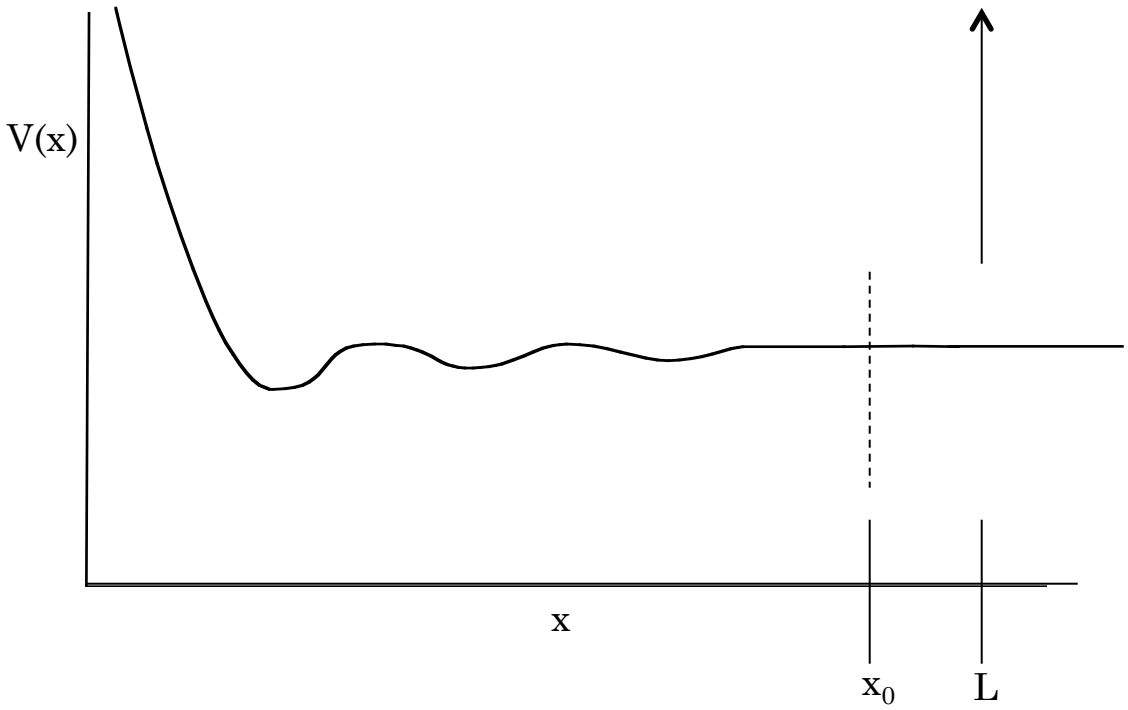
Want $P(x,\theta)$. Has dimensions $L^{-1}\,\theta^{-1}$ (as shown for $\psi_{\delta,\delta E}$, $\psi_{\delta_{p},pE}$, and $\psi_{\delta_{k},kE}$)

$$P(x,\theta) = \psi^{*}_{\delta\theta,\theta}(x)\psi_{\delta\theta,\theta}(x) \qquad !$$

There is another less abstract way to get this kind of information. "Discretize the continuum" by adding an infinite barrier at x = L and taking the limit $L\to\infty$. This way we can use box-normalized states, and actually count the states.

The WKB quantization condition (will be derived in Lecture #7) gives

$$\frac{dn}{dE} = \frac{(2m)^{1/2}}{h}\int_{x_(E)}^{x_{+}(E)} dx\left(E - V(x)\right)^{-1/2}$$



We have a complicated V(x) for $x < x_0$ and constant for $x > x_0$.

In the region where V(x) is constant at $V(x_0) = V_0$.

$$\int_{x_0}^{L} \left[E - V(x)\right]^{-1/2} dx = \left[E - V_0\right]^{-1/2} \left(L - x_0\right) \propto L$$

$$\text{and box normalization causes } |\psi|^2 \propto \frac{1}{L}$$

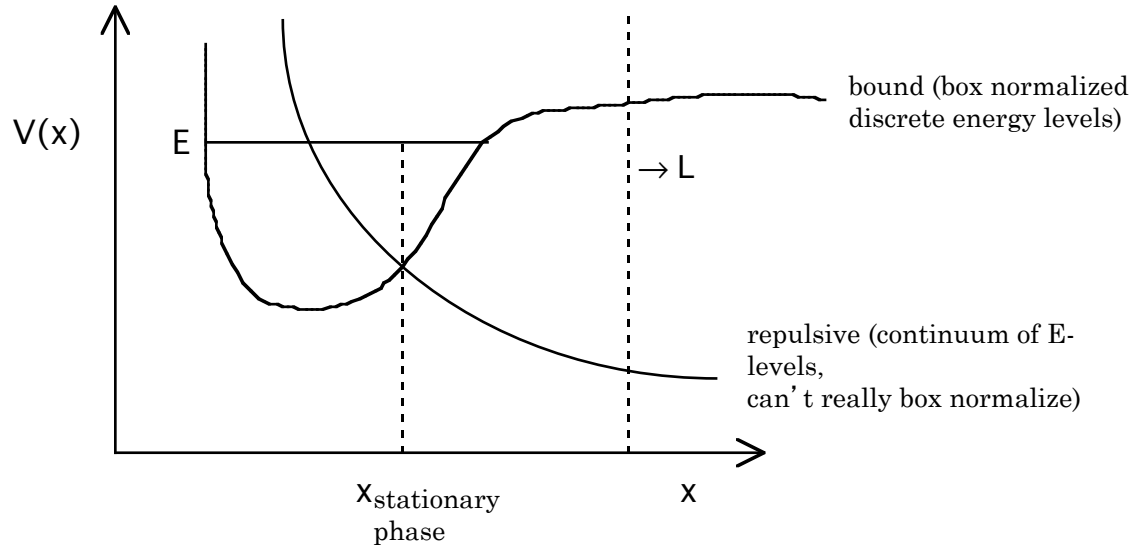
$$\text{so we get } \underbrace{P(x,E)}_{\substack{\text{dimension} \\ L^{-1}E^{-1}}} = \lim_{L\to\infty} \underbrace{\left(\frac{dn_L}{dE}\right)}_{\substack{\text{dimension} \\ E^{-1}}} \underbrace{\psi^*_{L,E}(x)\psi_{L,E}(x)}_{\substack{\text{dimension} \\ L^{-1}}}$$

---

2 Schematic Examples

* Bound → free transition probabilities
* Constant spectral density across a dissociation or ionization limit.

Bound-Free Transition (predissociation)



At t = 0, system is prepared in $\Psi(x,0) = \psi_{bound}(x)$

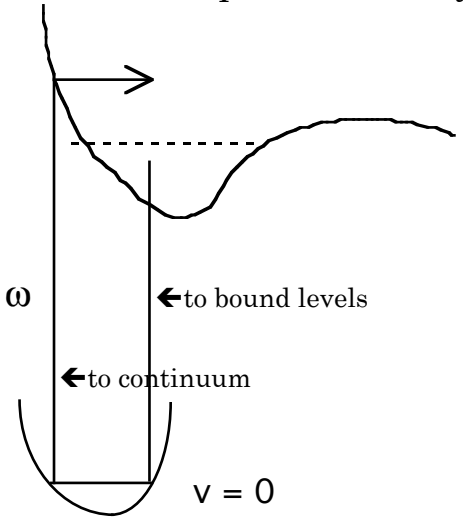
Fermi's Golden Rule:

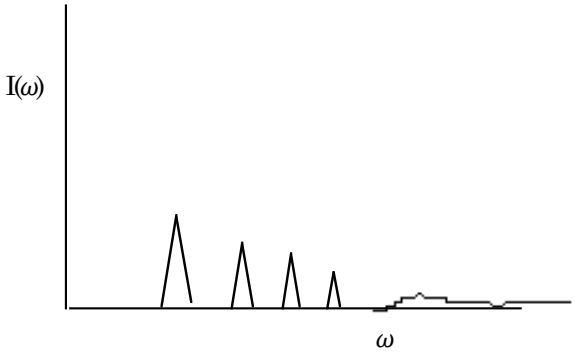
$$Rate = \Gamma_{bound\to free} = \frac{2\pi}{\hbar}\left|\int \psi_{\delta E}^{free*}(E)\hat{\mathbf{H}}\psi_{L,E}^{bound}dx\right|^2 \rho_{\delta E}(E)$$

$$\rho_{\delta E} = \frac{dn_{\delta E}(E)}{dE}$$ derive this key quantity by box normalizing repulsive state and taking $\lim_{L\to\infty}\left(\frac{1}{L}\frac{dn_L}{dE}\right)$

Then compute the $\hat{\mathbf{H}}$ integral using two box normalized functions.

Constant spectral density on both sides of a bound/free limit





$\frac{\text{Intensity}(\omega)}{\Delta E} \sim$ smooth function of ω, no discontinuity at onset of continuum

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Lecture #6: Linear V(x). JWKB Approximation and Quantization

JWKB: Jeffreys, Wentzel, Kramers, Brillouin.

Last time: Normalization schemes for eigenfunctions which belong to continuously variable eigenvalues.

1. identities
2. $\psi_{\delta k}, \psi_{\delta p}, \psi_{\delta E}, \psi_{\text{box}}$: different normalization schemes
3. trick using box normalization $(\theta \text{ is } k, p, E)$

$$\left(\frac{\#\text{ states}}{\underset{\propto L}{\delta\theta}}\right)\left(\frac{\#\text{ particles}}{\underset{\propto 1/L}{\delta x}}\right)$$ for box normalization

4. $\frac{dn}{dE}$ ("density of states") often needed - alternate method via JWKB next lecture

1. V(x) = α x linear potential
   solve in momentum representation, $\phi(p)$, and take F.T. to $\psi(x) \rightarrow$ Airy functions
2. Semi-classical (JWKB) approx. for $\psi(x)$

* $p(x) = \left[(E - V(x))2m\right]^{1/2}$ Classical mechanical momentum function dependence on x.

* $\psi(x) = \underset{\text{envelope}}{|p(x)|^{-1/2}} \exp\left[\pm \frac{i}{\hbar}\int_c^x p(x')dx'\right]$

* visualize $\psi(x)$ as plane wave with x-dependent wave vector

* useful for evaluating stationary phase integrals (localization, causality)

**** splicing across boundary between classical (E > V) and forbidden (E < V) regions] Next lecture

WKB Quantization Condition

$$\int_{x_-(E)}^{x_+(E)} p(x')dx' = \frac{h}{2}\left(n + 1/2\right) \quad n = 0,1,\ldots$$

Linear Potential. $V(x) = \alpha x$

$$\hat{H} = \frac{\hat{p}^2}{2m} + \alpha\hat{x}$$

coordinate representation	momentum representation
$\hat{x} \to x$	$\hat{p} \to p$
$\hat{p} \to \frac{\hbar}{i}\frac{\partial}{\partial x}$	$\hat{x} \to i\hbar\frac{\partial}{\partial p}$

(note $[\hat{x},\hat{p}] = i\hbar$ in both representations - prove this?)

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + \alpha x$$

2nd order differential equation

$$\hat{H} = \frac{p^2}{2m} + i\hbar\alpha\frac{d}{dp}$$

1st order - much easier!

$$0 = (H - E)\phi(p)$$

$$0 = \left(\frac{p^2}{2m} + i\hbar\alpha\frac{d}{dp} - E\right)\phi(p)$$

Solve in momentum representation (a sometimes useful trick)

Schr. Eq. $\boxed{\frac{d\phi(p)}{dp} = -\frac{i}{\hbar\alpha}\left(E - p^2/2m\right)\phi(p)}$

Form of Solution $\phi(p) = Ne^{ap+bp^3}$

$\frac{d\phi}{dp}$ gives p^2 times $\phi(p)$

$\frac{d\phi}{dp}$ gives constant times $\phi(p)$

when you take $\frac{d}{dp}$

Must solve for a and b

plug into Schr. Eq. and identify correspondences, term-by-term, to get

$$a = -\frac{iE}{\hbar\alpha}$$

$$b = \frac{i}{6\hbar\alpha m}$$

$$\boxed{\phi(p) = N\exp\left[-\frac{i}{\hbar\alpha}\left(Ep - p^3/6m\right)\right]}$$

easy? Note that, if p is real, $\phi(p)$ is oscillatory

$$\phi^*(p)\phi(p) = 1! \qquad \therefore \quad N = 1!$$

Now p is an observable, so it must be real. Thus $\phi(p)$ is defined for all (real) p and is oscillatory in p for all p. $\phi(p)$ is NEVER exponentially increasing or decreasing if p is real!

IT IS STRANGE THAT $\phi(p)$ does not distinguish between classically allowed and forbidden regions. IS THIS REALLY STRANGE? If we allow p to be imaginary in order to deal with classically forbidden regions, $\phi(p)$ becomes an increasing or decreasing exponential. When we extend the solution to the Schrödinger equation into the classically forbidden region, p is imaginary and $\phi(p)$ is exponentially increasing or decreasing.

If we insist on working in the $\psi(x)$ picture, we must perform a Fourier Transform.

$$\psi(x) = N' \int_{-\infty}^{\infty} e^{ipx/\hbar} \phi(p) dp$$

$$\psi(x) = N' \int_{-\infty}^{\infty} \exp\left[\frac{i}{\hbar\alpha} \left\{ \underbrace{p(\alpha x - E) + p^3/6m}_{\text{odd function of p: O(p)}} \right\} \right] dp$$

$$e^{i\theta} = \underbrace{\cos\theta}_{even} + i \underbrace{\sin\theta}_{odd}$$

$$\int_{-\infty}^{\infty} \sin O(p) dp = 0 \qquad \text{since sin } O(p) \text{ is odd wrt p} \rightarrow -\text{p.}$$

$$\psi(x) = N' \int_{-\infty}^{\infty} \cos\left[\frac{(\alpha x - E)p + p^3/6m}{\hbar\alpha} \right] dp$$

Solution!

$$Ai(z) = \pi^{-1/2} \int_0^{\infty} \cos\left(s^3/3 + sz \right) ds$$

$E = V(x) = \alpha x_{tp}$

$x_{tp} = E/\alpha$

Surprise! This is a *named* (Airy) function and a *tabulated* integral

* numerical tables for x near turning point i.e., $x \approx E/\alpha$
* analytic "asymptotic" functions for x far from turning point.
  - useful for deriving energy levels as an explicit function of quantum numbers and for matching wave functions across boundaries.
* zeroes of Airy functions [$Ai(z_i)=0$] and of derivatives of Airy functions [$Ai'(z'_i)=0$] are tabulated. (Useful for matching across center symmetry-point of potentials with definite even or odd symmetry.) [Two kinds of Airy functions, Ai and Bi.]

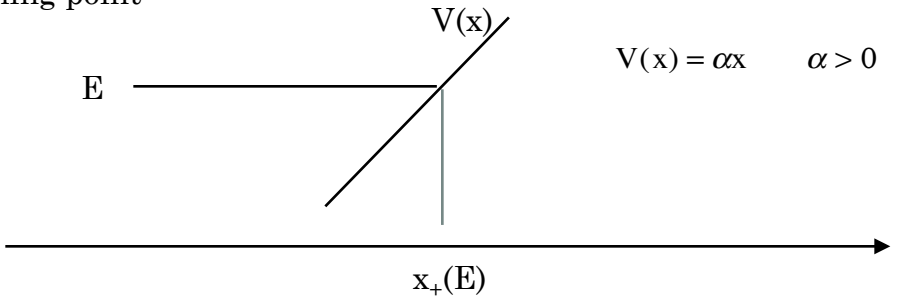
$$\mathrm{Ai}(z) = \pi^{-1/2}\int_0^{\infty} \cos\left(\frac{s^3}{3} + sz\right) ds$$

for our specific problem

$$s \equiv p(2m\hbar\alpha)^{-1/3} \qquad (if\ \alpha > 0)$$

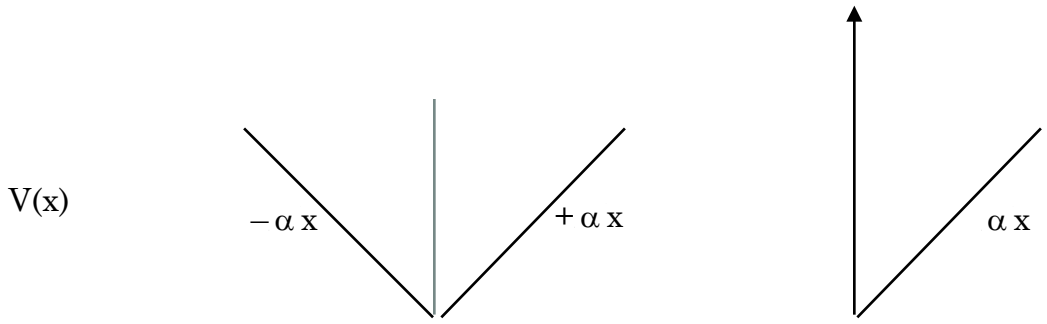
$$z \equiv \frac{(\alpha x - E)}{\alpha}\left[2m\alpha/\hbar^2\right]^{1/3}$$

Turning point



At a turning point $E = V(x_+) = \alpha x_+ \therefore \boxed{x_+(E) = E/\alpha}$

<u>Problems with linear potentials:</u> boundary conditions

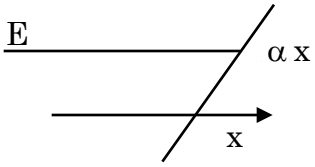


When there is symmetry (or 1/2 symmetry) we need to know the locations of the zeroes of $\underbrace{d\psi/dx}_{\text{for even functions}}$ and $\underbrace{\psi(x)}_{\text{for odd functions or } \infty \text{ barrier}}$.

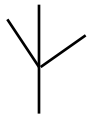
tables of zeroes of Ai(z) and Ai′(z)
"z_n" "z'_n"

See Handout

When there is no symmetry, must match or join Ai (or, more precisely, a linear combination of Ai and Bi) and Ai′ across boundaries, but we do not need to actually look at the Airy function itself near the joining point.



This is not as bad as it seems because we are usually far from the *turning* point at an internal *joining* point and can use *analytic asymptotic expressions* for Ai(z).



2 linear potentials of different |slope|.

For $\alpha > 0$ there are 2 cases (classical and non-classical regions)

(i) $z \ll 0, E > V(x)$ classically allowed region

$$\mathrm{Ai}(z) \to \pi^{-1/2}(\underbrace{-z}_{\text{positive}})^{-1/4}\sin\left[\frac{2}{3}(\underbrace{-z}_{\text{x is in here}})^{3/2} + \underbrace{\pi/4}_{\text{phase shift}}\right]$$

asymptotic form for $z \ll 0$.

* oscillatory, but wave vector, k, varies with x
* Ai vanishes as $x \to -\infty$ because of $(-z)^{-1/4}$ factor
* Bi is needed for case where Airy function must vanish as $x \to +\infty$ in classical region

(ii) well behaved

$z >> 0$, $E < V(x)$ forbidden region

$$\mathrm{Ai}(z) \to \left(\pi^{-1/2}/2\right)\underbrace{z^{-1/4}}_{\text{positive}}\underbrace{e^{-(2/3)z^{3/2}}}_{\text{decreasing exponential}}$$

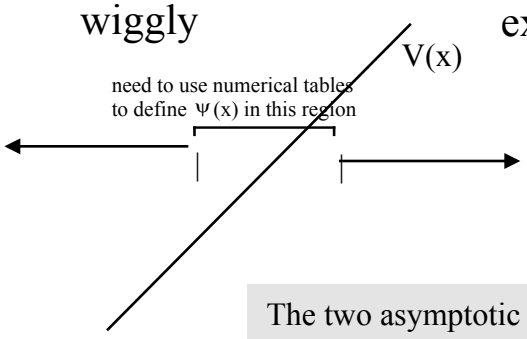
asymptotic form for $z \gg 0$.

* not oscillatory, monotonic
* Ai vanishes as $x \to +\infty$
* Bi vanishes as $x \to -\infty$ in forbidden region

Cartoon

damped wiggly

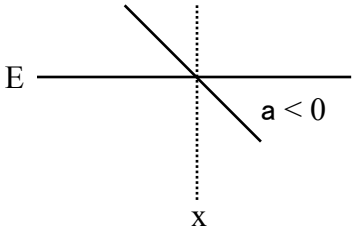
damped exponential



The two asymptotic forms of Ai(z) are not normalized, but their amplitudes (& phases) can and must be matched. This links boundary condition at $x \to +\infty$ to boundary condition at $x \to -\infty$.

NonLecture

OTHER CASE: $\alpha < 0 \rightarrow z \equiv -\frac{(|\alpha| x + E)}{|\alpha|}\left[2m|\alpha|/\hbar^2\right]^{1/3}$



for this case, we need Bi(z) instead of Ai(z)

$$Bi(z) \rightarrow \left(\pi^{-1/2}/2\right)|z|^{-1/4} \exp\left[-\frac{2}{3}|z|^{3/2}\right] \qquad \text{(forbidden region, } z \ll 0.\text{)}$$

$$Bi(z) \rightarrow \pi^{-1/2}|z|^{-1/4} \cos\left[\frac{2}{3}|z|^{3/2} + \frac{\pi}{4}\right] \quad \text{(allowed region, } z \gg 0.\text{)}$$

What is so great about $V(x) = \alpha x$? $\Psi(x)$ seems ugly — need lookup tables, complicated solutions!

But Ai(z) turns out to be key to generalization of quantization of *all* (well behaved) V(x)!

These are semi-classical JWKB $\Psi(x)$ functions — They blow up near turning points (i.e. on both sides). The Ai(z)'s permit matching of JWKB $\Psi(x)$s across the large gap where Ψ_{JWKB} is invalid, ill-defined.

(**J**EFFREYS)

WENTZEL

KRAMERS

BRILLOUIN

JWKB provides a way to get $\Psi_n(x)$ and E_n without solving differential equations or performing a Fourier Transform.

But actually, the differential equations are easy to solve numerically. The reason we care about JWKB is that it provides a basis for:

* physical interpretation (semi-classical)
* RKR inversion from $E_{vJ} \rightarrow V_J(R)$. [Rydberg, Klein, Rees]
* semi-classical quantization.
* the link to classical mechanics is essential for wavepacket pictures.

(generalize on e^{ikx} for free particle by letting $k = p(x)/\hbar$ depend explicitly on x (why does this not violate $[x,p]=i\hbar$?)

$$\psi_{JWKB} = \underbrace{|p(x)|^{-1/2}}_{\text{classical envelope}} \exp\left[\pm\frac{i}{\hbar}\int_c^x p(x')dx'\right]$$

$$p(x) = [2m(E - V(x)]^{1/2}$$

No violation because k(x) and p(x) are classical mechanical functions of x, not QM operators.

phase factor: choose c to satisfy boundary conditions

p(x) is pure real (classically allowed) or pure imaginary (classically forbidden). p(x) is not the Q.M. momentum. It is a classically motivated function of x, which has the form of the classical mechanical momentum and has the property that the $\lambda = \frac{h}{p}$ varies with x in a reasonable way.

* $|p(x)|^{-1/2}$ is probability amplitude envelope because

probability $\propto \frac{1}{v}$ so amplitude $\propto \sqrt{\frac{1}{v}}$ (v is velocity)

* $\exp-\left[\frac{i}{\hbar}\int_c^x p(x')dx'\right]$ is the generalization of $e^{ipx/\hbar}$ to non-constant V(x).

*node spacing $\frac{\lambda(x)}{2} = \frac{h}{2p(x)}$

**gives* easily identifiable stationary phase region for many wiggly integrands.

(Both ψ's have same λ at stationary phase point $x_{s.p.}$)

Long Nonlecture derivation/motivation of the JWKB splice across the turning point, even though the JWKB functions are not valid near the turning point.

Try $\psi(x) = N(x)\exp\left[\pm\frac{i}{\hbar}\int_c^x p(x')dx'\right]$

plug into Schr. Eq. and get a new differential equation that N(x) must satisfy

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - V(x))\psi = 0$$

$$\frac{d^2\psi}{dx^2} + \frac{1}{\hbar^2}p(x)^2\psi = 0 \qquad **$$

* derived in box below

$$0 = \left[N'' \pm \frac{2ip(x)}{\hbar}N' \pm \frac{ip'(x)}{\hbar}N\right]\exp\left[\pm\frac{i}{\hbar}\int_c^x p(x')dx'\right]$$

This is a new Schr. Eq. for N(x). Now make an approximation, to be tested later, that N″ is negligible everywhere. This is based on the expectation that a slowly varying V(x) will lead to a slowly varying N(x).

*
$$\frac{d\psi}{dx} = \left[N' \pm \frac{i}{\hbar}p(x)\right]\exp\left[\pm\frac{i}{\hbar}\int_c^x p(x')dx'\right]$$

$$\frac{d^2\psi}{dx^2} = \left[N'' \pm \frac{i}{\hbar}N'p \pm \frac{i}{\hbar}Np' \pm \frac{ip}{\hbar}\left(N' \pm \frac{i}{\hbar}Np\right)\right]\exp\left[\pm\frac{i}{\hbar}\int_c^x p(x')dx'\right]$$

$$= \left[N'' \pm \frac{2i}{\hbar}N'p \pm \frac{ip'}{\hbar}N - \frac{p^2}{\hbar^2}N\right]\exp\left[\pm\frac{i}{\hbar}\int_c^x p(x')dx'\right]$$

$$0 = \frac{d^2\psi}{dx^2} + \frac{p^2}{\hbar^2}N\exp\left[\pm\frac{i}{\hbar}\int_c^x p(x')dx'\right]$$

$$0 = \left[N'' \pm \frac{2ip(x)}{\hbar}N' \pm \frac{ip'}{\hbar}N\right]\exp\left[\pm\frac{i}{\hbar}\int_c^x p(x')dx'\right]$$

so, if we neglect N″, we get for the first term in [ ]

$2pN'+p'N=0$

if $p \neq 0$, then $2p^{1/2}\left[p^{1/2}N' + \frac{1}{2}p^{-1/2}p'N \right] = 0$

$$\frac{d\left(Np^{1/2}\right)}{dx} = \left[N'p^{1/2} + \frac{1}{2}p^{-1/2}p'N \right]$$

$$\therefore \frac{d\left(Np^{1/2}\right)}{dx} = 0$$

$$N(x)p^{1/2}(x) = \text{constant}$$

$$\boxed{\therefore\ \mathrm{N(x) = cp(x)^{-1/2}}}$$

OK, now we have a form for N(x) that we can use to tell us what conditions must be satisfied so that N″(x) is negligible everywhere.

$$N = cp^{-1/2}$$

$$\frac{dp^{-1/2}}{dx} = -\frac{1}{2}p^{-3/2}\frac{dp}{dx} \qquad p(x) = \left[2m\left(E - V(x)\right)\right]^{1/2}$$

$$\frac{dp}{dx} = \frac{1}{2}\left[2m\left(E - V(x)\right)\right]^{-1/2}(-2m)\frac{dV}{dx}$$

$$= \frac{1}{2}p^{-1}(-2m)\frac{dV}{dx} = -mp^{-1}\frac{dV}{dx}$$

$$\therefore \frac{dp^{-1/2}}{dx} = p^{-5/2}\frac{m}{2}\frac{dV}{dx}$$

$$\frac{d^2p^{-1/2}}{dx^2} = \frac{m}{2}\frac{dV}{dx}\left(-\frac{5}{2}\right)p^{-7/2}\left[-\frac{m}{p}\frac{dV}{dx}\right] + p^{-5/2}\frac{m}{2}\underbrace{\frac{d^2V}{dx^2}}_{\text{ignore}}$$

$$\therefore \boxed{N'' = c\frac{5}{4}m^2 p^{-9/2}\left(\frac{dV}{dx}\right)^2}$$

But we have made several assumptions about N″:

$$*\left|N''\right| \ll \left|\frac{2ip}{\hbar}N'\right| = \left|+\frac{icm}{\hbar}p^{-3/2}\frac{dV}{dx}\right|$$

$$*\left|N''\right| \ll \left|\frac{ip'}{\hbar}N\right| = \left|-\frac{icm}{\hbar}p^{-3/2}\frac{dV}{dx}\right|$$

$$*\left|N''\right| \ll \frac{p^2}{\hbar^2}N = \frac{c}{\hbar^2}p^{+3/2}$$

all of this is satisfied if

$$\left|\frac{5}{4}\frac{m\hbar}{i}\left(\frac{dV}{dx}\right)p^{-3}\right| \ll 1$$

Is this the JWKB validity condition? If it is, what does it mean?

Spirit of JWKB: if initial JWKB approximation is not sufficiently accurate, iterate:

$p(x) \rightarrow \psi_0(x)$ (ordinary JWKB)

$\psi_0(x) \rightarrow p_1(x)$

$p_1(x) \rightarrow \psi_1(x)$ (first order JWKB)

e.g. $\frac{d^2\psi_0}{dx^2} + \frac{p_1^2}{\hbar^2}\psi_0 = 0 \rightarrow p_1(x) = \left[-\frac{\hbar^2}{\psi_0(x)}\frac{d^2\psi_0}{dx^2}\right]^{1/2}$

see ** Eq. on p. 6-8

$$\psi_1(x) = \left|p_1(x)\right|^{-1/2}\exp\left[\pm\frac{i}{\hbar}\int_c^x p_1(x')dx'\right]$$

iterative improvement of accuracy

$p_1(x)$ is not smaller than $p_0(x)$, but it has more nearly correct wiggles in it.

END OF NONLECTURE

Resume Lecture

$$\psi(x) \approx \underbrace{|p(x)|}_{\text{envelope}}^{-1/2} \exp\left[\pm\frac{i}{\hbar}\int_c^x p(x')\,dx'\right]$$

adjustable phase shift.

provided that $\frac{d^2V}{dx^2}$ is negligible

AND

$$\underbrace{\frac{\hbar m}{|p|^3}\frac{dV}{dx} \ll 1}_{\text{required for } N''(x) \text{ to be negligible}} \left(\text{satisfied by } \lambda(x)\left|\frac{dp}{dx}\right| < |p(x)| \text{ or } \frac{d\lambda}{dx} \ll 1\right)$$

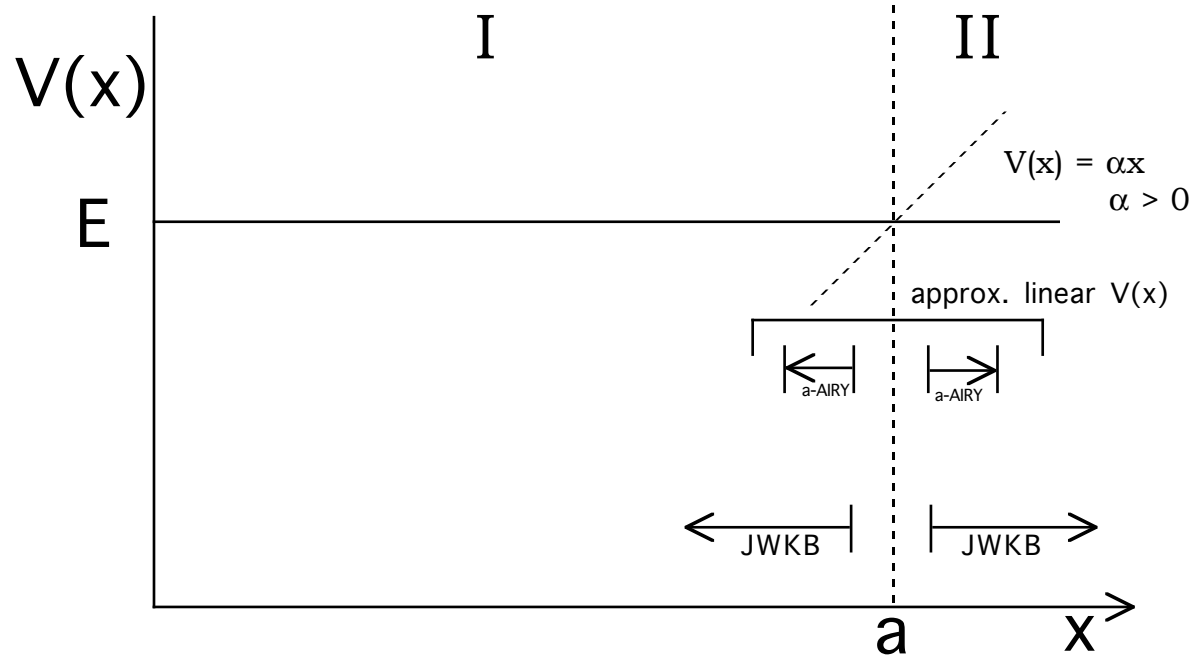
Next need to work out connection of $\Psi_{JWKB}(x)$ functions across region of x where the JWKB approx. breaks down (at turning points!).

$\left|\frac{d\lambda}{dx}\right| \to \infty$ at turning point because $p(x) \to 0$ Looks BAD!

BUT ALL IS NOT LOST — near enough to a turning point <u>all potentials</u> V(x) look like $V(x) = \alpha x$! We have Airy functions that are solutions to the Schrödinger Equation for this linear potential.

Now our job is to show that <u>asymptotic – AIRY</u> and <u>JWKB</u> are identical for a small region not too close and not too far on both sides of each turning point.

THIS PERMITS ACCURATE SPLICING OF $\Psi(x)$ ACROSS TURNING POINT REGION!



Region I E > V(x) classical

$$\psi^{I}_{a-AIRY} \sim \pi^{-1/2}(-z)^{-1/4}\sin\left[\frac{2}{3}(-z)^{3/2}+\pi/4\right]$$

First use Airy to splice across I,II junction

$$z=\frac{(\alpha x-E)}{\alpha}\left[\frac{2m\alpha}{\hbar^2}\right]^{1/3}$$

at turning point $E = V(a) = \alpha a$ so $\left[\frac{\alpha x - E}{\alpha}\right] = (x-a)$

$$\boxed{z=(x-a)\left(\frac{2m\alpha}{\hbar^2}\right)^{1/3} \ll 0 \qquad \text{when } x \ll a}$$

Region I/II splice using a-Airy.

Region II E < V(x) forbidden region, $z \gg 0$

$$\psi^{II}_{a-AIRY} \sim \frac{\pi^{-1/2}}{2} z^{-1/4} e^{-(2/3)z^{3/2}}$$

Now consider ψ_{JWKB} for a linear potential and show that it is identical to a-Airy!

$$\psi_{JWKB} \sim c_{\pm}|p(x)|^{-1/2}\exp\left[\pm\frac{i}{\hbar}\int_a^x p(x')dx'\right]$$

Then use WKB

both c_+ and c_- additive terms could be present

$$p(x) \equiv \left[2m(E-V(x))\right]^{1/2}$$

x < a	classical ,	p is real ,	ψ_{JWKM}	oscillates
x > a	forbidden ,	p is imaginary ,	ψ_{JWKB}	is exponential

pretend V(x) looks linear near x = a (ℓ-JWKB)

linear

$$p(x) = \left[2m\alpha(a-x)\right]^{1/2}$$

$$\int_a^x p(x')\,dx' = (2m\alpha)^{1/2}\int_a^x (a-x')^{1/2}\,dx'$$

$$= (2m\alpha)^{1/2}\left(-\frac{2}{3}\right)(a-x')^{3/2}\Big|_a^x$$

$$= -(2m\alpha)^{1/2}\frac{2}{3}(a-x)^{3/2}$$

Region I

$$\psi^I_{l-JWKB}(x) \sim |p(x)|^{-1/2}\left[Ae^{i\theta} + Be^{-i\theta}\right]$$

$$= |p(x)|^{-1/2} C\sin(\theta+\phi)$$

Define the JWKB phase factor, θ(x):

$$\theta = \frac{1}{\hbar}\int_a^x p(x')\,dx' = -\left(\frac{2m\alpha}{\hbar^2}\right)^{1/2}\frac{2}{3}(a-x)^{3/2}$$

Now compare θ(x) to z(x)

but, earlier, $z = (x-a)\left(\frac{2m\alpha}{\hbar^2}\right)^{1/3}$ $\quad\therefore \theta = -\frac{2}{3}(-z)^{3/2}$

$$\therefore p = (2m\alpha\hbar)^{1/3}(-z)^{1/2}$$ for exponential factor

$$|p|^{-1/2} = (2m\alpha\hbar)^{-1/6}(-z)^{-1/4}$$ for pre-exponential factor

Thus, putting all of the pieces together

$$\psi^{I}_{\ell-\mathrm{JWKB}} = \overbrace{-(2m\alpha\hbar)^{-1/6}(-z)^{-1/4}}^{-|p|^{-1/2}} C \sin\left[\overbrace{\frac{2}{3}(-z)^{3/2}}^{-\theta} - \phi\right]$$

$$= \psi^{I}_{a-\mathrm{AIRY}} \qquad \text{If } C = -(2m\alpha\hbar)^{1/6}\pi^{-1/2}, \quad \phi = -\pi/4$$

$\psi^{I}_{\ell-\mathrm{JWKB}}$ exactly splices onto $\psi^{I}_{a-\mathrm{AIRY}}$ with a $\boxed{\pi/4 \text{ phase factor}}$ (shifted from what the argument of sine would have been if one had started the phase integral at x = a

Similar result in Region II

$$\psi^{II}_{\mathrm{JWKB}} \sim Ae^{-f(x)} + Be^{+f(x)}$$

at $x \to +\infty$ $\quad f(x) \to \infty$ $\quad \therefore B = 0$

$$\therefore \psi^{II}_{\ell-\mathrm{JWKB}} = A(2m\alpha)^{-1/4}(x-a)^{-1/4}\exp\left[-\left(\frac{2m\alpha}{\hbar^2}\right)^{1/2}\frac{2}{3}(x-a)^{3/2}\right]$$

which is equal to $\psi^{II}_{a-\mathrm{AIRY}}$ if $A = (2m\alpha\hbar)^{+1/6}\pi^{-1/2}/2$

Final step: $\psi^{I}_{\mathrm{JWKB}} \leftrightarrow \psi^{I}_{a-\mathrm{AIRY}}$, $\psi^{II}_{\mathrm{JWKB}} \leftrightarrow \psi^{II}_{a-\mathrm{AIRY}}$

require A = –C/2

perfect match on opposite sides of turning point.

Ai(z) is valid in region where ψ_{JWKB} is invalid.

The logic is complicated, but the analysis assures that matching of Ψ^{I}_{JWKB} to $\Psi^{II}_{\mathrm{JWKB}}$ is valid and that one gets an extra π/4 phase factor at each turning point in the classically allowed region. This corresponds to the extra phase accumulated in the non-classical region so that $\Psi(\pm\infty) \to 0$. The energy levels are lowered below where they would have been if the wavefunctions in the classically allowed region were zero at turning points.

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JWKB QUANTIZATION CONDITION

Last time:

1. $V(x) = \alpha x$ $\quad \phi(p) = N \exp\left[-\frac{i}{\hbar\alpha}\left(Ep - p^3/6m\right)\right]$

$\psi(x) = Ai(z)$ * zeroes of Ai, Ai′ (and Bi, Bi′)
* tables of Ai (and Bi)
* asymptotic forms far from turning points

2. Semi-Classical Approximation for $\psi(x)$

* $p(x) = \left[(E - V(x))2m\right]^{1/2}$ — modifies classical to make it QM wavefunction (oscillatory)

"classical wavefunction" (nodeless)

* $\psi(x) = \underbrace{|p(x)|^{-1/2}}_{\text{envelope}} \exp\left[\underbrace{\pm\frac{i}{\hbar}\int_c^x p(x')dx'}_{\text{wiggly-variable } \phi(x)}\right]$ — adjustable phase to satisfy boundary conditions

* ψ without differential equation
* qualitative behavior of integrals (stationary phase)

* *validity*: $\frac{d\lambda}{dx} \ll 1$ *valid* when not too near a turning point.

[One reason for using semi-classical wavefunctions is that we often need to evaluate integrals of the type $\int \psi_i^* \hat{O}p \psi_j \, dx$. If $\hat{O}p$ is a slow function of x, the phase factor is $\exp\frac{i}{\hbar}\left[p_j(x') - p_i(x')\right]dx'$. Take $\frac{d}{dx}\left[\quad\right] = 0$ to find the stationary phase point $x_{s.p.}$.

δx is range about $x_{s.p.}$ over which phase changes by $\pm\pi/2$. Integral is equal to $I\left(x_{s.p.}\right)\delta x$.]

Logical Structure of pages 6-11 to 6-14 (not covered in lecture):

1. ψ_{JWKB} is not valid (it blows up) near turning point — ∴ we can't use ψ_{JWKB} to match ψ's on either side of turning point.

2. However, near a turning point, $x_\pm(E)$, every well-behaved V(x) looks like a linear poteintal

$$V(x) \approx V\left(x_\pm(E)\right) + \left.\frac{dV}{dx}\right|_{x=x_\pm}\left(x - x_\pm\right)$$ first term in a Taylor series.

This makes it possible to use Airy functions for *any* V(x) near turning point.

3. asymptotic-Airy functions have matched amplitudes (and phase) across the JWKB validity-gap that straddles the turning point.

4. Ψ_{JWKB} for a linear V(x) is identical to asymptotic-Airy!

It may be grubby, but it works!

TODAY

1. Summary of regions of validity for Airy, a-Airy, ℓ-JWKB, and JWKB on both sides of turning point. This seems complicated, but it leads to a result that will be exceptionally useful!

2. WKB quantization condition: energy levels without wavefunctions!

3. compute density of states dn_E/dE: (for box normalization — can then convert to any other kind of normalization)

4. trivial solution of Harmonic Oscillator
   $E_v = \hbar\omega\ (v + 1/2)\ \ v = 0, 1, 2\ldots$

Non-lecture (from pages 6-12 to 6-14)

classical $$\psi_{a-AIRY} = \pi^{-1/12}\left(\frac{2m\alpha}{\hbar^2}\right)^{-1/12}(a-x)^{-1/4}\sin\left[\frac{2}{3}\left(\frac{2m\alpha}{\hbar^2}\right)^{1/2}(a-x)^{3/2}+\frac{\pi}{4}\right]$$

forbidden $$\psi_{a-AIRY} = \frac{\pi}{2}^{-1/12}\left(\frac{2m\alpha}{\hbar^2}\right)^{-1/12}(x-a)^{-1/4}\exp\left[-\frac{2}{3}\left(\frac{2m\alpha}{\hbar^2}\right)^{1/2}(x-a)^{3/2}\right]$$

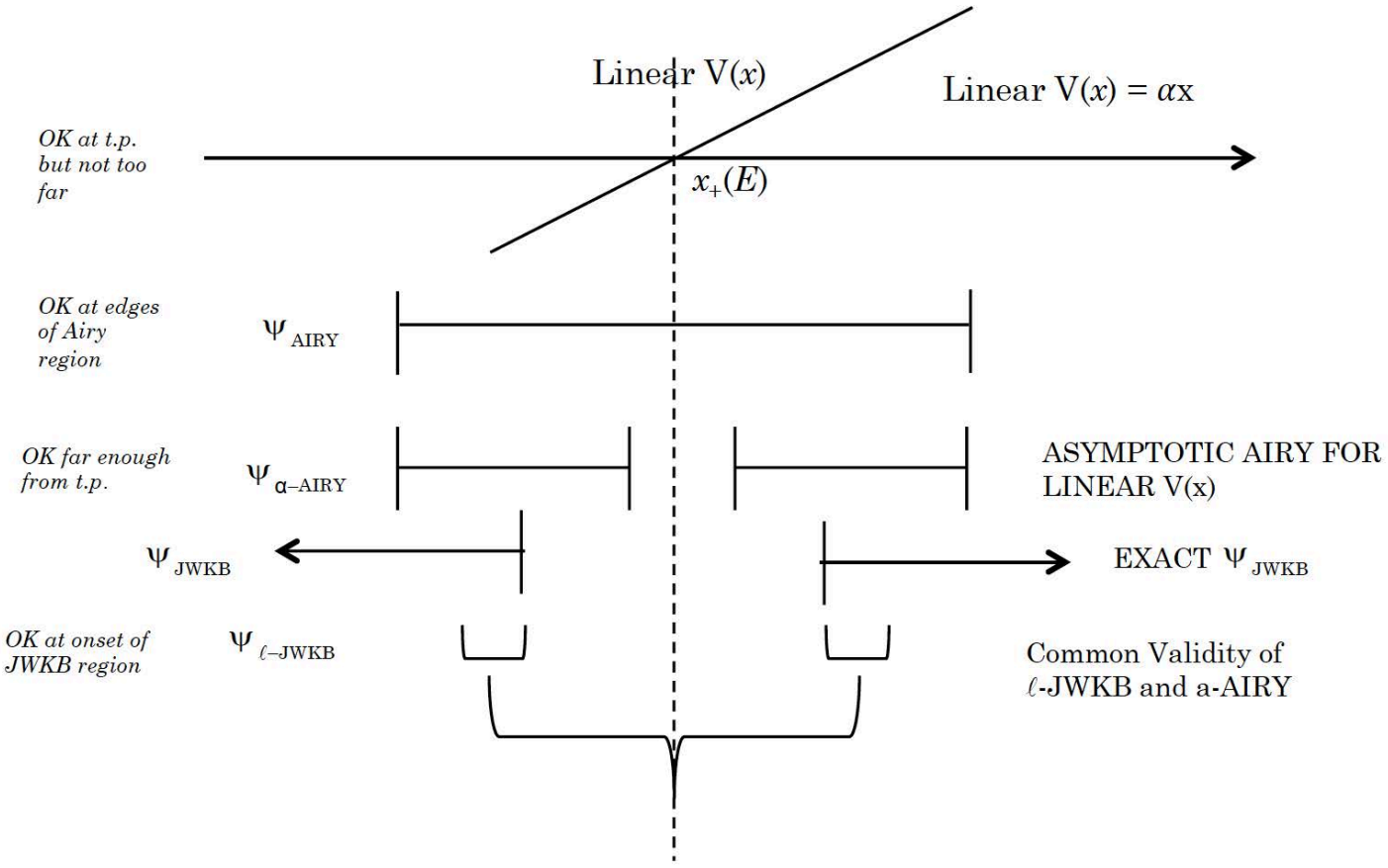
classical $$\psi_{\ell-JWKB} = \quad C \quad (a-x)^{-1/4}\sin\left[\frac{2}{3}\left(\frac{2m\alpha}{\hbar^2}\right)^{1/2}(a-x)^{3/2}+\phi\right]$$

forbidden $$\psi_{\ell-JWKB} = \quad D \quad (x-a)^{-1/4}\exp\left[-\frac{2}{3}\left(\frac{2m\alpha}{\hbar^2}\right)^{1/2}(x-a)^{3/2}\right]$$

C, D, and ϕ are determined by matching.

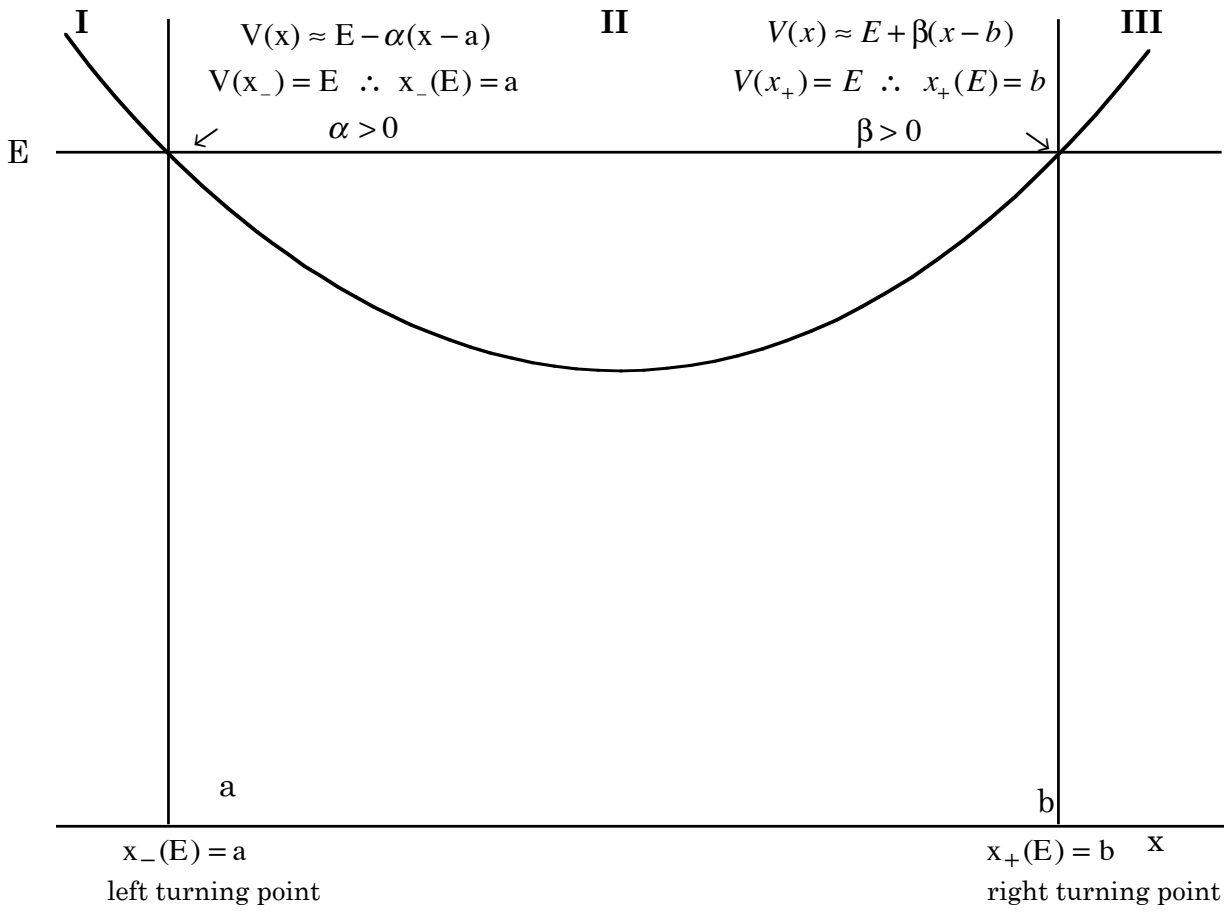
These Airy functions are not normalized, but each pair has the correct relative amplitude on opposite sides of the turning point. ℓ-JWKB has same functional form as a-Airy. This permits us to link pairs of JWKB functions across invalid-JWKB region and then use JWKB to extend $\psi(x)$ into regions further from turning point where the linear approximation to V(x) is no longer valid (and no longer required).

Regions of Validity Near Turning Point E = V($x_{\pm}$(E))



Common regions of validity for $\psi_{a\text{-AIRY}}$ and $\psi_{\ell\text{-JWKB}}$ — same functional form, specify amplitude and phase for $\psi_{JWKB}(x)$ valid far from turning point for exact V(x)!

Quantization of E in Arbitrary Shaped Wells



We already know how to splice across I, II and II, III regions, but how do we match ψ's in the entire a < x < b region? (ψ propagated inward from $x_-(E)$ must join smoothly onto ψ propagated inward from $x_+(E)$.)

Region I $$\psi^{I}_{JWKB}(x) = \frac{C}{2}|p(x)|^{-1/2}\, e^{-\frac{1}{\hbar}\int_x^a |p(x')dx'|} \qquad x < a$$ (forbidden region)

(ψ real, no oscillations)

Note carefully that the argument of $\exp\left[-\frac{1}{\hbar}\int_x^a |p(x')|\,dx'\right]$ goes to $-\infty$ as $x \rightarrow -\infty$, thus $\psi_I(-\infty) \rightarrow 0$.

Note also that $|\psi^I/C|$ increases monotonically as x increases up to x = a.

When you are doing matching for the first time, it is very important to verify that the phase of ψ varies with x in the way you expect it to vary.

Region II $\psi_{JWKB}^{IIa}(x) = C|p(x)|^{-1/2} \sin\left[\frac{1}{\hbar}\int_a^x p(x')dx' + \frac{\pi}{4}\right]$ $a < x < b$

The *first* zero is located at an accumulated phase of (3/4)π inside x = a because (3/4 + 1/4)π = π and sin π = 0. Why is this the first zero?

It does not matter that is invalid near x = and x = b.

Note that phase increases as x increases – as it must. The π/4 is the extra phase required by the AIRY splice across I,II. It reflects the tunneling of $\psi(x)$ into the forbidden region. *This means the real state with tunneling lies at lower energy than one that satisfies the incorrect $\psi(x_\pm) = 0$ boundary condition.*

|PHASE| starts at π/4 in classical region and always increases as one moves (further into the classical region) away from the turning point.

NEVER FORGET THIS!

Region III $\psi_{JWKB}^{III}(x) = \frac{C'}{2}|p(x)|^{-1/2} e^{-\frac{1}{\hbar}\int_b^x |p(x')|dx'}$ $x > b$

Note that phase advances monotonically (i.e. the phase integral gets more positive) as $x \to \infty$.

$\therefore |\psi_{JWKB}^{III}|$ decreases monotonically to 0 as $x \to +\infty$.

Region II again $\psi_{JWKB}^{IIb}(x) = C'|p(x)|^{-1/2} \sin\left[\frac{1}{\hbar}\int_x^b p(x')dx' + \frac{\pi}{4}\right]$

note: the argument of sine <u>starts at π/4</u> and <u>increases</u> as one goes from x = b inward. In other words, opposite to ψ^{IIa}, the argument decreases from left to right!

But it must be true that $\psi^{IIa}(x) = \psi^{IIb}(x)$ for all a < x < b !

There are 2 ways, C = C′,and C = –C′,to satisfy this requirement.

1. $\sin(\underbrace{\theta(x)}_{\text{argument of } \psi^{IIa}}) = \sin[\underbrace{(-\theta(x))}_{\text{argument of } \psi^{IIb}} + (2n+1)\pi]$ AND C = C′ $\left[\psi^{I}\right.$ and $\left.\psi^{II}\right.$ have the same sign.$\left.\right]$

$$[\sin\theta = -\sin(-\theta), \qquad \left[\sin\left(\theta + (2n+1)\right)\pi\right] = -\sin\theta,$$
$$\therefore \sin\theta = \sin\left(-\theta + (2n+1)\pi\right)]$$

2. $\sin(\theta(x)) = -\sin\left[-\theta(x) + 2n\pi\right]$ *if* $C = -C'$ $\left[\psi^{I}$ and ψ^{II} have opposite signs$\right]$

now look at what the 2 cases require for the arguments

1. For $C = C'$ $\left[\frac{1}{\hbar}\int_a^x p\,dx + \frac{\pi}{4}\right] = -\left[\frac{1}{\hbar}\int_x^b p\,dx + \frac{\pi}{4}\right] + (2n+1)\pi$ $n = 0,1,2\ldots$

ψ_a^{II} ψ_b^{II}

$$\therefore \frac{1}{\hbar}\left(\int_a^x + \int_x^b p\,dx\right) = (2n+1)\pi - \frac{\pi}{4} - \frac{\pi}{4}$$

$$\int_a^b p(x')dx' = \hbar\pi\left[2n + 1/2\right]$$ Quantization: $\frac{1}{2}, \frac{5}{2}, \frac{9}{2}, \ldots$

2. For $C = -C'$ we get $\int_a^b p(x')dx' = \hbar\pi\left[2n - 1/2\right]$ Quantization: $\frac{3}{2}, \frac{7}{2}, \frac{11}{2}, \ldots$

$\boxed{\hbar\pi = h/2}$

combine the two:

$$\int_a^b p(x')\,dx' = h/2\left(n + 1/2\right)$$
$$n = 0,1,2,\ldots$$
$$C' = C(-1)^n$$

** WKB quantization condition. Most important result of this lecture.

note: n = 0 is lowest possible value

n is # of internal nodes because the argument always starts at π/4 and increases inward to (n + 3/4) π at the other turning point.

		inner t.p.	outer t.p.	
for	n = 0	sin(π/r)	⟶ sin(3π/4)	NO INTERNAL NODE
	n = 1	sin(π/4)	⟶ sin(7π/4)	1 internal node
		etc.		

Node count tells what level it is. ∫ pdx at arbitrary E_{probe} tells how many levels there are at $E \le E_{probe}$!

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"Density of States" $\frac{dn}{dE}$ Always a crucial quantity $\left[h\frac{dn}{dE} \text{ is the } \underbrace{\text{classical mechanical period}}_{\text{large } \frac{dn}{dE}\text{, slow oscillation}} \text{ of oscillation.} \right]$

$$n(E)=\frac{2}{\mathrm{h}}\int_{x_-(E)}^{x_+(E)} p_E(x')\,dx' - \frac{1}{2}$$

$$\frac{dn}{dE}=\frac{2}{\mathrm{h}}\left[p_E\left(x_+\right)\frac{dx_+}{dE} - p_E\left(x_-\right)\frac{dx_-}{dE} + \int_{x_-}^{x_+}\frac{dp_E}{dE}\,dx \right]$$

(must take derivatives of the limits of integration as well as the integrand)

but $p_E\left(x_\pm\right)\equiv 0$

$$\therefore \frac{dn}{dE}=\frac{2}{\mathrm{h}}\int_{x_-}^{x_+}\frac{d}{dE}\left[2m\left(E-V(x')\right)\right]^{1/2}dx'$$

$$\frac{dn}{dE}=\frac{2}{h}\frac{1}{2}(2m)\int_{x_-}^{x_+}\left[2m\left(E-V(x')\right)\right]^{-1/2}dx'$$ A very widely useful quantity!

you show that, for harmonic oscillator,

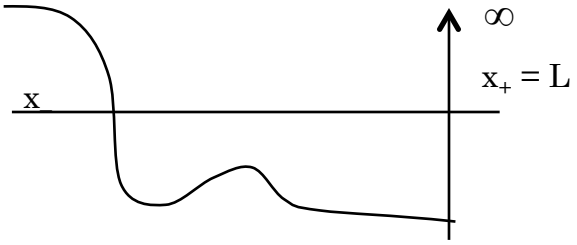
$$V(x)=\frac{1}{2}kx^2$$

$$\omega\equiv\left(k/m\right)^{1/2}$$

that $\frac{dn}{dE}=\frac{1}{\hbar\omega}$ independent of E, thus the oscillation period of the Harmonic Oscillator is independent of E.

Non-lecture

for general box normalization



can still use this to compute $\frac{\mathrm{dn}}{\mathrm{dE}}$ because

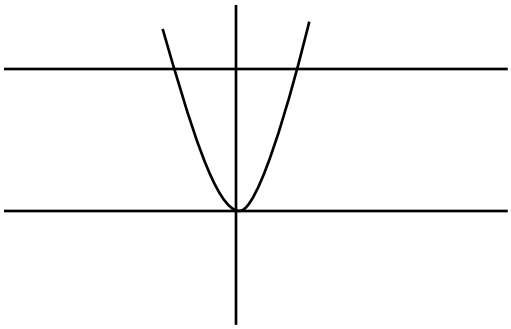
$\frac{\mathrm{dx}_+}{dE}=0$ (even though $p_E\left(x_+\right)\neq 0$).

↑ location of right hand turning point is independent of E.

Can always use WKB quantization to compute density of box-normalized Ψ_E's, provided that E > V(x) everywhere except at the 2 turning points.

Use WKB to solve a few "standard" problems. Since WKB is "semi-classical", we expect it to work in the $n \to \infty$ limit. There could be some errors for a few of the lowest-n E_n's.

Harmonic Oscillator $V(x) = kx^2/2$ (k is force constant, not wave vector)



$$p(x) = \left[2m\left(E - \frac{1}{2}kx^2 \right) \right]^{1/2}$$

At turning points, $V\left(x_{tp}\right) = E$ and $p\left(x_{tp}\right) = 0$,

thus, at turning points $x_{\pm} = \pm\left[2E_n/k\right]^{1/2}$

because $E_n = \frac{1}{2}kx_{\pm}^2$

$$\hbar\pi\left(n+1/2\right) = \int_{x_-=-\left[2E_n/k\right]^{1/2}}^{x_+=\left[2E_n/k\right]^{1/2}} \left[2m\left(E_n - kx^2/2\right) \right]^{1/2} dx$$

Non-lecture: Dwight Integral Table #350.01 $\quad t \equiv \left[a^2 - x^2\right]^{1/2}$

$$\int t\,dx = \frac{xt}{2} + \frac{a^2}{2}\sin^{-1}(x/a)$$

here $t = 0$ at both x_+ and x_-

$$I = \left(2mk/2\right)^{1/2} \int_{-\left[2E_n/k\right]^{1/2}}^{\left[2E_n/k\right]^{1/2}} \left[2E_n/k - x^2 \right]^{1/2} dx$$

$$I = \left(2mk/2\right)^{1/2} \left(\frac{2E_n}{k} \right) \left[\underbrace{\sin^{-1}1}_{\pi/2} - \underbrace{\sin^{-1}(-1)}_{-\pi/2} \right]$$

$$I = \left(\frac{m}{k} \right)^{1/2} E_n \left((\pi/2) - (-\pi/2) \right) = \pi\left(\frac{m}{k} \right)^{1/2} E_n$$

---

use the nonlecture result: $\quad \hbar\pi(n+1/2) = \pi\left(\frac{m}{k} \right)^{1/2} E_n$

$$E_n = \underbrace{\hbar\left(\frac{k}{m} \right)^{1/2}}_{\omega} (n+1/2)$$

I suggest you apply WKB Quantization Condition to the following problems: See Shankar pages 454-457.

Vee	$V(x) = a\|x\|$	$E_n \propto (n+1/2)^{2/3}$
quartic	$V(x) = bx^4$	$E_n \propto (n+1/2)^{4/3}$
$\ell = 0$, H atom	$V(x) = cx^{-1}$	$E_n \propto n^{-2}$
harmonic	$V(x) = \frac{1}{2}kx^2$	$E_n \propto (n+1/2)^1$

What does this tell you about the relationship between the exponents m and α in $V_m \propto x^m$ and $E_n \propto n^\alpha$?

Power of x in $V(x)$	Power of n in $E(n)$	
–1	-2	$\ell = 0$ H atom
1	2/3	Vee
2	1	Harmonic oscillator
4	4/3	Quartic oscillator

As power of x increases, power of n increases but slower.

Validity limits of WKB? surprisingly robust!

* splicing of ψ^{IIa}, ψ^{IIb} ? $\frac{d^2V}{dx^2}$ can't be too large near the splice region

* ψ_{JWKB} is bad when $\frac{d\lambda}{dx} \gtrsim 1$ (λ changes by more than itself for $\Delta x = \lambda$)
  bad near turning points and near the minimum of V(x)

* can't use WKB QC if there are more than 2 turning points

* bad near bottom of well $\frac{d^2V}{dx^2}$ is not small and $\frac{d\lambda}{dx} > 1$
  (near both turning points). However, most wells look harmonic near minimum and WKB gives exact result for harmonic oscillator - should be more OK near the minimum of V(x) than one has any right to expect.

* semi-classical: should be good in high-n limit. If exact E_n has same form as WKB QC at low-n, WKB E_n is valid for all n.

H.O., Morse Oscillator…

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5.73 Quantum Mechanics I
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Last time: WKB quantization condition for bound eigenstates of <u>almost general</u> V(x) — Connections into bound region from left and right

WKB Quantization Condition

$$\int_{x_-(E)}^{x_+(E)} p_E(x')dx' = \frac{h}{2}(n+1/2)$$

$$p_E(x) = \left[2m\left(E - V(x)\right)\right]^{1/2}$$

Error: no more than one extra level, raise E, p↑, nodes get closer together

E_n without ψ_n!

But where do we get V(x)?
Certainly not from femtochemistry!
From FREQUENCY DOMAIN SPECTROSCOPY

timing of w.p. as it moves on V(x)?

other stuff too

$E_{v,J} \rightarrow V(x)$? Energy levels as function of quantum numbers → potential energy as function of coordinate, $V(x)$.

RKR method

Next time: Numerical Integration of 1-D Schr. Eq. — See handouts.
Then we will begin working toward matrix picture

We will need background in Chapter 2 of CTDL,
pages 94-121 soon, apges 121-144 by next week

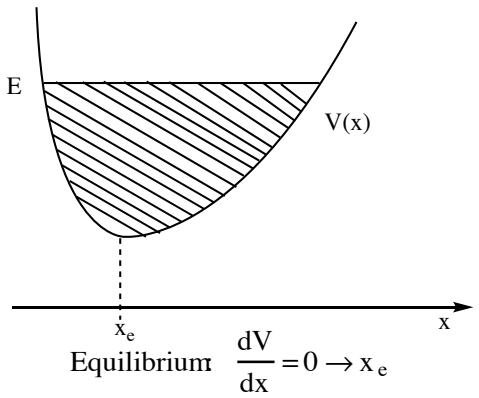
Postulates and Theorems will not be covered except as needed for solving problems

---

Today: $E_{v,J} \rightarrow$ spectroscopic notation : cm^{-1} units,

T electronic
G vibrational
F rotational

Area: $A(E,J) =$



Equilibrium $\frac{dV}{dx} = 0 \rightarrow x_e$

$\frac{\partial A}{\partial E}, \frac{\partial A}{\partial J}$ respectively determine $x_+(E) - x_-(E)$ and $\frac{1}{x_+(E)} - \frac{1}{x_-(E)}$

WKB Quantum Condition applied to $\frac{\partial A}{\partial E}, \frac{\partial A}{\partial J} \leftarrow G(v), B(v)$ used to determine $x_\pm(E)$.

($G(v)$: vibrational constants; $B(v)$: rotational constants)

Long Range Theory: Ultra Cold Collisions: Atom in Molecule

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Someday you will discover that the energy levels of a diatomic molecule are given by

$$E_{evJ}/hc = \underset{\text{electronic}}{T_e} + \underset{\text{vibration}}{G(v)} + \underset{\text{rotation}}{F_v(J)} \qquad \text{cm}^{-1}$$

$$= \nu_e + \underbrace{\left[Y_{00} + \omega_e(v+1/2) - \omega_e x_e (v+1/2)^2 + \ldots\right]}_{G(v)}$$

$$+ \underbrace{\left[B_e - \alpha_e(v+1/2) + \ldots\right]}_{B(v)} J(J+1) - DJ^2(J+1)^2$$

RKR requires only G(v) and B(v) spectroscopic data to get $V_J(x)$

$$\text{where } \underset{\substack{\text{J-dependent}\\ \text{effective}\\ \text{potential}}}{V_J(x)} = \underset{\substack{\text{bare}\\ \text{potential}}}{U(x)} + \underbrace{\frac{\hbar^2 J(J+1)}{2\mu x^2}}_{\substack{\text{centrifugal barrier}\\ \text{(actually rotational}\\ \text{kinetic energy)}}} \qquad x \equiv R - R_e$$

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$

We are going to derive $V_0(x)$ directly from G(v), B(v) data. This is the **only direct spectrum to potential energy function inversion method**! WKB quantization is the basis for this. It is easy to go from $V_0(x)$ to G(v) and B(v), but RKR is special. Many methods work in the opposite direction to get G(v) and B(v) from $V_0(x)$. We start with the WKB quantization condition:

$$\int_{x_-(E_v)}^{x_+(E_v)} p_{E_v}(x')dx' = (h/2)(v+1/2) \qquad v = 0,1,\ldots \#\text{ of nodes}$$

In this equation, what we know (E_v) and what we want (V(x) and x at turning points) are hopelessly intermixed. There is a trick to disentangle them!

$$A(E,J) \equiv \int_{x_-(E,J)}^{x_+(E,J)} \left[E - V_J(x')\right]dx'$$

area at E

but, still, we know neither $V_J(x)$ nor $x_{\pm}(E,J)$!

Roadmap: 1. Show that $\frac{\partial A}{\partial E}$ and $\frac{\partial A}{\partial J}$ are numerically evaluable integrals (via WKB QC) involving only $E_{v,J}$ information

data input here

2. independently, $\frac{\partial A}{\partial E}$ and $\frac{\partial A}{\partial J}$ determine

$[x_+(E,J) - x_-(E,J)]$ and $\left[\frac{1}{x_+(E,J)} - \frac{1}{x_-(E,J)}\right]$

2 eqs. in 2 unknowns give turning points

Do #2 first because it is so easy

$$\frac{\partial A}{\partial E} = \frac{\partial}{\partial E}\left[\int_{x_-(E,J)}^{x_+(E,J)} \left[E - U(x') - \frac{\hbar^2 J(J+1)}{2\mu x'^2}\right] dx'\right]$$

$$= \int_{x_-(E,J)}^{x_+(E,J)} 1\,dx' + \underbrace{0 + 0}_{\text{contributions from } \frac{\partial x_{\pm}(E,J)}{\partial E} \text{ are zero because integrand is 0 at both turning points}}$$

$$\therefore \boxed{\frac{\partial A}{\partial E} = x_+(E,J) - x_-(E,J)}!$$

$$\frac{\partial A}{\partial J} = \frac{\partial}{\partial J}\left[\int_{x_-(E,J)}^{x_+(E,J)} \left[E - U(x') - \frac{\hbar^2 J(J+1)}{2\mu x'^2}\right] dx'\right]$$

$$= -\frac{\hbar^2}{2\mu}\int_{x_-(E,J)}^{x_+(E,J)} \frac{2J+1}{x'^2}\,dx' + \underbrace{0 + 0}_{\text{integrand} = 0 \text{ at } x_{\pm}}$$

$$\boxed{\frac{\partial A}{\partial J} = -\frac{\hbar^2(2J+1)}{2\mu}\left[\frac{1}{x_+(E,J)} - \frac{1}{x_-(E,J)}\right]}$$

So, if we can evaluate these derivatives from E_{vJ} data, we have $V_J(x)$!

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some clever manipulations to put A(E,J) into convenient form
(see nonlecture notes on pages 8-5,6,7)

$$A(E,J) = \int_{x_-(E,J)}^{x_+(E,J)} \left[E - V_J(x')\right] dx' \quad \text{(change variables from x to v)}$$

$$A(E,J) = 2\pi \int_{\underbrace{v(E_{\min},J)}_{data}}^{v(E,J)} \left[E - \underbrace{E'_{vJ}}_{data} \right]^{1/2} dv$$

skipped steps are shown on pages 8-5, 6, 7.

this integral could be evaluated at any E, but we really only want $\frac{\partial A}{\partial E}$ and $\frac{\partial A}{\partial J}$. Evaluate these derivatives at J = 0.

$$\frac{\partial A}{\partial E} = \pi \int_{v(E_{\min},J)}^{v(E,J)} \left[E - E'_{vJ} \right]^{-1/2} dv + 0 + 0$$

lower limit independent of E

integrand = 0 at upper limit

$$v(E_{min},J) = -\frac{1}{2} - \frac{Y_{00}}{\omega_e}$$

defined so that $G(v_{min}) = 0$ this occurs at $v_{min} \neq -1/2$

$$\begin{aligned} G(v) &= Y_{00} + \omega_e (v + 1/2) \\ 0 = G(v_{min}) &= Y_{00} + \omega_e (v_{min} + 1/2) \\ -\frac{Y_{00}}{\omega_e} &= v_{min} + 1/2 \\ v_{min} &= -\frac{Y_{00}}{\omega_e} - \frac{1}{2} \end{aligned}$$

[v_{min} is slightly different from –1/2]

for J = 0 $\quad E'_{v,J} = G(v)$

$$Y_{00} = \frac{B_e - \omega_e x_e}{4} + \frac{\alpha_e \omega_e}{12 B_e} + \left[\frac{\alpha_e \omega_e}{12 B_e} \right]^2 \frac{1}{B_e}$$

$$\frac{\partial A}{\partial E} = \pi \int_{-1/2 - Y_{00}/\omega_e}^{v(E)} \left[E - G(v) \right]^{-1/2} dv \equiv 2 f(E)$$

data (upper limit of integration)

data (G(v))

evaluate this integral numerically at any E.

[Singularity at upper limit of integration fixed by change of variable: Zeleznik *JCP* **42**, 2836 (1965).]

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$$\left.\frac{\partial A}{\partial J}\right|_{J=0} = \pi\int_{-1/2-Y_{00}/\omega_e}^{v(E)} \left[E-G(v)\right]^{-1/2}\frac{\partial E}{\partial J}dv + 0 + 0$$

$$E = B_v J(J+1)$$

$$\frac{\partial E}{\partial J} = B_v(2J+1) \qquad \left.\frac{\partial E}{\partial J}\right|_{J=0} = B_v$$

data ← Use G(v) to get v as a continuous function of E.

$$\therefore \left.\frac{\partial A}{\partial J}\right|_{J=0} = \pi\int_{-1/2-Y_{00}/\omega_e}^{v(E)} \left[E-G(v)\right]^{-1/2} B_v\, dv \equiv -\pi 2 g(E)$$

data

(again, a nonfatal singularity at upper limit of integration)
f(E) and g(E) are "Klein action integrals" which are jointly determined by empirical G(v) and B(v) functions.

Nonlecture derivation of this useful form of $A(E,J)$:

$$A(E,J) = 2\pi\int_{v(E_{\min},J)}^{v(E,J)} \left[E - E'_{vJ}\right]^{1/2} dv$$

Begin here: $$A(E,J) = \int_{x_-(E,J)}^{x_+(E,J)} \left[E - V_J(x')\right]dx'$$

an integral identity: $$b-a = \frac{2}{\pi}\int_a^b \left(\frac{x-a}{b-x}\right)^{1/2} dx$$

let $b = E$

$a = V_J(x)$

$x = E'_{vJ}$

so that $$\left(\frac{x-a}{b-x}\right) = \frac{E_{vJ} - V_J(x)}{E - E_{vJ}}$$

$$\therefore \quad A(E,J) = \int_{x_-(E,J)}^{x_+(E,J)} [b-a]dx'$$

Now insert the integral identity

$$A(E,J)=\int_{x_-(E,J)}^{x_+(E,J)}\left(\frac{2}{\pi}\int_a^b\left[\frac{x-a}{b-x}\right]^{1/2}dx\right)dx' \qquad \text{put in values of a, b, and x}$$

$$=\int_{x_-(E,J)}^{x_+(E,J)}\left(\frac{2}{\pi}\int_{V_J(x)}^{E}\left[\frac{E'_{vJ}-V_J(x')}{E-E'_{vJ}}\right]^{1/2}dE'_{vJ}\right)dx'$$

Reverse order of integration and recognize the WKB QC in disguise:

$$=\frac{2}{\pi}\int_{V_J(x)}^{E}\left(\int_{x_-(E,J)}^{x_+(E,J)}\left[\frac{E'_{vJ}-V_J(x')}{E-E'_{vJ}}\right]^{1/2}dx'\right)dE'_{vJ}$$

Numerator of dx′ integral is QC — insert QC and then integrate by parts.
Denominator is independent of x′, so insert QC

$$\int_{x_-(E,J)}^{x_+(E,J)}\left[E'-V(x')\right]^{1/2}dx' = (2\mu)^{-1/2}\int_{x_-}^{x_+}p(x')dx'$$

$$=(2\mu)^{-1/2}\frac{h}{2}(v+1/2)$$

$$\therefore\quad A(E,J)=\left(\frac{2}{\pi}\right)(2\mu)^{-1/2}\frac{h}{2}\int_{E_{min}}^{E}\left[\frac{v(E',J)+1/2}{\left(E-E'_{vJ}\right)^{1/2}}\right]dE'_{vJ} \qquad **$$

** integrate by parts

$$f'=\left(E-E'_{vJ}\right)^{-1/2}$$

$$f=-2\left(E-E'_{vJ}\right)^{1/2}$$

$$g=\left[v\left(E'_{vJ},J\right)+1/2\right]$$

$$g'=\frac{dv}{dE'}, \qquad \text{which is known from } E_{vJ} \quad \left(\text{not a type because the variable is } E'_{vJ}, \text{ not } E\right)$$

$$A(E,J)=\underbrace{fg\Big|_{E'=E_{min}}^{E'=E}}_{=0 \text{ at both limits}}+\left(\frac{2h^2}{\mu}\right)^{1/2}\int_{E_{min}}^{E}2(E-E')^{1/2}\frac{dv}{dE'}dE'$$

(caution: f and g here are not Klein's action integrals)

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*** Change variables from dE' to dv'

$$dv = \frac{dv}{dE'} dE'$$

limits of integration become $\int_{v(E_{min},J)}^{v(E,J)}$

finished: $A(E,J) = 2\pi \int_{v(E_{min},J)}^{v(E,J)} \left[E - E'_{v',J} \right]^{1/2} dv'$

---

we have independent evaluations of both f(E) and g(E) from G(v) and B(v) data

one leads to $x_+(E,0) - x_-(E,0) = 2f(E)$

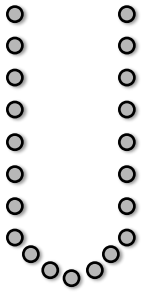
$$\frac{1}{x_+(E,0)} - \frac{1}{x_-(E,0)} = -2g(E)$$

pair of turning points $x_\pm(E,0) = \left[f(E)/g(E) + f(E)^2 \right]^{1/2} \pm f(E)$ from quadratic formula

so we get a pair of turning points at each E. Not restricted to E's with integer v's!

can use very fine grid of E's.

V(x)



connect the dots!

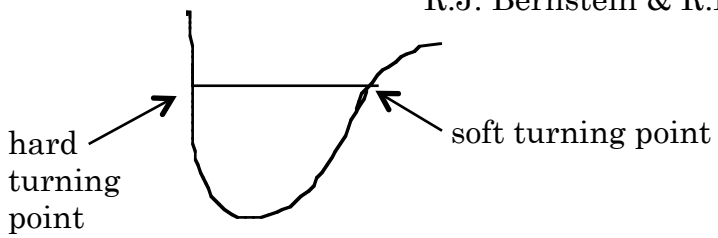
Robert LeRoy: modern, n-th generation RKR program at

http://leroy.uwaterloo.ca/programs.html

Download program, instructions, and sample data.

RKR does not work for polyatomic molecules because $E - V(\underset{\sim}{Q})$ does not determine the multicomponent vector $\vec{\mathbf{P}}$ (one component for each normal mode).

Long-Range Molecules
- W.C. Stwalley CPL 6, 241 (1970) (weak perturbation of atomic properties)
- R.J. Bernstein & R.LeRoy JCP 52, 3869 (1970)



What does $\psi(x)$ look like at very high v?

* lots of nodes (v nodes)
* *small* lobe at inner turning point. Why?
* *large* lobe at outer turning point. Why?

Hint: Force $= -\frac{dV(x)}{dx}$

At sufficiently large v, it is certain that $\psi(x)$ is dominated by outer-most lobe and any expectation value of a function of x, such as V(x), will be dominated by the region of the outer turning point. Since the vibrational Schrödinger equation contains V(x), it is evident that E_v at high v should be determined primarily by the long range part of V(x) (and should be insensitive to details near both x_e and the inner turning point).

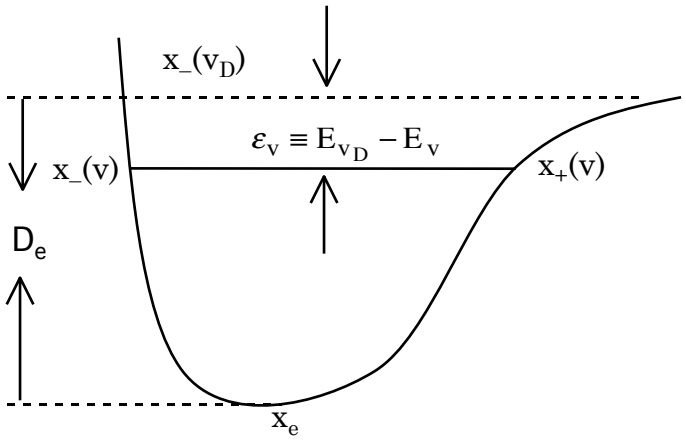
$$\therefore \qquad A(E,J) = \int_{x_-(E,J)}^{x_+(E,J)} [b-a]\,dx' \qquad \text{(See page 8-5)}$$

What do we know about covalent bonding?
ATOMIC ORBITAL OVERLAP IS REQUIRED!
NEGLIGIBLE OVERLAP at large x, V(x) at large x is determined by properties of isolated atoms: dipole moment, polarizability — return to this later when we do perturbation theory.

The lobe of $\psi(x)$ that we use to sample V(x) sees nearly pure atomic electronic properties.

It is always possible to predict what is the longest range term in $V(x) = C_n x^{-n}$ where the longest range term is the one with SMALLEST n.

Quick review of the Long-Range Theory



v_D is non-integer value of v of the "Level" at the "top" of the potential

ε_v is the binding energy of v-th level, $\varepsilon_v \equiv E_{v_D} - E_v$

at J = 0 $V_0(x) = U(x) = -C_n x^{-n}$ at long range (large x)

$U(\infty) = 0 \equiv E_{v_D}$

$U(x_e) = -D_e$

$x_+(v) = (-C_n / E_v)^{1/n}$ $\left[E_v = V(x_+(v)) = -C_n x_+^{-n}\right]$

$x_+(v_D) = \infty$

binding energy: $\varepsilon_v = E_{v_D} - E_v = C_n x_+^{-n}$

How many bound levels are there in the potential? WKB Quantization Condition:

$$\frac{h}{2}(v_D + 1/2) = \int_{x_-(v_D)}^{x_+(v_D)=\infty} p_D(x')dx'.$$

Now we do not know v_D, C_n, or D_e, but we do know n and know that E_v will be primarily determined by the long-range part of V(x) near v_D. So, for any E_v we expect that it will be possible to derive a relationship between

$(v_D - v)$ # of levels below highest bound level

and

$(E_{v_D} - E_v)$ binding energy.

By some clever tricks you may discover on Problem Sets #4 and #5, we find

$$v_D - v = a_n \varepsilon_v^{\frac{n-2}{2n}}.$$

This equation tells us how to plot E_v vs. v to extrapolate to v_D and then to obtain an accurate value of D_e from a linear plot near dissociation.

$V(x) = C_n x^{-n}$ for 2 Atoms [CTDL, page 283]

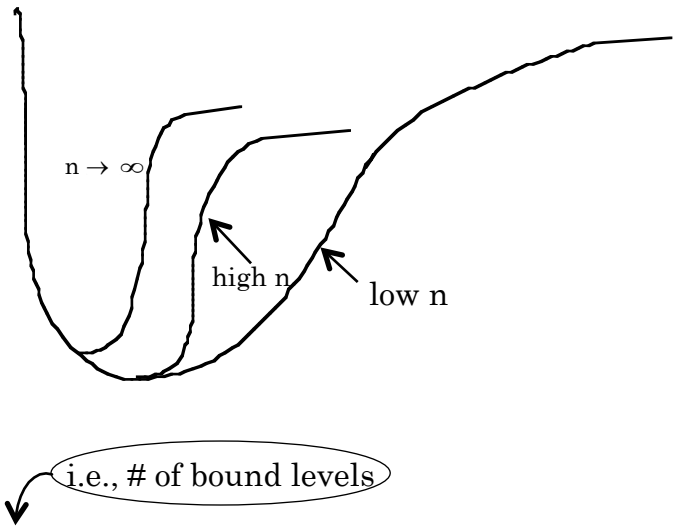
n=1 Both atoms are charged
2 One Atom is charged, the other atom has permanent electric dipole
3 2 identical uncharged atoms with transition dipole moments e.g. Na(^{2}P) + Na(^{2}S)
4 charge-induced dipole and dipole-quadrupole
5 quadrupole- quadrupole
6 induced dipole-induced dipole (London dispersion)

The interaction with smallest n is dominant at long-range

Not only is the limiting value of n known, but also C_n is known *because it is calculable from a measurable property of the free atom*. Many molecular states are described at long range by the same C_n's! Ultra-cold collisions are now used to determine V(x) to very large x. This has now become the best route to the properties of separated atoms!

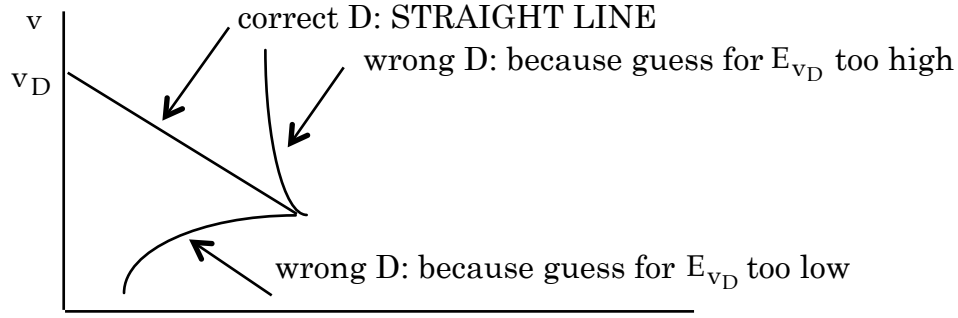
Mostly, long-range theory has been used as a guide to extrapolation to accurate dissociation energy (relevant to determination of ΔH_f°). Now Bose condensates. Molecule trapping.

x^{-1} and x^{-2} potentials have ∞ number of bound levels. x^{-3}, x^{-5}, and x^{-6} potentials have finite number, and the number of levels breaks off more abruptly as n increases.



action integral affected more by wider classical Δx region than by deeper ΔE binding region because $p \propto (E - V(x))^{1/2}$

This means (equation at bottom of page 8-9) that if we plot (given that we can predict n with certainty) as shown below.



0

$$\left(E_{v_D} - E_v\right)^{\frac{n-2}{2n}}$$

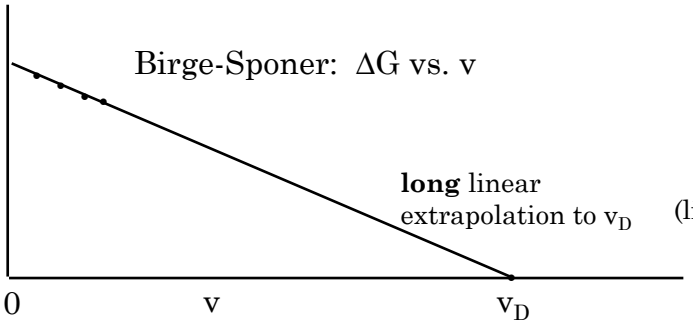
guessed known

n	$\frac{n-2}{2n}$
3	1/6
5	3/10
6	1/3
7	5/4

it is possible to determine D and v_D very accurately

much better than Birge – Sponer plot, which is valid only for a Morse potential

$\Delta G(v+1/2)=G(v+1) - G(v)$



(likely to be wrong!)

for Morse potential

Morse Potential

$$V_0(x) = D_e\left[1 - e^{-Ax}\right]^2$$

$$G(v) = \omega_e(v+1/2) - \omega_e x_e(v+1/2)^2$$

$\Delta G(v+1/2) = G(v+1) - G(v) = \omega_e - \omega_e x_e(2v+2)$ decreasing to 0 as v increases

when $\Delta G(v+1/2) = 0, \omega_e = \omega_e x_e(2v+2)$

$$v_D = \frac{\omega_e}{2\omega_e x_e} - 1$$

v_D is noninteger # of bound vibrational levels

$$D_e = G(v_D) = \frac{1}{4}\left[\frac{\omega_e^2}{\omega_e x_e} - \omega_e x_e\right]$$

$$= (v_D+1)\frac{\omega_e}{2} - \frac{\omega_e x_e}{4} \approx (v_D+1)\frac{\omega_e}{2}$$

But Morse potential inevitably has incorrect long-range form!

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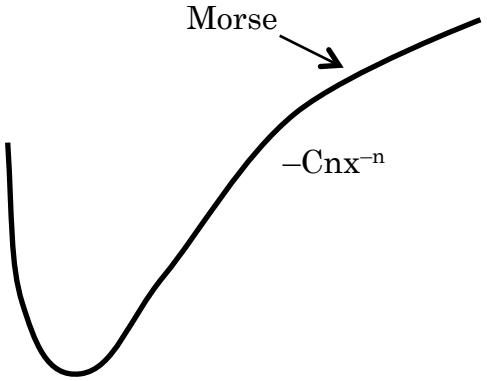
Which potential curve is longer range? Morse or $C_n x^{-n}$? Take ratio of binding energy at large x.

$$\lim_{x\to\infty} \frac{-C_n x^{-n}}{D_e\left[1-e^{-Ax}\right]^2 - D_e} = \lim_{x\to\infty} \frac{-C_n x^{-n}}{D_e\left[e^{-2Ax} - e^{-Ax}\right]}$$

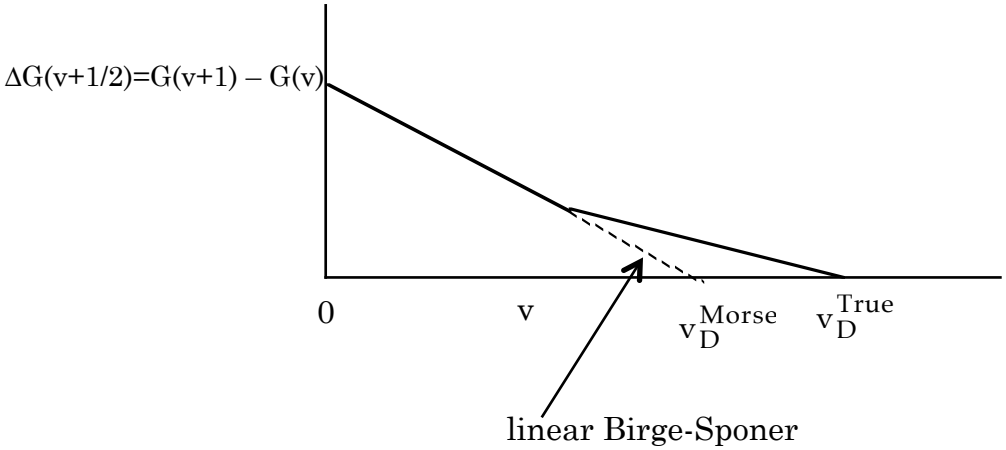
$$= \lim_{x\to\infty} \frac{-C_n x^{-n} e^{2Ax}}{\boxed{D_e - 2D_e e^{Ax}}} \leftarrow \text{dominant term}$$

$$= \lim_{x\to\infty} \frac{C_n}{2D_e} x^{-n} e^{Ax} \to \infty$$

This means that Morse potential binding energy gets small faster than $C_n x^{-n}$ for *any* value of n.



G(v+1) – G(v) will get small faster for Morse potential. Plot ΔG(v + 1/2) vs. v.



Dissociation energy is usually underestimated by linear Birge-Sponer extrapolation. Long-range plot of correct (*a priori* known) power of $E_{v_D} - E_v$ gives more accurate dissociation energy.

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5.73 Quantum Mechanics I
Fall 2018

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Numerov-Cooley Method : 1-D Schr. Eq.

Last time: Rydberg, Klein, Rees Method and Long-Range Model
G(v), B(v) rotation-vibration constants
↓
$V_J(x)$ potential energy curve
↓ $x = R - R_e$
$E_{v,J}, \psi_{v,J}$, all conceivable experiments including wavepackets, $wp(x,t)$

$$wp(x,t) = \sum_i a_i \psi_i e^{-E_i t/\hbar}$$

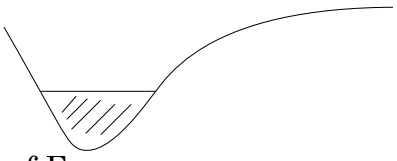
Treated in Lectures #19 and #20

a_i: Initial preparation of wp: $a_i = \int \psi_i^* \left[wp(x,0) \right] dx$

$e^{-E_i t/\hbar}$: determined by $V_J(x)$ free evolution of wp

Completeness: Any arbitrary wavefunction may be expanded as a linear combination of *eigenstates* of **H**.

Method for RKR derivation: A(E,J) = area of V(x) below E:



used WKB Quantization Condition

obtained $x_\pm(E,J)$ − a series of pairs of turning points on a grid of E.

Today: What do we do when we have a potential energy curve, $V_J(x)$, (especially when V(x) is not suited for WKB)?

Solve Schr. Eq. numerically!

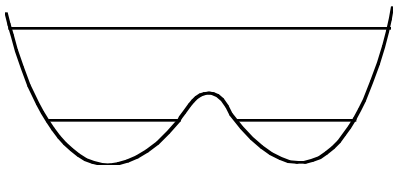
- No models
- 15 digit reproducibility
- cheap

This is the final tool we will develop for use in the Schrödinger representation. To summarize the classes of 1–D problem we are able to solve:

* piecewise constant potentials (matrix approach for joining at $\psi(x)$ boundaries)
* Airy functions (linear potential and joining JWKB across turning point) **[Not done in 2018]**
* JWKB (quantization condition and semi-classical wavefunctions)
* numerical integration (today)

See Merzbacher Quantum Mechanics pp. 92-103

Numerical Integration of the 1-D Schrödinger Equation



widely used
incredibly accurate
no restrictions on V(x) or on E–V(x) [e.g. nonclassical region, near turning points, double minimum potential, kinks in V(x).]
For most 1-D problems, where all one cares about is a set of $\{E_i, \psi_i\}$, where ψ_i is defined on a grid of points x_i, one uses Numerov-Cooley

leroy.uwaterloo.ca/programs.html

LEVEL: A computer program for solving the radial Schrödinger equation for bound and quasibound levels
R.J. Le Roy, *Journal of Quantitative Spectroscopy and Radiative Transfer* **186**, 167-178 (2016).
RKR1 – A Computer Program implementing the first-order RKR method for determining diatomic molecule potential energy functions
R.J. Le Roy, *Journal of Quantitative Spectroscopy and Radiative Transfer* **186**, 158-166 (2016).

J. A.Tellinghuisen, "Potential Fitting RKR Method: Semiclassical vs. Quantal Comparisons". *Journal of Molecular Spectroscopy***. 330**, 20-27 (2016).

Supplements for Lecture #9:

1. Cooley, Math. Comput. 15, 363 (1961).
2. Press et. al., Numerical Recipes, Chapters 16 and 17
3. Classic unpublished paper by Zare and Cashion with listing of Fortran program [now better to see LeRoy web site]
4. Tests of Numerov-Cooley vs. other methods by Tellinghuisen
5. LeRoy JQSRT paper

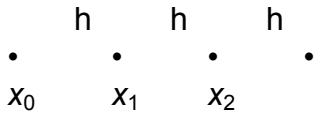
Basic Idea: grid method

* solve differential equation by starting at some x_i and propagating trial solution from one grid point to the next

* apply $\Psi(x) = 0$ boundary conditions at $x = 0$ and ∞ by two different tricks and then force agreement between $\psi_{left}(x)$ and $\psi_{right}(x)$ at some intermediate point by adjusting E.

Euler's Method

want $\psi(x)$ at a series of grid points $x_0, x_1, \ldots x_n = x_0 + nh$



call these $\psi_i = \psi(x_i)$

Need a "generating function," $f(x_n, \psi_n)$

$$\psi_{n+1} = \psi_n + h f(x_n, \psi_n)$$

prescription for going $n \to n+1$ must depend on both x_n and ψ_n. x_n samples potential, ψ_n samples previous value of ψ.

h ↑ increment in x

$x_{n+1} - x_n = h$ [NOT Planck's constant]

(ψ_n is a number, not the entire wavefunction.)
For the Euler method, the generating function is simply:

$$f(x_n, \psi_n) = \left.\frac{d\psi}{dx}\right|_{x_n} \approx \frac{\psi_{n+1} - \psi_n}{x_{n+1} - x_n} = \frac{\psi_{n+1} - \psi_n}{h}$$

The value of this derivative actually comes from the differential equation that ψ must satisfy, not from prior knowledge of $\psi(x)$ (which we do not yet have!)

For the Schrödinger Eqn. $\frac{d^2\psi}{dx^2} = -\frac{2\mu}{\hbar^2}(E - U(x))\psi$

All constants absorbed in V(x). V(x) must be in units of Å^{-2}.

note that $V(x) = -k(x)^2 = \frac{-p(x)^2}{\hbar^2}$

$$\frac{d^2\psi}{dx^2} = V(x)\psi(x)$$

$V(x) \equiv C[U(x) - E]$ $U(x)$ is the potential.

$C = 10^{-16}\left(8\pi^2 c\mu/h\right)$ h here is *not* increment of distance, — Planck

$= 0.0593203146\mu_A$ h and x are in Å. E and $U(x)$ are in

$$\left.\frac{d\psi}{dx}\right|_{x_i} \equiv \frac{\psi_{i+1} - \psi_i}{h}$$ — not Planck

$\left(\mu_A \text{ in } amu = \frac{\text{gram}}{mole}, {}^{12}C\right)$ cm^{-1} units (E/hc)

$$\left.\frac{d^2\psi}{dx^2}\right|_{x_i} = \left\{\left[\frac{\psi_{i+1} - \psi_i}{h}\right] - \left[\frac{\psi_i - \psi_{i-1}}{h}\right]\right\} \Big/ h$$

$$= h^{-2}\left[\psi_{i+1} - 2\psi_i + \psi_{i-1}\right]$$

$\mu_A = \frac{m_1 m_2}{m_1 + m_2}$

(Planck's constant h is in italics, grid spacing h is not italic.)

Schr. Eq. tells us the rule for propagating Ψ. Employing Euler's method (h is not Planck's constant):

$$h^{-2}\left[\psi_{i+1} - 2\psi_i + \psi_{i-1}\right] = V_i\psi_i$$ rearrange to solve for ψ_{i+1}

$$\psi_{i+1} - 2\psi_i + \psi_{i-1} = h^2 V_i \psi_i$$

$$\psi_{i+1} = 2\psi_i - \psi_{i-1} + h^2 V_i \psi_i$$ a recursion relationship. Need *both* ψ_i and ψ_{i-1} to get ψ_{i+1}.

In order to *get things started* we need two values of Ψ starting at either edge of the region where Ψ is defined and Ψ starts out very small.

[See Press et. al. handout (Numerical Recipes, Chapters 16 and 17) for discussion of nth-order Runge-Kutta method. The **generator** is chosen more cleverly than in the Euler method so that stepping errors are minimized by taking more derivatives at intermediate points in the x_i, x_{i+1} interval.]

Cooley specifies

$$y_{i+1} = 2y_i - y_{i-1} + h^2 V_i \psi_i$$

$$y_i = \left[1 - \left(h^2/12\right)V_i\right]\psi_i \text{ (and vice versa)}$$

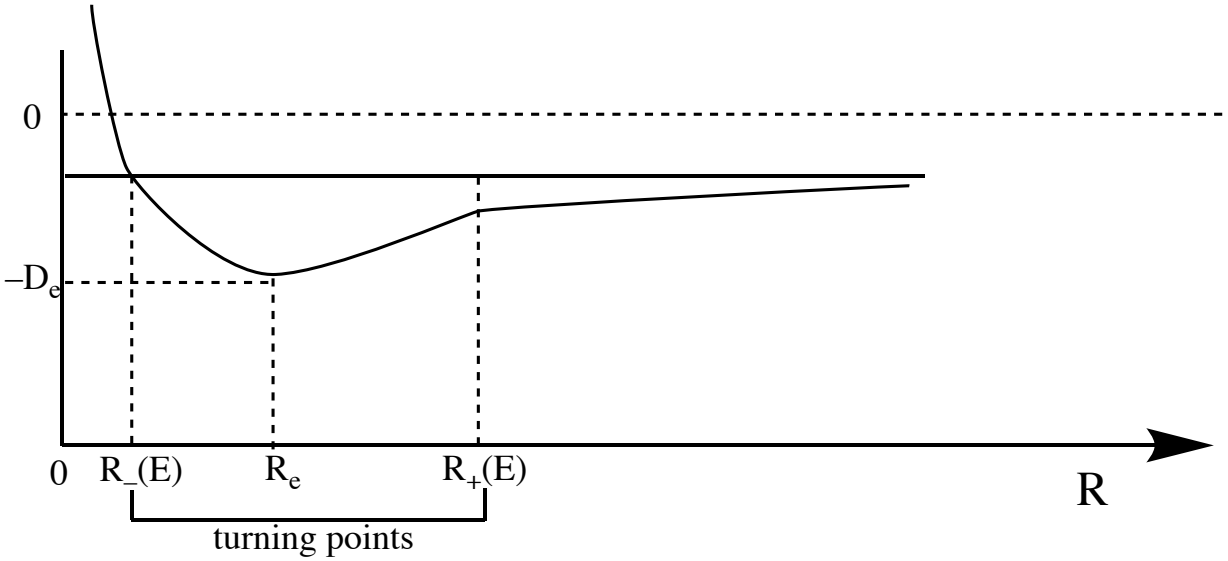
* use ψ_i to get y_i
* use ψ_i and y_i (and y_{i-1}) to get y_{i+1}
* use y_{i+1} to get ψ_{i+1}

The result is that the error in ψ_{i+1} is on the order of

$$\frac{h^6}{240}\psi_i V_i$$ — smaller error if h is smaller

(Cooley's method is much better than Euler's, which converges as a smaller power of h)

So what do we do?



$$\text{e.g., } V_{MORSE}(R) = D_e\left[1 - e^{-\beta(R-R_e)}\right]^2 - D_e \quad \text{[zero of V at dissociation]}$$

$$V(\infty) = 0, \; V(R_e) = -D_e$$

or, more generally than Morse, use $V_{J,RKR}$.

$x = R - R_e$ (displacement from equilibrium)

at $R = 0$ $\quad x = -R_e \quad \psi(-R_e) = 0$

$R = \infty \quad \psi(\infty) = 0$ } boundary conditions

The two boundary conditions are handled differently, because we want to define a finite # of equally spaced grid points (not actually necessary — see Press: he uses a variable grid spacing, which is needed to sample an infinite range of *x* with a finite number of grid points)

experience — doesn't really matter for ψ_1

* at $R = 0$ $\quad \psi_0 \equiv 0$ (required)

$\psi_1 = 10^{-20}$ (arbitrarily chosen small number, which will be corrected later upon normalization)

Use this to start the integration outward. When we discover later that we made a wrong choice for ψ_1, this is corrected merely by dividing *all* ψ_i $i \geq 1$ by an i-independent correction factor.

At large R (in the classically forbidden region), choose Ψ_n at the last grid point, x_n, to be small and use WKB *only once* to compute the *next to last grid point*. We do this because we have no reason to extend the calculation to $x \to \infty$.

$\Psi_n = 10^{-30}$ (the final grid point) (based on experience)

$$\Psi_{\text{WKB}} = |p|^{-1/2} \exp\left[-\frac{1}{\hbar} \int_{R_+(E)}^{x} |p| dx \right]$$

integrating inward from outer turning point.

$$V(x) \equiv -\frac{2m}{\hbar^2}\left(E - U(x)\right) = \frac{p^2}{\hbar^2}$$

$$|p_n| = \left[2m\left(E - U_n\right)\right]^{1/2} \equiv \hbar V_n^{1/2}$$

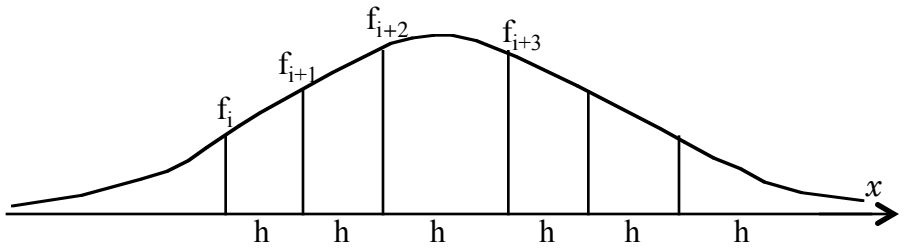
$$U_n = U\left(x_n\right)$$

WKB gives us the ratio

$$\frac{\Psi_{n-1}}{\Psi_n} = \frac{|p_{n-1}|^{-1/2} \exp\left[-\frac{1}{\hbar} \int_{R_+(E)}^{x_{n-1}} p_{n-1} dx \right]}{|p_n|^{-1/2} \exp\left[-\frac{1}{\hbar} \int_{R_+(E)}^{x_n} p_n dx \right]}$$

$p_{n-1} \approx p_n$ so pre-exponential terms cancel. Evaluate integral in exp as a sum.

$$\int_{x_-}^{x_+} f(x) dx = \sum_{i(x_-)}^{i(x_+)} \text{h} f(x_i)$$



and $e^{a+b+c} = e^a e^b e^c$, and $\dfrac{e^{a+b+c}}{e^{a+b+c+d}} = \dfrac{1}{e^d}$.

Thus, for $\dfrac{\psi_{n-1}}{\psi_n} = \dfrac{1}{\exp\left[-\dfrac{1}{\hbar}|p_n|\hbar\right]} = \exp\left[V_n^{1/2}\hbar\right]$ all of the corresponding terms in the sum cancel, leaving only the p_n term.

But we chose $\psi_n = 10^{-30}$

thus $\psi_{n-1} = 10^{-30}\exp\left[V_n^{1/2}\mathrm{h}\right]$

So we have ψ_n and ψ_{n-1} and we are off to the races.

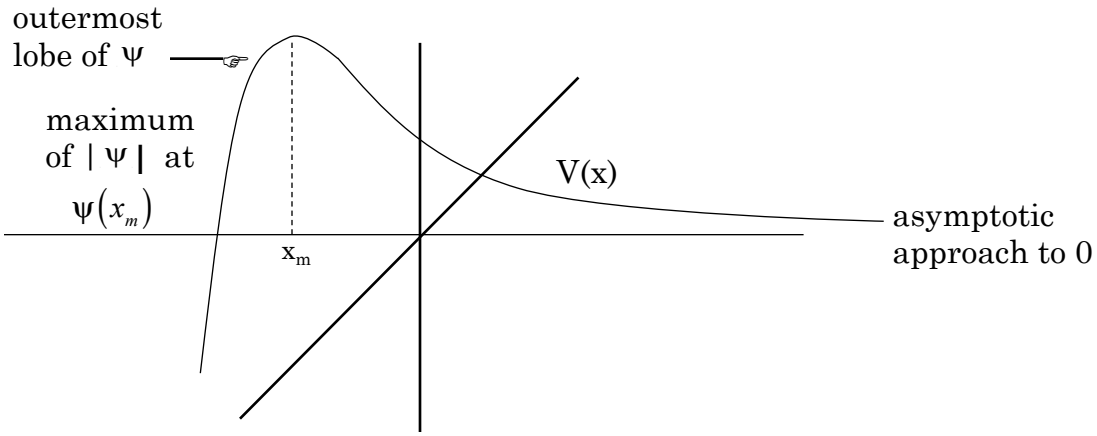
$$\psi_0 \rightarrow \psi_1 \rightarrow \psi_2 \rightarrow\rightarrow\rightarrow \psi_m \quad \Big| \quad \psi_m \leftarrow\leftarrow\leftarrow \psi_{n-1} \leftarrow \psi_n$$

m

meet in the middle at ψ_m.

Stop the inward propagation of ψ when a point is reached where, for the first time, $|\psi_m| > |\psi_{m+1}|$. This locates the *first maximum* of ψ inside the outer turning point.

Since $|\psi_i|$ is exponentially increasing from 10^{-30} at i=n until it reaches its *first maximum* inside the classically allowed region, this outer lobe of ψ is also the most important feature of ψ (because most of the probability resides in it).



Use the outermost lobe because this is the global maximum of Ψ(x). This choice minimizes the problem of precision being limited by the finite number of significant figures in the computer.

Set value of $\Psi_m = 1.0$ by renormalizing both functions

* ψ from n, n − 1, … m (from the right)	⋮	replace each ψ_i by ψ_i/ψ_m for all i down to m.
* ψ from i = 0, 1, … m (from the left)	⋮	replace each ψ_i by ψ_i/ψ_m for all i up to m.

The renormalized ψ's are denoted by Ψ'.

$$\psi'_i = \frac{\psi_i}{\psi_m} \qquad \psi'_m = 1$$

ψ′ must be continuous, especially at the joining grid point, m.

This ensures that $\psi(x)$ is continuous everywhere and that it satisfies grid form of Schr. Eq. everywhere *except* at $i = m$.

$$0 = \left(-\psi_{i+1} + 2\psi_i - \psi_{i-1}\right) + h^2 V_i \psi_i$$

In order to satisfy Schr. Eq. for $i = m$, it is necessary to adjust E. The above equation can be viewed as a nonlinear requirement on E. At the crucial grid point $i = m$, define an error function, F(E).

$$F(E) = \left(-\psi_{m+1}^E + 2\psi_m^E - \psi_{m-1}^E\right) + h^2 V_m^E \psi_m^E$$

where, by varying E, we want to search for the zeroes of F(E).

Assume that F(E) can be expanded about E_1 (E_1 is the initial, randomly chosen value of E.)

$$F(E) = F(E_1) + \left.\frac{dF}{dE}\right|_{E_1} (E - E_1) + \quad \text{discard higher terms (linearization approximation): Newton-Raphson}$$

and solve for the value of E where F(E) = 0.

Call this E_2

$$0 = F(E_1) + \left.\frac{dF}{dE}\right|_{E_1} (E_2 - E_1)$$

$$E_2 = \underbrace{-\frac{F(E_1)}{(dF/dE)_{E_1}}}_{\text{Correction to } E_1} + E_1$$

This gives an *estimate* of where the zero of F(E) nearest E_1 is located.

Usual approach: compute $\left.\frac{dF}{dE}\right|_{E_1} = \frac{F(E_1+\delta)-F(E_1)}{\delta}$

Once the derivative is known, use it to compute the trial correction to E_1 (assuming linearity).

Newton-Raphson method for solving nonlinear equation

$$E_2 = E_1 + \Delta \qquad \Delta \equiv -\frac{F(E_1)}{(dF/dE)_{E_1}}$$

Are we done? Not quite done if $F(E)$ is not sufficiently linear near E_i. Keep iterating!

Iterate until the correction, Δ, to E is smaller than a pre-set convergence criterion, ε. This seems like a lot of work, but the computer does not complain.

Now we have one not-yet-normalized eigenfunction of **H** and the corresponding eigenvalue, E.

Normalize ψ_E by dividing by $\left|\int \psi^* \psi \, dx\right|^{1/2} = N_E$

$$\int \psi^* \psi \, dx = \sum_{i=0}^{n} \left|\psi_i\right|^2 \mathrm{h}$$

Recall that normalization integral is evaluated by *summation* over grid points.

ψ real for bound 1-D function

box normalized: $\psi_E(x_i) = \dfrac{\psi_i}{\left[\sum_j \psi_j^2 \mathrm{h}\right]^{1/2}}$

This procedure has been used and tested by many workers. The latest and extensively documented version, "Level 16" is obtainable at Robert LeRoy's web site:

http://dx.doi.org/10.1016/j.jqsrt.2016.05.028

LEVEL: A computer program for solving the radial Schrödinger equation for bound and quasi-bound levels, Robert J. LeRoy.

I will assign some problems based on Numerov-Cooley method for integrating the 1-D Schr. Eq.

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5.73 Quantum Mechanics I
Fall 2018

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Matrix Mechanics

should have read CDTL pages 94-121
read CTDL pages 121-144 ASAP

Last time: * Numerov-Cooley Integration of 1-D Schr. Eqn. Defined on a Grid.
* 2-sided boundary conditions (two different kinds of boundary condition)
* nonlinear system - iterate to eigenenergies (Newton-Raphson)

So far focussed on $\psi(x)$ and Schr. Eq. as differential equation.
Variety of methods $\{E_i, \psi_i(x)\} \leftrightarrow V(x)$

Often we want to evaluate integrals of the form

overlap of special ψ with standard functions $\{\phi\}$

$$\int \psi^*(x)\phi_i(x)dx = a_i$$

a is "mixing coefficient"
ϕ_i is a member of a "complete" set of basis functions, $\{\phi\}$

OR

expectation values and transition moments

$$\int \phi_i^* \hat{x}^n \phi_j dx \equiv \left(x^n\right)_{ij}$$

There are going to be elegant tricks for evaluating these integrals and relating one integral to others that are already known. Also "selection" rules for knowing automatically which integrals are zero: symmetry, commutation rules

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Today: begin matrix mechanics - deal with matrices composed of these integrals - focus on manipulating these matrices rather than solving a differential equation - find eigenvalues and eigenvectors of *matrices* instead (COMPUTER "DIAGONALIZATION"). LINEAR ALGEBRA.

HIGHLIGHTS

* Perturbation Theory: tricks to find approximate eigenvalues of <u>infinite</u> matrices
* Wigner-Eckart Theorem and 3-j coefficients: use symmetry to identify and inter-relate values of nonzero integrals
* Density Matrices: information about "state of system" as separate from "measurement operators"

First Goal: Dirac notation as convenient NOTATIONAL simplification
It is actually a new abstract picture
(vector spaces) — but we will stress the *utility* of $\psi \leftrightarrow |\rangle$ relationships rather than the *philosophy*!

Find equivalent matrix form of standard $\psi(x)$ concepts and methods.

1. Orthonormality $\int \psi_i^* \psi_j dx = \delta_{ij}$

2. Completeness $\psi(x)$ is an arbitrary function

(expand ψ) A. Always possible to expand $\psi(x)$ uniquely in a COMPLETE BASIS SET $\{\phi\}$

$$\psi(x) = \sum_i a_i \phi_i(x)$$

a_i: mixing coefficient — how to get it?

* $a_i = \int \phi_i^* \psi dx$ left multiply by ϕ^*_i and integrate over x

(expand $\hat{B}\psi$) B. Always possible to expand $\hat{B}\psi$ in $\{\phi\}$ since we can write ψ in terms of $\{\phi\}$.
So simplify the question we are asking to $\hat{B}\phi_i = \sum_j b_j \phi_j$

What are the $\{b_j\}$? Multiply by $\int \phi_j^*$

$$b_j = \int \phi_j^* \hat{B} \phi_i \, dx \equiv B_{ji}$$

$$\hat{B}\phi_i = \sum_j \underline{B_{ji}} \phi_j$$

note counter-intuitive pattern of indices. We will return to this.

* The effect of any operator on ψ_i is to give a linear combination of ψ_j's.

3. Products of Operators

$$\left(\hat{A}\hat{B}\right)\phi_i = \hat{A}\left(\hat{B}\phi_i\right) = \hat{A}\sum_j B_{ji}\phi_j$$

can move numbers (but not operators) around freely

$$= \sum_j B_{ji}\hat{A}\phi_j = \sum_j \sum_k B_{ji}A_{kj}\phi_k \qquad \text{note repeated j-index}$$

$$= \sum_{j,k}\left(A_{kj}B_{ji}\right)\phi_k = \sum_k (\mathbf{AB})_{ki}\phi_k \qquad \text{note repeated k-index}$$

* Thus the product of 2 operators follows the rules of matrix multiplication:

$\hat{A}\hat{B}$ acts like **A B**

Recall rules for matrix multiplication:

$\left(\quad\right)\left(\quad\right)$ indices of a matrix are $A_{row,column}$

must match # of columns on left to # of rows on right

the order of matrices matters!

$$\left(N\times N\right) \otimes \left(N\times N\right) \rightarrow \left(N\times N\right) \quad \text{a matrix}$$

$$\left(1\times N\right) \otimes \left(N\times 1\right) \rightarrow \left(1\times 1\right) \quad \text{a number}$$

" row vector" " column vector"

$$\left(N\times 1\right) \otimes \left(1\times N\right) \rightarrow \left(N\times N\right) \quad \text{a matrix!}$$

column vector row vector

Need a notation that accomplishes all of this *memorably* and compactly.

Dirac's bra and ket notation

Heisenberg's matrix mechanics

ket $|\ \rangle$ is a column matrix , i.e. a vector $\begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_N \end{pmatrix}$

The ket contains all of the "mixing coefficients" for ψ expressed in some (implicit) basis set.
[These are projections onto unit vectors in N-dimensional vector space.]
Must be clear what state is being expanded in what basis

$$\psi(x) = \sum_i \overbrace{\left[\int \phi_i^* \psi dx\right]}^{a_i} \phi_i(x)$$ express $\{\psi\}$ basis in $\{\phi_i\}$ basis

$$|\psi\rangle = \begin{pmatrix} \int \phi_1^* \psi dx \\ \int \phi_2^* \psi dx \\ \vdots \\ \int \phi_N^* \psi dx \end{pmatrix}_{\phi} = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_N \end{pmatrix}$$

* ψ expressed in ϕ basis
* a column of complex #s
* nothing here is a function of x

ϕ ← bookkeeping device (RARELY USED) to specify basis set

OR, a pure state in its own basis

$$|\phi_2\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}_{\phi}$$ one 1, all others 0 (often expressed as $|2\rangle$)

$$|\psi\rangle = a_1 \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} + a_2 \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} + \ldots a_N \begin{pmatrix} 0 \\ \vdots \\ \vdots \\ 1 \end{pmatrix}$$

a weighted sum of unit vectors

bra $\langle\ |$ is a row matrix $(b_1, b_2 \ldots b_N)$

$\langle\ |$ contains all mixing coefficients for ψ^* in $\{\phi^*\}$ basis set

$$\psi^*(x) = \sum_i \left[\int \phi_i \psi^* dx\right] \phi_i^*(x)$$ (This is * of $\psi(x)$ above)

The * stuff is needed to make sure $\langle\psi|\psi\rangle = 1$ even though $\langle\phi_i|\psi\rangle$ is complex.

The symbol $\langle a|b\rangle$, a bra–ket, is defined in the sense of a product of $(1 \times N) \otimes (N \times 1)$ matrices $\rightarrow$ a 1×1 matrix: a number!

Box Normalization in both Ψ and $\langle\ |\ \rangle$ pictures

$$1 = \int \psi^* \psi \, dx$$

$$\psi = \sum_i \left(\int \phi_i^* \psi \, dx \right) \phi_i$$

take complex conjugate of ψ equation

$$\psi^* = \sum_j \left(\int \phi_j \psi^* \, dx \right) \phi_j^*$$

expand both in ortho-normal ϕ basis

$$1 = \int \psi^* \psi \, dx = \sum_{i,j} \left(\int \phi_j \psi^* \, dx \right) \left(\int \phi_i^* \psi \, dx \right) \int \phi_j^* \phi_i \, dx$$

c.c. $\qquad \delta_{ij}$

forces the 2 sums (over i and j) to collapse into 1 sum (over j)

$$1 = \sum_j \left| \int \phi_j^* \psi \, dx \right|^2$$

real, positive #'s

We have proved that sum of |mixing coefficients|^2 = 1. These mixing coefficients "squared" are called "mixing fractions" or "fractional characters".

now in $\langle\ |\ \rangle$ picture

$$\langle \psi | \psi \rangle = \underbrace{\left(\int \phi_1 \psi^* \, dx \quad \int \phi_2 \psi^* \, dx \quad \cdots \right)}_{\text{row vector: "bra"}} \underbrace{\begin{pmatrix} \int \phi_1^* \psi \, dx \\ \int \phi_2^* \psi \, dx \\ \vdots \end{pmatrix}}_{\text{column vector "ket"}} = 1 \times 1 \text{ matrix}$$

$$= \sum_j \left| \int \phi_j^* \psi \, dx \right|^2$$ same result as in wavefunction representation

[CTDL talks about "dual vector spaces" — best to walk before you run. Always translate $\langle\ \rangle$ into Ψ picture until you are sure you understand the notation.]

Any symbol $\langle\ \rangle$ is a complex number.
Any symbol $|\ \rangle\langle\ |$ is a square matrix.

$$\text{again } \langle\psi|\psi\rangle = \left(\langle\psi|\phi_1\rangle \quad \langle\psi|\phi_2\rangle \ldots\right)\begin{pmatrix} \langle\phi_1|\psi\rangle \\ \langle\phi_2|\psi\rangle \\ \vdots \end{pmatrix}$$

$$= \sum_i \langle\psi|\underbrace{\phi_i\rangle\langle\phi_i}|\psi\rangle = \langle\psi|\psi\rangle = 1$$

unit matrix $\mathbb{1}$

$$\text{what is } |\phi_1\rangle\langle\phi_1| = \left(1\,0\ldots 0\right)\begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & \cdots \\ 0 & 0 & \\ \vdots & & \ddots \end{pmatrix}$$

three dots are shorthand for specifying only the important part of an infinite matrix

$$\text{what is } \sum_i |\phi_i\rangle\langle\phi_i| = \begin{pmatrix} 1 & & & 0 \\ & 1 & 0 & \\ & 0 & 1 & \\ 0 & & & \ddots \end{pmatrix}$$

unit or identity matrix = $\mathbb{1}$

Large zero (0) denotes a lot of zeroes.

"completeness" or "closure" involves insertion of $\mathbb{1}$ between any two symbols.

Use **1** to evaluate the matrix elements of the product of 2 operators, **AB** (we know how to do this in ψ picture).

$$\langle \phi_i | \mathbf{A} | \phi_j \rangle = (0 \ldots 1 \ldots 0)(\mathbf{A}) \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}$$

i-th; square matrix; j-th position – picks out j-th column of **A**

$$= (0 \ldots 1 \ldots 0) \begin{pmatrix} A_{1j} \\ A_{2j} \\ \vdots \end{pmatrix} = A_{ij}$$

i-th; picks out the i-th element of a column vector

$$\langle \phi_i | \mathbf{AB} | \phi_j \rangle = \sum_k \langle \phi_i | \mathbf{A} \underbrace{| \phi_k \rangle \langle \phi_k |}_{\mathbb{1}} \mathbf{B} | \phi_j \rangle$$

$$= \sum_k A_{ik} B_{kj} = (\mathbf{AB})_{ij}$$

a number (obtained by matrix multiplication)

In the Heisenberg picture, how do we get an exact equivalent of $\psi(x)$? Use basis set $\delta(x,x_0)$ for all x_0 – this is a "complete" basis (eigenbasis for $\hat{x}$, eigenvalue x_0) - perfect localization at any x_0

This $\langle x | \Psi \rangle$ symbol is the same thing as $\Psi(x)$

↑ $\left(\text{i.e., } \int \delta(x, x') * \psi(x') dx' = \psi(x) \right)$

x is continuously variable ↔ $\delta(x)$

Overlap of state vector Ψ with $\delta(x)$ – a complex number. $\Psi(x)$ is a complex function of a real variable.

other $\Psi \leftrightarrow \langle \; | \; \rangle$ relationships

1. All observable quantities are represented by a Hemitian operator (Why – because the expectation values of a Hermitian operator are always real). Definition of Hermitian operator:

   For a matrix: $A_{ij} = A^*_{ji}$ or $\mathbf{A} = \mathbf{A}^\dagger$

   Easy to prove that if all expectation values of **A** are real, then $\mathbf{A} = \mathbf{A}^\dagger$ and vice-versa

2. <u>Change of basis set</u>

   $\mathbf{A}^\phi \leftrightarrow \mathbf{A}^u$ $\{\phi\}$ to $\{u\}$

$$A^\phi_{ij} \equiv \left\langle \phi_i \middle| A \middle| \phi_j \right\rangle = \left\langle \phi_i \middle| \mathbf{1}A\mathbf{1} \middle| \phi_j \right\rangle$$

$$= \sum_{k,\ell} \left\langle \phi_i \middle| u_k \right\rangle \left\langle u_k \middle| A \middle| u_\ell \right\rangle \left\langle u_\ell \middle| \phi_j \right\rangle$$

$S^*_{ki} \equiv S^\dagger_{ik}$ (k: $\uparrow$ u, i: ϕ)

$S_{\ell j} \leftarrow \phi$ (ℓ: $\uparrow$ u)

S is Frequently used to denote an "overlap" integral

This is the j-th column of **S**

$$= \sum_{k,\ell} S^\dagger_{ik} A^u_{k\ell} S_{\ell j} = \left(\mathbf{S}^\dagger A^u \mathbf{S} \right)_{ij} \equiv A^\phi_{ij}$$

(u basis ↓ k,ℓ; ϕ basis ↓ ij)

$$A^\phi = S^\dagger A^u S$$

$\mathbf{S}^\dagger \cdots \mathbf{S}$ is a special kind of transformation (unitary)

(different from more-familiar $\mathbf{T}^{-1}\mathbf{AT}$ "similarity" transformation)

For a state vector (ket):

$$\left|\phi_j\right\rangle = \sum_{\ell} \left|u_\ell\right\rangle\left\langle u_\ell\middle|\phi_j\right\rangle = \sum_{\ell=1}^{N} \left|u_\ell\right\rangle S_{\ell j} = \begin{pmatrix} S_{1j} \\ \vdots \\ S_{Nj} \end{pmatrix}$$

$$\left|\phi_j\right\rangle = \begin{pmatrix} S_{\ell j} \\ \vdots \\ S_{Nj} \end{pmatrix}$$ This is the j-th column of **S**.

The linear combination of $\left|u_i\right\rangle$ for each $\left|\phi_j\right\rangle$ is the j-th column of S. Also, the linear combination of $\left|\phi_j\right\rangle$ for each $\left|u_i\right\rangle$ is the i-th column of $S^\dagger$. [This is a very useful thing to remember.]

i-th

$$\left|u_i\right\rangle = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}_u = \begin{pmatrix} S^\dagger_{1i} \\ \vdots \\ S^\dagger_{\mathbf{N}i} \end{pmatrix}_\phi = S^\dagger_{1i}\begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} + S^\dagger_{2i}\begin{pmatrix} 0 \\ 1 \\ \vdots \\ \vdots \end{pmatrix} + \ldots S^\dagger_{\mathbf{N}i}\begin{pmatrix} 0 \\ \vdots \\ \vdots \\ 1 \end{pmatrix}$$

pure state in $\{u\}$ basis

mixed state in $\{\phi\}$ basis

pure state in $\{\phi\}$ basis

What kind of matrix is **S**?

$$S_{\ell j} = \left\langle u_\ell \,|\, \phi_j \right\rangle$$

$$S^*_{\ell j} = \left[\left\langle u_\ell \,|\, \phi_j \right\rangle \right]^* = \left\langle \phi_j \,|\, u_\ell \right\rangle \equiv S^\dagger_{j\ell}$$

† means take complex conjugate and interchange indices.

Using the definitions of S and $S^\dagger$:

$$S_{\ell j} S^\dagger_{jk} = \left\langle u_\ell \middle| \phi_j \right\rangle \left\langle \phi_j \middle| u_k \right\rangle$$

$$\sum_j S_{\ell j} S^\dagger_{jk} = \sum_j \left\langle u_\ell \middle| \phi_j \right\rangle \left\langle \phi_j \middle| u_k \right\rangle = \left\langle u_\ell \middle| \mathbb{1} \middle| u_k \right\rangle = \delta_{\ell k}$$

$$= \left\langle u_\ell \middle| u_k \right\rangle = \delta_{\ell k} = \mathbb{1}_{\ell k}$$

identity matrix

$\mathbf{S}\mathbf{S}^\dagger = \mathbb{1}$ OR $\mathbf{S}^\dagger = \mathbf{S}^{-1}$ "Unitary"

a very special and convenient property.

$\mathbf{S}^\dagger$ is inverse of **S**!

Unitary transformations preserve both normalization and orthogonality.

$$\mathbf{A}^{\phi} = \mathbf{S}^{\dagger}\mathbf{A}^{u}\mathbf{S}$$

$$\mathbf{S}\mathbf{A}^{\phi}\mathbf{S}^{\dagger} = \mathbf{S}\mathbf{S}^{\dagger}\mathbf{A}^{u}\mathbf{S}\mathbf{S}^{\dagger} = \mathbf{A}^{u}$$

$$\mathbf{A}^{u} = \mathbf{S}\mathbf{A}^{\phi}\mathbf{S}^{\dagger}$$

Take matrix element of both sides of equation:

$$A^{u}_{ij} = \langle u_i \,|\, \mathbf{A} \,|\, u_j \rangle = \left(\mathbf{S}\mathbf{A}^{\phi}\mathbf{S}^{\dagger}\right)_{ij}$$

$$= \sum_{k,\ell} \mathbf{S}_{ik} \langle \phi_k \,|\, \mathbf{A} \,|\, \phi_\ell \rangle \mathbf{S}^{\dagger}_{\ell j}$$

$$\therefore |u_j\rangle = \sum_{\ell} |\phi_\ell\rangle \mathbf{S}^{\dagger}_{\ell j} \qquad |u_j\rangle \text{ is j-th column of } \mathbf{S}^{\dagger}$$

$$\phi \rightarrow u \; via \; \mathbf{S}^{\dagger}, \mathbf{S} : |u_j\rangle \text{ is j-th column of } \mathbf{S}^{\dagger}$$

Thus,

$$\text{j-th} \quad \left|u_j\right\rangle = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}_u = \begin{pmatrix} S^\dagger_{1j} \\ S^\dagger_{2j} \\ \vdots \\ \vdots \\ S^\dagger_{nj} \end{pmatrix}_\phi$$

Similarly,

$$\mathrm{A}^{\phi}_{\mathrm{pq}} = \left\langle \phi_p \middle| \mathrm{A} \middle| \phi_q \right\rangle = \left(\mathrm{S}^\dagger \mathrm{A}^u \mathrm{S}\right)_{pq}$$

$$= \sum_{mn} S^\dagger_{pm} \left\langle u_m \underbrace{\middle| \mathrm{A} \middle| u_n \right\rangle}_{\left|\phi_q\right\rangle} S_{nq}$$

$$\therefore \left|\phi_q\right\rangle = \sum_n \left|u_n\right\rangle S_{nq} \qquad q\text{-th column of S}$$

$$\left|\phi_{\mathrm{q}}\right\rangle = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}_\phi = \begin{pmatrix} S_{1q} \\ S_{2q} \\ \vdots \\ S_{nq} \end{pmatrix}_u \quad \text{q-th}$$

Commutation Rules

* $\left[\hat{A},\hat{B}\right] = \hat{A}\hat{B} - \hat{B}\hat{A}$

e.g. $\left[\hat{x},\hat{p}\right] = i\hbar$ means $\left(\hat{x}\hat{p}\text{-}\hat{p}\hat{x}\right)\phi = x\frac{\hbar}{i}\frac{d\phi}{dx} - \frac{\hbar}{i}\left(\phi + x\frac{d\phi}{dx}\right) = -\frac{\hbar}{i}\phi = \hbar i\phi$

$= i\hbar\phi$

* If $\hat{A}$ and $\hat{B}$ are Hermitian, is $\hat{A}\hat{B}$ Hermitian?

$$\left(AB\right)_{ij} = \sum_k A_{ik}B_{kj} = \sum \overbrace{A^*_{ki}B^*_{jk}}^{\text{Hermitian } \mathbf{A} \text{ and } \mathbf{B}} = \sum B^*_{jk}A^*_{ki} = \left(BA\right)^*_{ji}$$

but this is **not** what we need to be able to show that **AB** is Hermitian:
That would be: $(\mathbf{AB})_{ij} = (\mathbf{AB})^*_{ji}$ *or* $\mathbf{AB} = (\mathbf{AB})^\dagger \neq (\mathbf{BA})^\dagger$

AB is Hermitian only if $\left[\mathbf{A},\mathbf{B}\right] = 0$

However, $\frac{1}{2}\left[\mathbf{AB} + \mathbf{BA}\right]$ is Hermitian if both **A** and **B** are Hermitian.

This is a foolproof way to construct a new Hermitian operator out of simpler Hermitian operators.

This is the standard prescription for implementing the Correspondence Principle for constructing a quantum mechanical equivalent of a classical mechanical quantity. Quantities that commute in classical mechanics do not always commute in quantum mechanics. *Almost everything that is not classical mechanical in quantum mechanics is derivable from* $\left[\hat{x},\hat{p}_x\right] \neq 0$!

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